

발 간 등 록 번 호

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2023 K-Innovation Partnership Program with Cambodia

Policy Consultation and Pilot Implementation for the Development of the National R&D Management System and the R&D Centre of Excellence in Cambodia

2023 K-Innovation Partnership Program with Cambodia

Title	Policy Consultation and Pilot Implementation for the Development of the National R&D Management System and the R&D Centre of Excellence in Cambodia
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Executive Summary

The main objective of the K-Innovation ODA Programme is to build and enhance the STI policy capacities of each partner country, ranging from design and implementation, to evaluation and feedback. The K-Innovation ODA Programme's wealth of professional and academic experience has enabled the Kingdom of Cambodia (Cambodia) project to produce important national strategies and technology roadmaps.

This is the first year of the Phase 3 project of the K-Innovation Programme with the Ministry of Industry, Science, Technology & Innovation (MISTI) of the Royal Government of Cambodia (RGC). The Phase 1 project, conducted from 2018 to 2019, established Cambodia's first national plan on science and technology and resulted in the launch of "STI Policy 2020-2030" as the first official STI Master Plan of the RGC. Subsequently, the Phase 2 project (2020 to 2022) developed six national technology roadmaps (Education, Health, Agriculture, Digital, Energy, and Tourism) related to the implementation of the National STI Master Plan. Despite the COVID-19 restrictions, the project was managed and conducted effectively thanks to the joint efforts of STEPI and MISTI. The Phase 3 project, which will be carried out from 2023 to 2025, is focused on the establishment of a national R&D management system in Cambodia.

The consultation process for the project in 2023 consisted of the following main activities:

As a first step, STEPI and MISTI held a kick-off meeting on 7 February 2023 and agreed on the key activities and schedule of the 2023 programme. Then, STEPI participated in Cambodia's first "National STI Day" event held on 24-31 March 2023. During this event, STEPI and MISTI held a kick-off workshop for the third phase of the programme and a launch ceremony for the three TRMs, which were endorsed during the opening ceremony by H.E. PM Mr. Hun Sen. He underlined the importance of the TRMs as a compass for Cambodia to plan and develop a strategy for acquiring the right

technologies and to develop technology products that will satisfy the demands and needs of Cambodia. STEPI also interviewed and established networks with key stakeholders relating to national R&D management from MISTI and universities.

The second step consisted of a training workshop, held from May 22 to 26, 2023, in Seoul. The Science and Technology Policy Institute (STEPI) invited nine officials from Cambodia to enhance their skills and competencies in policymaking for the national R&D management system, including NTIS and COMPAS from the Korea Institute of Science and Technology Information (KISTI), as well as R&D funding and economic development experiences through science, technology, and innovation (STI) with the National Research Foundation (NRF) of Korea and the Korea Institute of Science and Technology Evaluation and Planning (KISTEP). The workshop program also included on-site visits to KISTI, the Global Knowledge Exchange & Development Centre of the Korea Development Institute (KDI), and STEPI. Additionally, a bilateral meeting was held between the Ministry of Science, Innovation, Technology, and Telecommunications (MISTI) and the Ministry of Science and ICT of Korea to discuss potential areas of collaboration.

As a third step, STEPI and MISTI surveyed the status of and demand for national R&D management in Cambodia, and the both parties signed a contract for the survey on 20 July 2023. On 13 September of the same year, STEPI and MISTI reviewed a draft of the survey questionnaire prepared by MISTI, and conducted the survey between 28 September and 31 October. Thereafter, MISTI submitted a draft report of the survey and revised the report to reflect the comments made by the STEPI team. The report identified the direction and key features of the national R&D management system envisioned by Cambodia's government and its research community. The report will be used as a reference material in the design of the national R&D management system for the RGC.

Finally, overall project efficiency and effectiveness were carefully managed throughout the project process. STEPI and MSITI have worked together very closely to implement the project's key activities, including the preliminary survey for the design of the national R&D management system in Cambodia.

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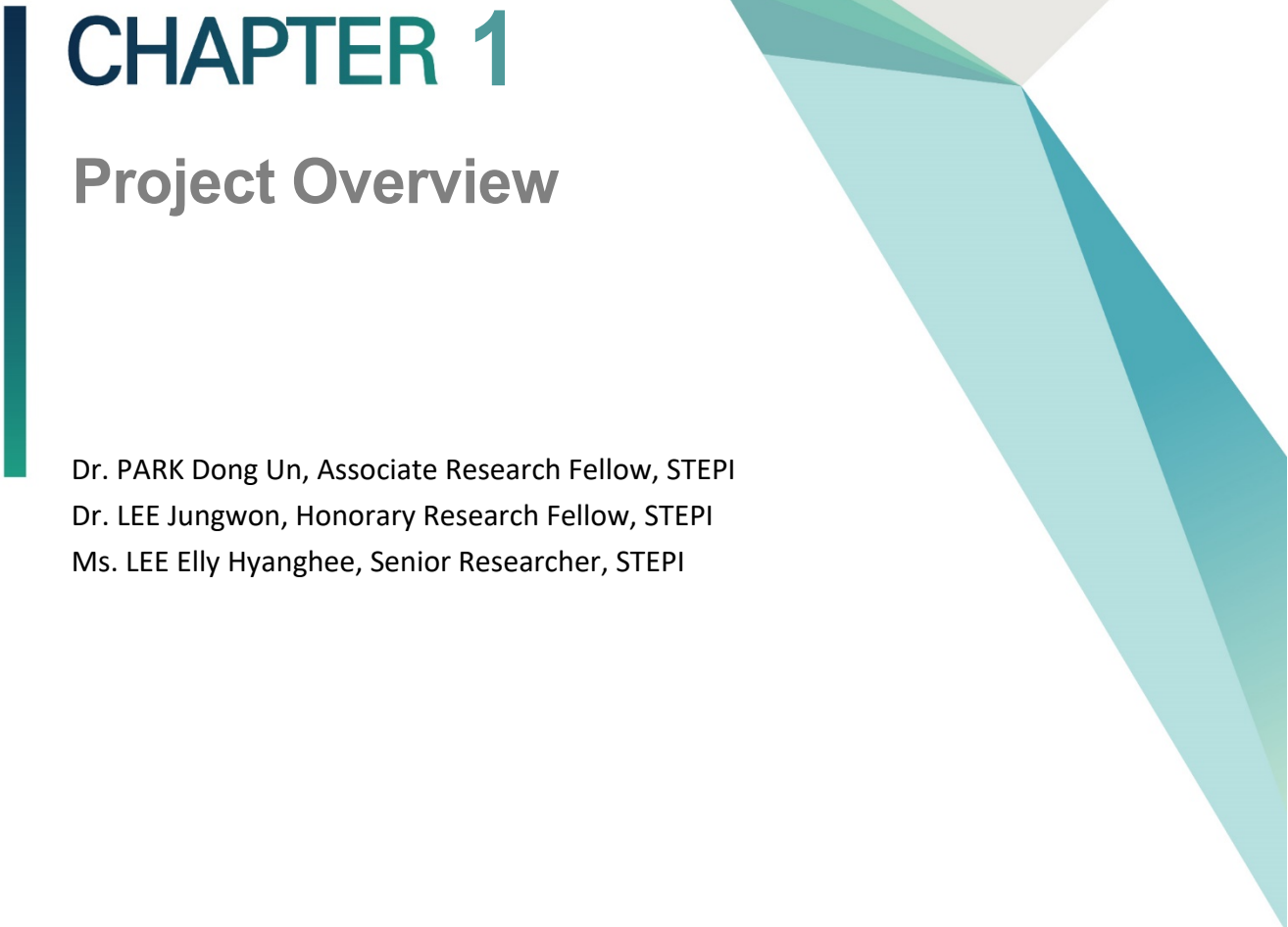
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CHAPTER 1

Project Overview

Dr. PARK Dong Un, Associate Research Fellow, STEPI

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Ms. LEE Elly Hyanghee, Senior Researcher, STEPI



Chapter 1. Project Overview

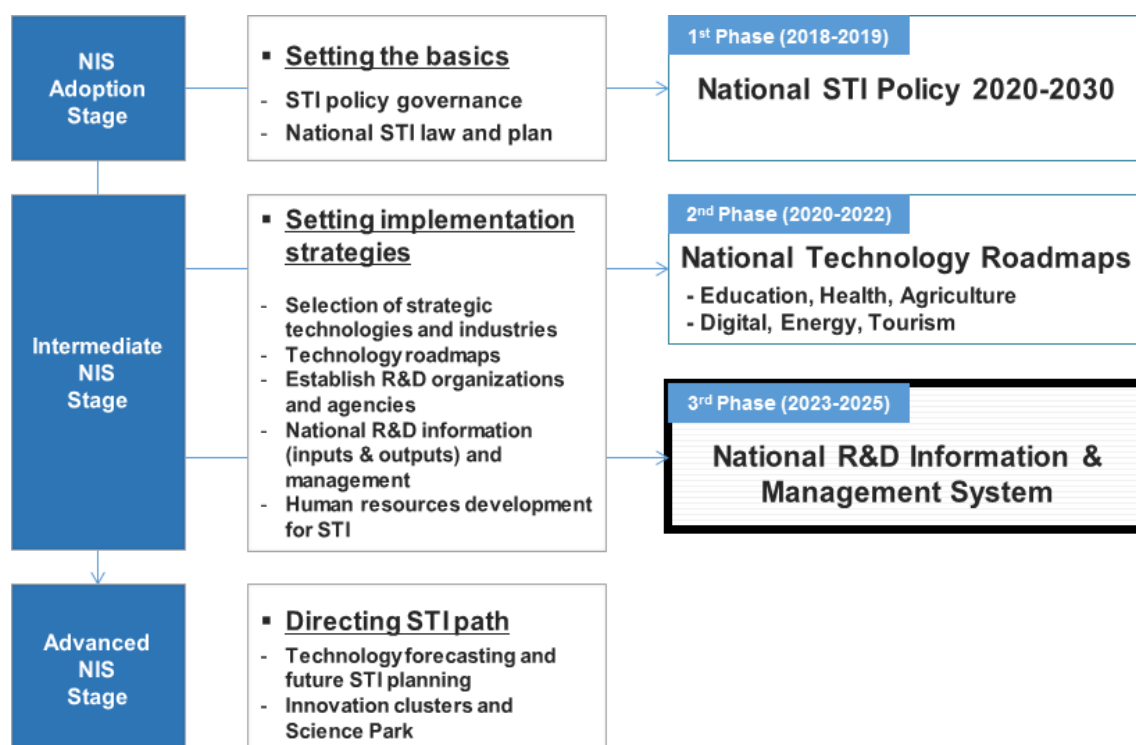
1. Introduction

This K-Innovation Partnership Programme with Cambodia is currently in the first year of its third phase (2023-2025), titled “Project Consultation and Pilot Implementation for the Development of the National R&D Management System and the R&D Centre of Excellence in Cambodia”. The partner organisation is the Ministry of Industry, Science, Technology & Innovation (MISTI).

During the first phase of the K-Innovation Programme (2018 to 2019), STEPI provided policy consulting on STI policy governance and National STI law and supported the Cambodian government in developing its first National STI Master Plan and National STI Policy 2030-2030. In the second phase (2020 to 2022), the STEPI team collaborated with Cambodian stakeholders to develop a national technology roadmap (TRMs) for each of six major technology areas, namely, education, health, agriculture, digital, energy, and tourism. Three technology roadmaps (education, health, and agriculture) were adopted by Cambodia’s National Council on Science, Technology & Innovation in July 2021, while the other three technology roadmaps were launched on Cambodia’s first “National STI Day” in March 2023. As part of the second phase of the K-Innovation Programme with Cambodia, equivalent to an intermediate “National Innovation System (NIS)” stage, the activities included selecting strategic technologies and industries, developing technology roadmaps, and improving capacity building for STI policy. The third phase of the programme is also a part of the intermediate NIS stage,

providing policy consultation and supporting capacity building for the development of the National R&D Management System for the Cambodian government, as shown in Figure 1-1 below.

[Figure 1-1] Overview of the K-Innovation Partnership Programme Phases 1-3



Source: Kim et al. (2023).

2. Objectives

The overall objectives of the third phase of K-Innovation Partnership Programme with Cambodia are to provide policy consulting aimed at advancing the National R&D Management System in Cambodia in order that its STI Master Plan and TRMs can be implemented fully, and at building the skills and capacities of the researchers and the related policymakers. In 2023, the programme focused on conducting a survey on the status of national R&D management and analysing the results. Regarding the topic of national R&D management, Korean experts from KISTI shared Korea's rich experience in developing and operating two national R&D management programmes, NTIS and COMPAS, whose roles and functions are explained in details in Chapters 2 and 3 respectively. For the next two years, 2024 to 2025, STEPI's research team will support and assist MISTI in designing the conceptual and implementation plan for the national R&D management system, which will be tailored to Cambodia's needs and its national innovation system.

3. Project Framework

The framework of the K-Innovation Partnership Programme has two major components: policy consulting and capacity building. Together with invited experts from the Korea Institute of Science and Technology Information (KISTI), the Korea Institute of Science & Technology Evaluation and Planning (KISTEP), and the National Research Foundation of Korea (NRF), STEPI provided policy consulting and organised a capacity building workshop for the personnel of MISTI who will operate the national R&D management system. The policy consulting mainly covered national R&D management governance and introduced exemplary Korean cases such as NTIS and COMPAS. In addition, in consultation with STEPI, MISTI conducted a survey and held a series of consultation meetings with local stakeholders to understand the status of, and identify demand for, the national R&D management system in Cambodia. The results of the survey form the main part of the policy consulting report, which will serve as a reference material when elaborating and implementing the activities for the next two years during the second phase of the programme.

4. Project Team

■ Korean Research Team

The STEPI research team was composed of three researchers, a PM, an advisor, and a project coordinator. To ensure the continuity of the multi-year development aid project, Dr. PARK Dong Un took the PM position after participating in the 2022 programme with Cambodia. Dr. LEE Jungwon, an honorary research fellow of STEPI and head of the first and second phases of the programme with Cambodia, participated as an advisor in the 2023 programme.

Two KISTI experts with expertise in the national R&D management system were invited to share South Korea's experience in operating the national R&D management system.

[Table 1-1] Korean research team

Category	Institute	Position	Name	Major role
Internal expert	STEPI	Associate Research Fellow	Dr. PARK Dong Un	Overall management of the 2023 Programme with Cambodia, research and surveys)
	STEPI	Honorary Research Fellow	Dr. LEE Jungwon	Advisor
	STEPI	Senior Researcher	Ms. LEE Elly Hyanghee	Project coordination, research and surveys
External expert	KISTI	Principal Researcher	Dr. LEE Hyuck Jai	Policy consulting on COMPAS
	KISTI	Senior Researcher	Dr. YU Eunji	Policy consulting on NTIS

■ Cambodian Research Team

The Cambodian research team was headed by Dr. HUL Seingheng, the Under-Secretary of MISTI. Under his leadership for the third phase of the programme, Dr. SRUN Pagnarith served as a project coordinator for the Cambodian team as well as a counterpart for the STEPI team. Three other officials from MISTI who are responsible

for STI policy and STI data management were also members of the team involved in the 2023 Programme and the national R&D management system.

[Table 1-2] Cambodian research team

Category	Organization	Position	Name	Major role
Cambodian research team	Ministry of Industry, Science, Technology and Innovation (MISTI)	Under-Secretary	Dr. HUL Seingheng	Overall project management on the Cambodian side
		Director of the STI Policy Department	Dr. SRUN Pagnarith	Project coordinator, research and surveys
		Director-General, General Department	Dr. TRY Sophal	Research and surveys
		Deputy Director of the STI Data Management Department, GDSTI	Dr. CHEN Sovann	Research and surveys
		Deputy Director	Dr. SENG Touch	Research and surveys

5. Project Activities

5.1 Programme adjustment and operation meetings (online)

- 7 February 2023: A kick-off meeting between STEPI and MISTI was held to discuss the activities and schedule of the 2023 programme.
- 16 February 2023: An online meeting was held to discuss the kick-off workshop for the 2023 programme and advocacy of the TRMs, and key deliverables of the second phase programme in Cambodia in March during Cambodia's first "National STI Day".

5.2 Kick-off workshop and interviews with Cambodian stakeholders to understand the status of national R&D management in Cambodia

- 24-31 March 2023: A kick-off workshop for the third phase of the programme and a launch ceremony for three TRMs were held during the national event to commemorate Cambodia's first national STI day, which was organised by MISTI. The Korean research team interviewed and established networks with key stakeholders relating to national R&D management from MISTI and universities.

5.3 Capacity building workshop for Cambodian officials

- 22-26 May 2023: STEPI invited nine officials to enhance their skills and competencies in policymaking related to the national R&D management system, R&D funding, and economic development through science, technology, and innovation (STI). The workshop program included lectures by Korean experts and on-site visits to the Korea Institute of Science and Technology Information (KISTI), the Global Knowledge Exchange and Development Centre of the Korea Development Institute (KDI), and STEPI itself. Additionally, a bilateral meeting between the Ministry of Science, Innovation, Technology, and Telecommunications (MISTI) and the Ministry of Science and ICT of Korea was held to discuss potential areas of collaboration.

- June-November 2023: A survey was conducted on the nine Cambodian participants' satisfaction with the workshop. The survey questionnaire included their overall satisfaction with the workshop programme, the contents of the lectures, and the methodologies used among other things. The respondents gave a rating of 'very satisfactory' or 'satisfactory' in 9 out of 10 questions. Notably, they rated it "very satisfactory" (93%) in two objective questions: Did the workshop meet the goals of the 2023 K-Innovation Programme?; and Were the presentation materials useful?

[Table 1-3] Summary of the results of the survey on satisfaction with the capacity-building workshop

Category	Questions	Very satisfactory	Satisfactory	Average	Total score (out of 45 points)	Satisfaction rate (%)
Programme	Were the contents of the workshop appropriate to the project objective?	67%	33%	-	42	93%
	Were the contents and level of the lectures appropriate for your understanding?	56%	44%	-	41	91%
Methodology	Were the methods used in the workshop (lecture and Q&A) helpful in understanding the contents?	56%	44%	-	41	91%
	Was the method used in the workshop suitable for achieving the goal of understanding Korea's National R&D management system (policies and governance)?	56%	44%	-	41	91%
Materials	Were the presentation materials used in the workshop helpful for understanding the contents?	67%	33%	-	42	93%
Self-efficacy	I have a better understanding of the elements and procedures necessary for developing the concept of the national R&D management system in Korea compared to before the workshop.	44%	56%	-	40	89%
	I am more fully aware of the current status of the national R&D management system than I was before the workshop.	33%	67%	-	39	87%
	I am more fully aware of the status of	67%	22%	11%	41	91%

Category	Questions	Very satisfactory	Satisfactory	Average	Total score (out of 45 points)	Satisfaction rate (%)
	the characteristics*(policies, governance, stakeholders) of Korea's national R&D management system than I was before the workshop.					
	I have better understanding of how to design the concept of the national R&D management system for Cambodia than I did before the workshop.	44%	56%	-	40	89%

5.4 Survey on the status of and demand for national R&D management in Cambodia

- 5 June 2023: An online meeting was held to review the outcomes of the capacity building workshop and the next steps, including the contract on the survey between STEPI and MISTI.
- June-July 2023: After discussing and adjusting the arrangements for the contract, STEPI signed the contract for the survey with MISTI on 20 July 2023.
- 13 September 2023: The draft survey questionnaire prepared by MISTI was reviewed, along with comments on the questionnaire and survey methodology, etc.
- November-December 2023: The draft report on the survey was submitted by MISTI and then revised according to the comments made by the STEPI team.

6. Project Schedule

The overall schedule of 2023 programme is as follows:

[Table 1-4] Project Implementation Schedule

Task/Month	1	2	3	4	5	6	7	8	9	10	11	12
Planning												
Revision of plan												
Kick-off workshop, onsite interviews, advocacy of the TRMs												
Capacity Building Workshop												
Mid-term Evaluation (STEPI internal)												
Signing of local service contracts for survey												
Development of questionnaire and methodology of the survey												
Conduct of the survey												
Draft report of the survey												
Final evaluation (STEPI internal)												
Final report of the survey												

7. Project Outputs

STEPI provided the following deliverables:

- Deliverable 1: Kick-off workshop presentation materials.
- Deliverable 2: Materials on the national R&D management system for the Capacity Building Workshop.
- Deliverable 3: Feedback on development of the questionnaire and survey methodology for the national R&D management system in Cambodia.

MISTI provided the following deliverables:

- Deliverable 1: Draft questionnaire and survey methodology for the national R&D management system in Cambodia.
- Deliverable 2: Draft and Final Report on the Survey to analyse the status of, and identify demand for, the national R&D management system in Cambodia.



CHAPTER 2

Korean Best Practice 1: National Science & Technology Information Service

Dr. YU Eunji, Senior Researcher, KISTI

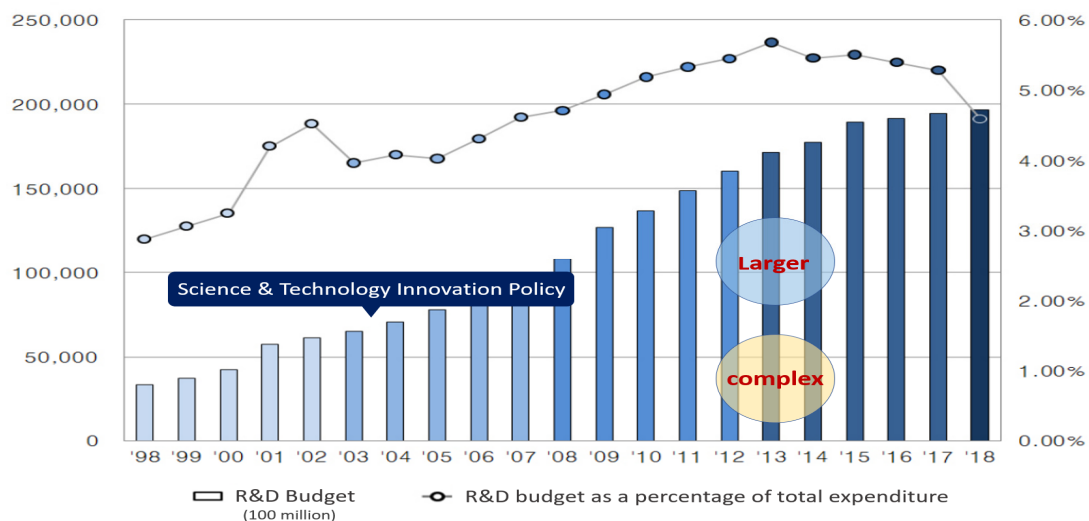
Chapter 2. Korean Best Practice 1: National Science & Technology Information Service

1. Overview of the NTIS

1.1 Background

During the early 2000s, the Korean government made significant efforts to promote policies aimed at fostering innovation in science and technology. This was driven by the government's recognition of the important role that science and technology could play in national development. As a result, the government's R&D budget was increased steadily during this period, as shown in Figure 2-1. With the increase of the national R&D budget, the scale and complexity of national R&D projects also tended to grow accordingly.

[Figure 2-1] Korea's R&D investment status



Source: Improvement of R&D Budget Coordination for Small-scale New Programmes (2018). Journal of Budget and Police.

Under these circumstances, managing R&D inevitably became increasingly challenging and led to a number of serious problems, the largest of which concerned the scattered distribution of national R&D information among different institutes, which made it difficult to gain a comprehensive overview of the overall condition of national R&D. This in turn led to overlapping investments, making it difficult to prioritise them effectively. Furthermore, negative consequences arose due to excessive and overlapping investments in research facilities and equipment. This issue has also been pointed out as a problem in government audits, as follows :

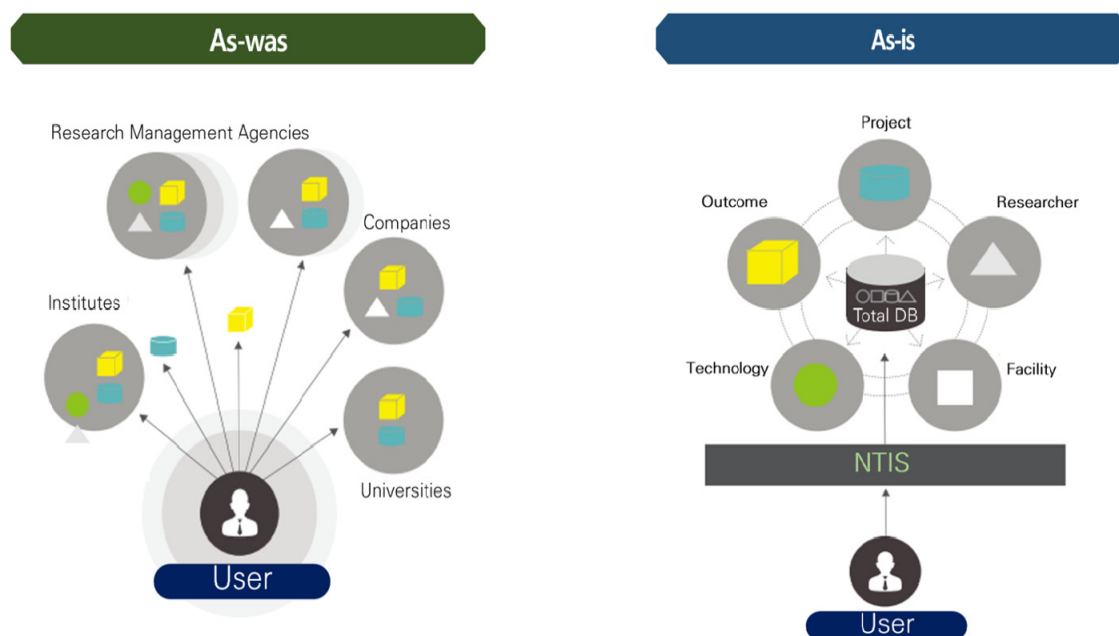
- “Due to the difficulty of researching and analysing the overall status of R&D in Korea, it is difficult to predict project feasibility in advance and to improve the efficiency of national R&D investment due to overlapping investments. It is also difficult to establish national-level policies and plan R&D.” [Parliamentary inspection of government offices, 2005]
- “It is difficult to grasp the construction status of research facilities and equipment built with the national budget. Excessive and overlapping investments are serious problems.” [Parliamentary inspection of government offices, 2006]

At the same time, the need to allocate the budget strategically so as to enhance the efficiency of national R&D investment, strengthen R&D capabilities, and exploit performance outcomes more efficiently has been recognized throughout society.

1.2 Progress

In accordance with these social needs, the president issued a directive for the establishment of an R&D efficiency plan in 2003, after which the 'National Science and Technology Comprehensive Information System Business Plan (draft)' was confirmed by the National Science and Technology Committee. Figure 2-2 shows a conceptual diagram of the NTIS included in the plan.

[Figure 2-2] Conceptual Diagram of the NTIS

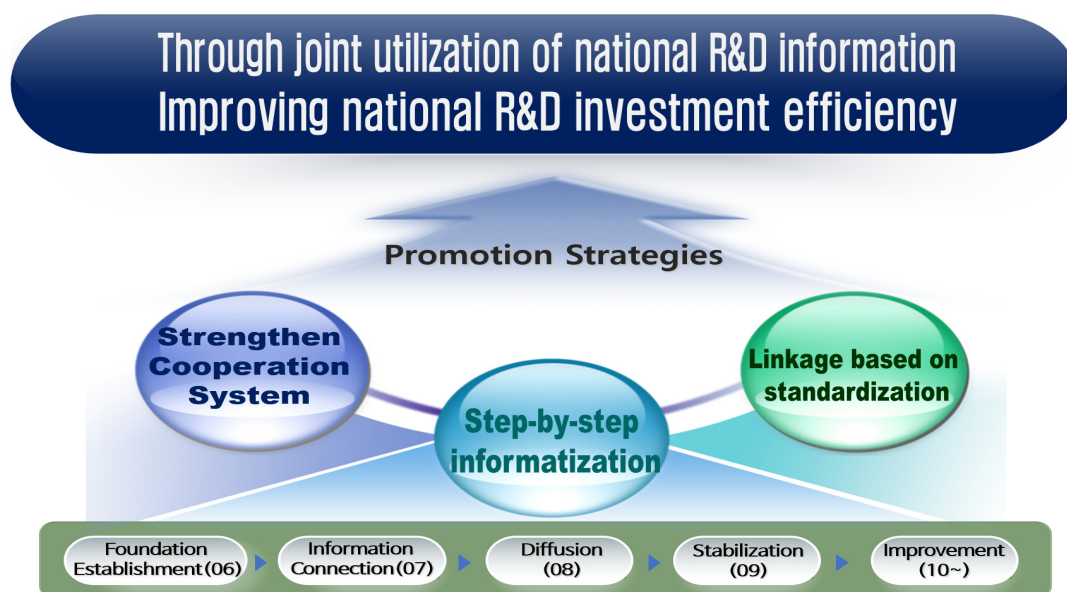


Source: National Science and Technology Commission (2004), NTIS Construction Plan.

As shown in the diagram on the left of Figure 2-2, science and technology information was previously stored and managed separately by different research institutes, project management institutes, and universities, making it very difficult for users to access the information they needed. In order to solve this problem, we wanted to establish the NTIS as an environment in which anyone from industry, academic institutions and research institutes could access science and technology information anytime, anywhere.

To build such a system, the Korea Institute of Science and Technology Information (KISTI) was selected as the managing institution in October 2005, and, in May 2007, the NTIS Project Promotion Plan (2007-2009) was finalised at a meeting of the Science and Technology Ministers. Also, three primary strategies were developed to achieve the objectives of R&D joint utilisation and investment efficiency through the NTIS. These strategies included strengthening the cooperation system with ministries, implementing step-by-step informatisation, and establishing standards for linkage.

[Figure 2-3] Goal and Strategies of the NTIS



Source: KISTI NTIS Centre, 「NTIS Introduction」 .

The specific promotional strategies were as follows: First, information related to national R&D was collected from ministries and research management institutions, and then an integrated information system and an integrated database were built. Second, the project was promoted step by step in order to improve the completeness of the project and prioritise the implementation of major projects. Third, a user-oriented comprehensive information portal site was established by referring to information on user demand and the information services of foreign information agencies.

To achieve the first and most important driving strategy, we formed a committee To implement a key strategic initiative, a committee was established with members from various ministries and R&D project management institutions. This group facilitated discussions on critical aspects of creating the National Technology Information Service (NTIS), ensuring that issues were collaboratively addressed and decisions were made to strengthen interconnections. The committee also played a vital role in overhauling legal and regulatory frameworks, which paved the way for an

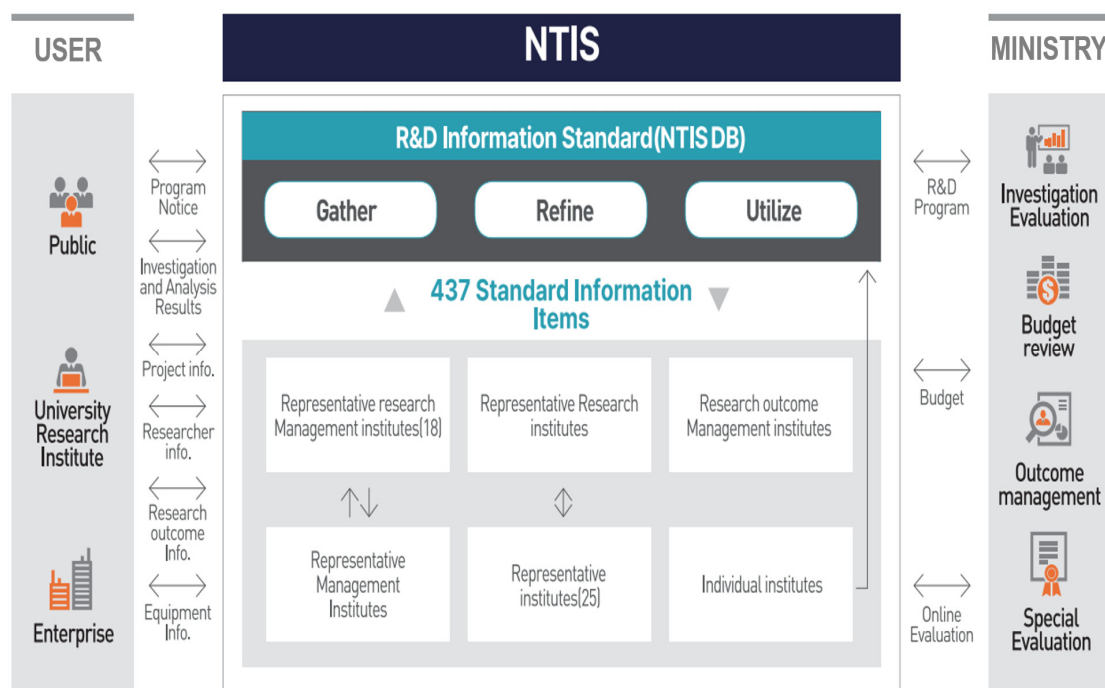
effective information collection system. This system connected national R&D data from various departments to the NTIS, managed by designated institutions from each ministry. In December 2007, a significant decision was made to gather comprehensive R&D data, covering 341 categories such as projects, researchers, results, research facilities, and equipment. This effort involved coordination with thirteen ministries and twelve key project management institutions, leading to the establishment of eight national R&D services by the end of 2007.

1.3 NTIS Service

The NTIS is the world's first national R&D information and knowledge portal to integrate and provide comprehensive R&D information including programmes, projects, outcomes, and human resources. It also provides 11.5 million datasets (as of August 2023), including science and technology information such as academic papers, patents, research reports, and policies. These data boast the best quality among all public institutes. The purpose of this service is to enhance the efficiency of R&D investment and the productivity of research by collecting and sharing information on national R&D, which until its establishment had been separately managed by diverse ministries and institutes. The NTIS supports evidence-based R&D planning by providing reference data for policymaking, provides transparent information on evaluations of the performance of national R&D programmes and institutes, and prevents duplicated expenditure on national R&D budgets.

Furthermore, it provides recommendations of user-customised information, supports the management of R&D outcomes, and provides information on the current status of national R&D, as well as providing customised information packages and information on the latest research trends and the real-time analytical environment of national R&D.

[Figure 2-4] National Science & Technology Information Service

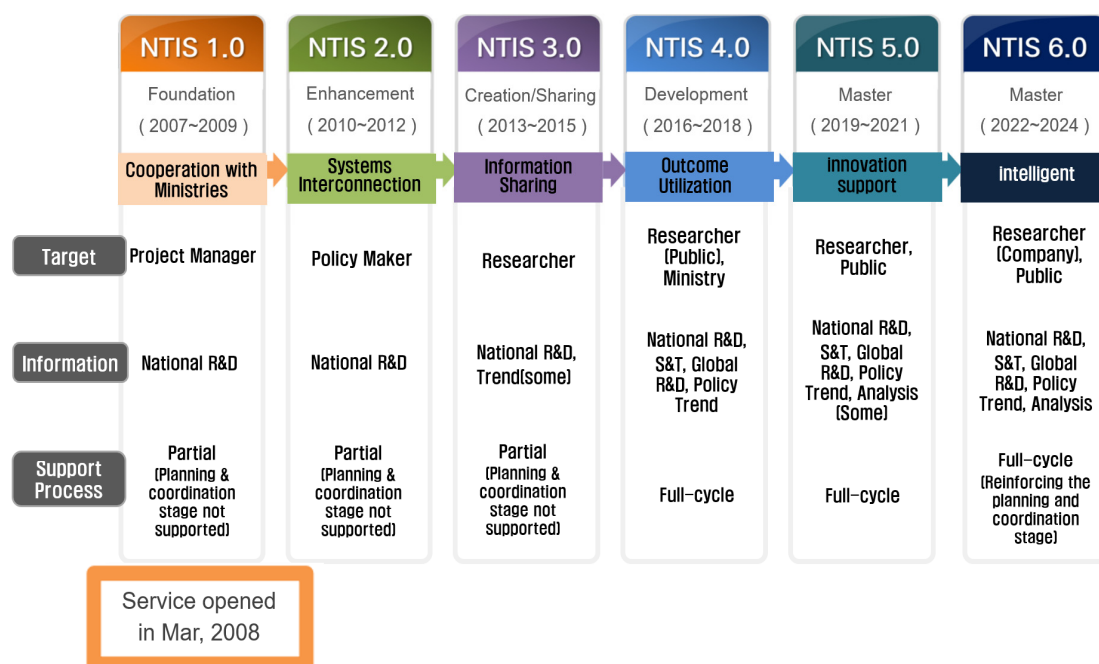


Source: KISTI NTIS Centre, 「Brochure and usage guide」.

1.4 History

Every three years, the National Technology Information Service (NTIS) develops a new plan through Information Strategy Planning (ISP) to set the future direction for its operations. Over time, these plans have led to significant changes in the NTIS's focus, the types of information it provides, and its R&D support processes. Initially, the NTIS was designed to assist project managers and policymakers. However, it has evolved to support a broader audience, including all types of researchers. Moreover, the NTIS has broadened its information services to include a diverse range of data on national and international R&D, policy trends, and science and technology developments. Today, the NTIS is committed to enhancing its offerings to support the entire R&D lifecycle, moving beyond its earlier focus on just segments of the R&D process.

[Figure 2-5] Progress and History of the NTIS

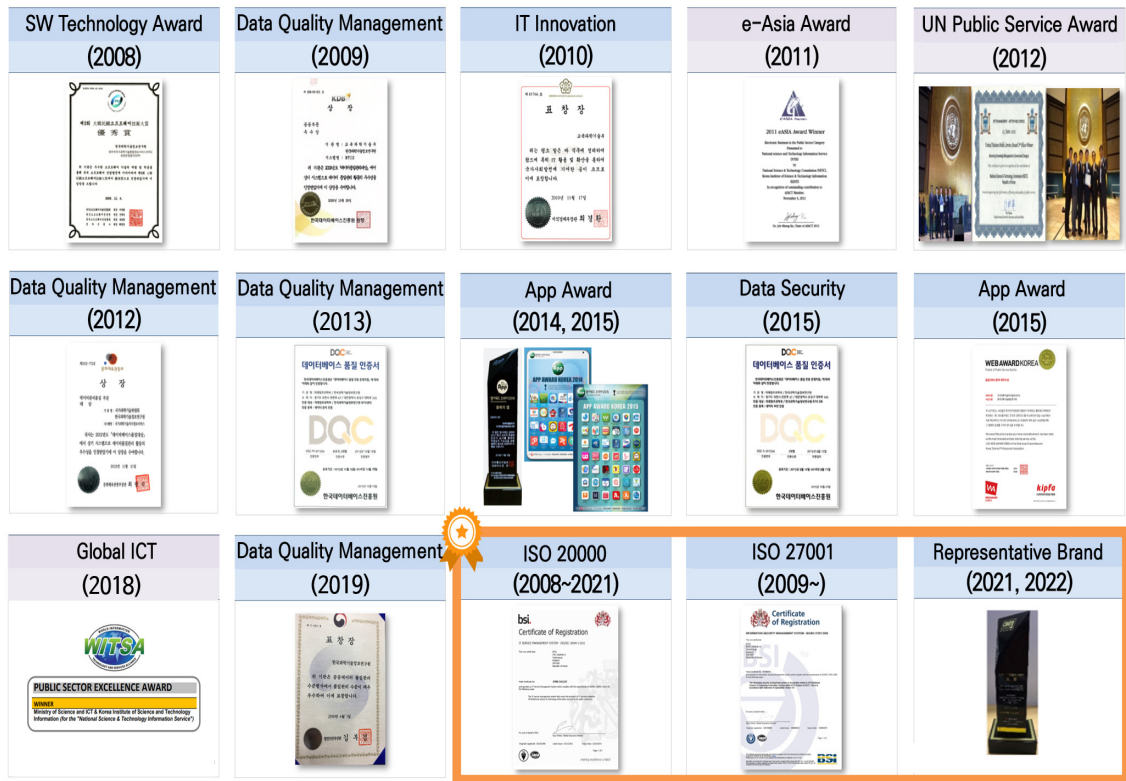


Source: KISTI NTIS Centre, 「NTIS Introduction」 .

1.5 Awards and Certifications

The NTIS service has earned significant recognition both within the country and globally for its excellence. Notably, it was awarded the United Nations (UN) Public Administration Award in 2011, and it also received the UN's Excellence Award for "improving the management of government knowledge in public administration." This prestigious category honors exemplary public policies and institutions worldwide. The NTIS was recognized for its efforts to enhance communication and collaboration among government entities, public institutions, universities, corporations, and research institutes, facilitating easy and shared access to national R&D information for these groups as well as the general public. Additionally, NTIS has established robust service management and data quality systems, underscored by achieving ISO 2000 and 27001 certifications, further testament to its commitment to maintaining high standards in its operations.

[Figure 2-6] Awards and Certifications of the NTIS



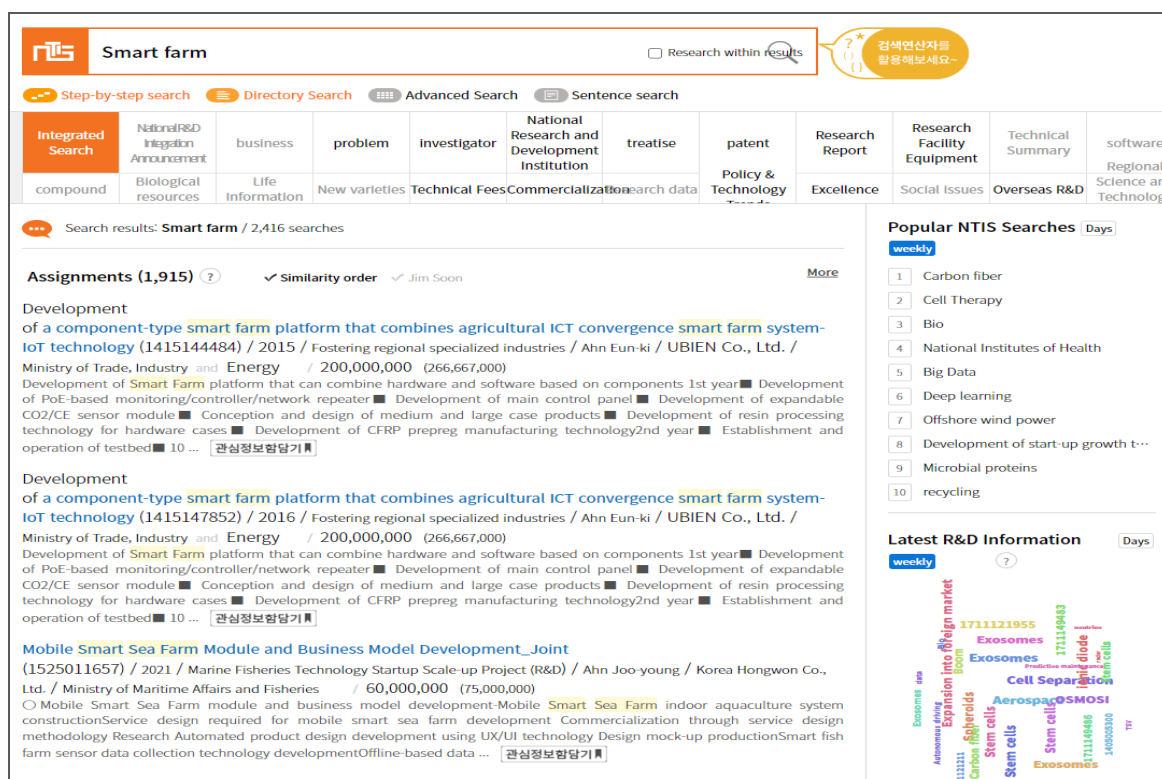
Source: KISTI NTIS Centre, 「NTIS Introduction」.

2. The Main Services of the NTIS

2.1 Integrated Search

The “Integrated Search” provides diverse information about integrated national R&D announcements, projects, researchers, overseas R&D information, and outcomes such as papers, patents, research reports, and allows the users to address any question to the NTIS. The NTIS has reduced the information search time dramatically by providing a simple and user-friendly search function and user-specific information using a smart search engine. Users can analyse search results on the web using the NTIS’s immediate analysis service.

[Figure 2-7] Integrated Search Service



Source: NTIS website.¹

¹ www.ntis.go.kr, accessed on November 8, 2023.

- **Keyword search:** Users can enter a keyword in the area they wish to search, whereupon it searches all contents available in the NTIS. Users can also use search operators, such as AND, OR and NOT, to obtain more accurate and detailed results.
- **Detailed search:** Users can specify keywords and conditions, such as the project name and year, in order to perform more detailed search queries.
- **Directory search:** Users can search for information in a directory format according to various classification criteria, such as the science and technology standard classification, national key science/technologies, national strategic technologies, and the 6 T's (six cutting-edge industrial technologies).
- **Download search result:** Users can download search result items in the Excel format.
- **Immediate analysis :** Users can analyse national R&D data on the Web retrieved by an integrated search engine, and check the results of analysis in a graph format, or download them.
- **Level-by-level search:** Users can perform any search easily step by step – from content selection to information filtering. They can also find desired information through interactions with the NTIS Chatbot. AI provides customized information by learning each user's fields of interest, keywords, information on projects in which they are participating, and search history.
- **Interactive search:** Users can find desired information through interactions with the NTIS Chatbot.
- **Recommendation of customized information:** The AI provides customized information by learning each user's fields of interest, keywords, information on projects in which they are participating, and search history.

2.2 Integrated National R&D Announcement

The Integrated National R&D Announcement service allows users to access national R&D announcements from various ministries and project management institutes. Users can customize their experience by registering with specific ministries or selecting

keywords that match their interests. This registration enables them to receive relevant public announcements directly in their email, ensuring they stay informed about updates conveniently and efficiently.

[Figure 2-8] Integrated National R&D Announcement Service

National R&D Integration Announcement

Search for public announcement agencies Search for attachment names

Announcement Information R&D Project Schedule MY Announcement Notification Request RSS Application

Announcement Form	whole	Consolidated Announcement	Individual Announcement	help	Announcement Status	whole	Submission Scheduled	Accepting	deadline
Buddha name	whole	Personal Information Protection Commission	Police agency	Ministry of Employment and Labor	Fair Trade Commission	Ministry of Science and ICT	Ministry	Department of Veterans Affairs	State Affairs Coordination Office
Pentagon	Ministry of Land, Infrastructure, and Transport	parliament	Observatory	Ministry of Planning and Finance	Ministry of Agriculture, Food and Rural Affairs	Rural Development Agency	Office of the President's Secret Service	Office of the President's Secretary	Cultural Heritage Administration
Ministry of Culture, Sports and Tourism	Korea Communications Commission	Defense Projects Agency	Justice	Legislative Affairs	Department of Health and Welfare	Forest Service	Ministry of Trade, Industry and Energy	Fire Agency	Ministry of Food and Drug Safety
Ministry of Gender Equality and Family	Foreign ministry	Nuclear Safety Commission	Ministry of SMEs and Startups	Korea Disease Control and Prevention Agency	Statistics	Ministry of Unification	Patent office	Maritime Police Agency	Ministry of Maritime Affairs and Fisheries
Ministry of Public Administration and Security	Ministry of Environment	Multi-Buddha	guitar	Reset Settings					

1 For national R&D integration announcements before 2018, information on the date of receipt, deadline for submission, type of announcement, type of announcement, size of announcement, contact information, and business name are not provided

1 For more information on the announcement, please check the attached file of the announcement

Announcement date ~ Announcement type whole Announcement scale whole Deadline

whole search

65,346 results Download the user manual Application for registration Download the list 10 pcs. application

Waiting	Status	Buddha name	Announcement Name	Submission date	Deadline	D-day
☐	65346	Accepting	Ministry of Science and ICT	2023 Korea-Finland (BF) Joint Research Project New Project Competition	2023.07.28	2023.09.07 D-37
☐	65345	Accepting	Ministry of Trade, Industry and Energy	Technology Demand Survey for New Business Planning of 'Fire Safety Technolo...	2023.07.31	2023.09.08 D-38
☐	65344	Accepting	Korea Disease Control and Prevention Agency	[Research Service] 2023 Operation of a monitoring system for causative agents of...	2023.07.31	2023.08.11 D-10

Source: NTIS website.²

- **Announcement information:** The NTIS interconnects, collects, and provides national R&D announcements scattered across diverse ministries and institutes. Users can categorise information according to their interests and search for announcements in a specific form with the selected keywords.
- **R&D project schedule:** The NTIS provides the annual schedule of the national R&D programmes by year, ministry, and area at the beginning of each year.
- **My Announcement:** Users can collect and view announcements related to their interests and announcements that they have registered separately.

² www.ntis.go.kr, accessed on 8 November 2023.

- **Alert request:** Users can receive an email notification about each new announcement.

2.3 Researcher Information

The NTIS collects and provides information on researchers who have been involved in national R&D projects and on candidates for the evaluation committee. When researchers are looking for experts, collaboration partners, or an evaluator who can evaluate their national R&D project objectively, they can search for the right person by major, affiliated institute, project, paper, patent, career, expertise, and so on.

[Figure 2-9] Researcher Information Service

The screenshot displays the 'Researcher Information' service interface. At the top, there's a search bar with 'whole' and 'Artificial Intelligence' selected, and a 'Research within results' checkbox. Below the search bar are three buttons: 'Specialization Directory Search', 'Find experts in your field of interest', and 'Search Help'. The main section shows 'Search results: Artificial Intelligence' with a filter icon. It indicates 'Number of 675 cases searches' and 'Navigation Artificial Intelligence'. Below this, there are buttons for 'Specialist Trap', 'Go to Expert Box', and 'Rounding up'. On the right, there are dropdowns for '10 pcs.' and 'Similarity order'. The results list two researchers: 'Cho Cho Bae' (Yonsei University) and 'Shinguibo' (Chung-Ang University/Chung-Ang University), each with a heart icon for favoriting. The results for Cho Cho Bae include 'Challenges (43)', 'Papers (768)', and 'Intellectual Property Rights (23)'. The results for Shinguibo include 'Challenges (29)', 'Papers (668)', and 'Intellectual Property Rights (30)'.

Source: NTIS website.³

- **Detailed search:** Users can find researchers who have participated in a national R&D project by expertise, major, paper title, intellectual property name, or educational background, etc.

³ www.ntis.go.kr, accessed on 8 November 2023.

- **Comparison search of experts:** Users can compare experts found through various searches and select the best partner for collaboration, taking into consideration such factors as region, age group, and other criteria.
- **Detailed search:** This service manages the research experience and outcomes of all researchers who have participated in a national R&D project in one place. The registered research outcomes of a national R&D project are automatically updated without manual entry. The NTIS enables lifetime experience management, which integrates and shares all the information on researchers' experience available from ministries and institutes.
- **Career History Management:** This service manages the research experience and outcomes of all researchers who have participated in a national R&D project in one place. The registered research outcomes of a national R&D project are automatically updated without manual entry. The NTIS enables lifetime experience management, which integrates and shares all the information on researchers' experience available from ministries and institutes. Committee Recommendation.
- **Committee Recommendations:** This section is open for recommendations of professional researchers to be commissioned as members of the Ministry of Science and ICT. Anyone can recommend either him/herself or other people.

2.4 Research Institute Information

The NTIS integrates external and national R&D data and provides specialised information on all institutes that have participated in national R&D projects from various perspectives. As such, users can find competitive or cooperation opportunities for an R&D project by analysing the participating R&D institutes. The provision of analytical information on institutes that have participated in national R&D projects in an individual user's areas of interest can help that user to plan a national R&D project.

[Figure 2-10] Research Institute Information Service

The screenshot displays the NTIS website interface. At the top, it says 'National Research and Development Institution' with a tagline 'Focusing on Information on Institutions and companies that have carried out national R&D projects and projects'. A search bar shows 'Smart City / 19 searches'. On the right, it indicates '4,34 cases As of 2023.04.08'. Below the search bar, there are filters for 'Institution Type' (Universities(8), Small and Medium-sized Business..., Public Enterprises(3), Cast (2), Conglomerates(1)) and 'area' (Seoul(7), Matches(3), Gangwon(2), South Chungcheong(2), Ulsan(1), Jeonbuk(1), Jeonnam(1)). The main content area lists search results for various institutions, including 'Korea Institute of Production Technology', 'Korea Institute of Science and Technology', 'Chonbuk National University', 'Gangwon', 'Ewha Womans University', 'Chung-Ang University, University Seoul', 'Korea Electric Power Corporation (KEPCO)', and 'Kookmin University'. Each result includes details like projects, funding, papers, and patents.

Source: NTIS website.⁴

- **Search:** Users can search for institutes that have executed a national R&D project.
- **By region:** Users can check the status of institutes that have executed a national R&D project by region.
- **By business type:** Users can check the status of institutes that have executed a national R&D project by standard industry category.
- **Comparison of institutes:** Users can select and compare various institutes.

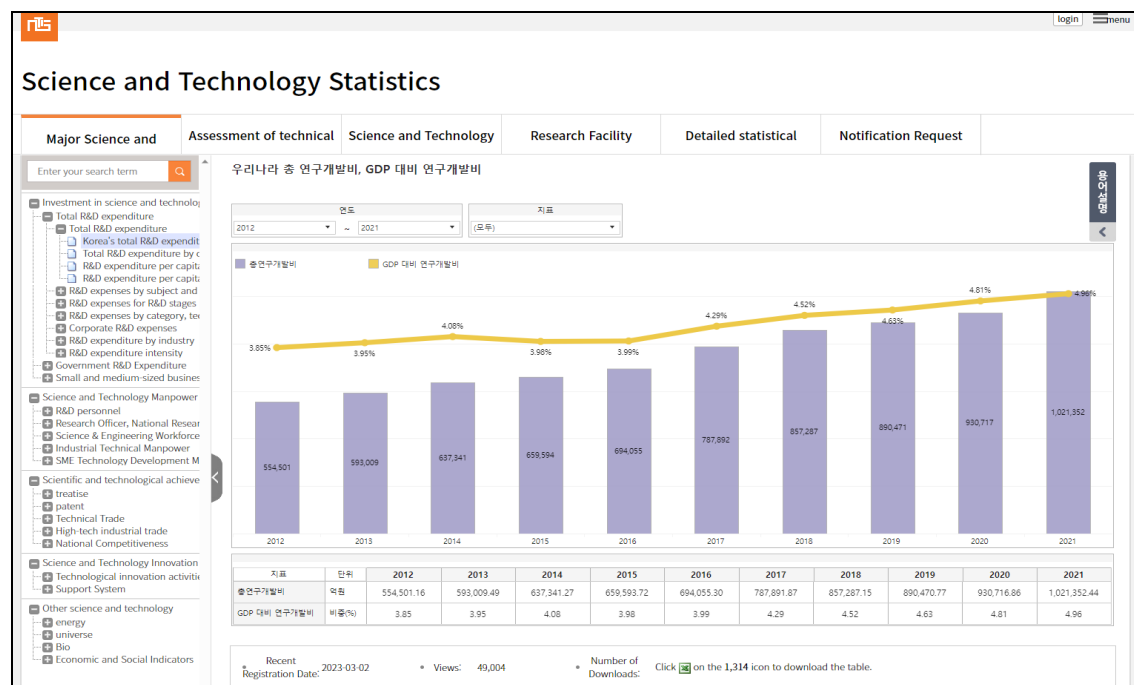
2.5 Science and Technology Statistic

The NTIS provides domestic and foreign science and technology statistical information, such as major science and technology statistics, technology capability evaluations, science and technology predictions, and various science and technology indicators related to research facilities and equipment. Users can check a wide range of indices, statistical data, and graphs in the field of science and technology, such as graphs

⁴ www.ntis.go.kr, accessed on 8 November 8 2023.

showing the R&D expenditure as a proportion of the GDP of diverse countries around the world, or South Korea's ranking in terms of the number of R&D researchers.

[Figure 2-11] Science and Technology Statistic Service



Source: NTIS website.⁵

- **Major science and technology statistics:** The NTIS analyses more than 120 statistical indices in science and technology in Korea and overseas in real time.
- **Technical competency evaluations:** The NTIS provides analyses of the level of science and technology in major advanced countries and the technology gap.
- **Science and technology forecasting:** The NTIS provides the results of analyses of science and technology forecasts every five years.
- **Research facility equipment:** The NTIS analyses the investment and utilisation status of national research facilities and equipment installed as the outcomes of national R&D programmes.

⁵ www.ntis.go.kr, accessed on 8 November 2023.

- **Detailed statistical analysis:** The NTIS provides multifaceted investigations and analyses of national R&D programmes and R&D activities.

2.6 Project and Performance Information Register (SIMS)

Users can register and manage national R&D project and programme information conveniently. The NTIS helps national R&D project managers and research managers to enter the standard information of national R&D projects anywhere at anytime, as well as helping the project managers and research managers at each ministry and project management institute to register and manage the standard information of national R&D projects and programmes in the SIMS (Standard Information Management Service). In addition, the NTIS reflects the entered information in real time to help users check the status of national R&D project and programmes accurately.

- **Programme registration:** Managers of national R&D programmes can enter and check information about the programmes they are currently managing.
- **Project registration:** Managers of national R&D programmes can enter and check information about the programmes they are managing.
- **Outcome registration:** Managers of national R&D projects and programmes can enter and check information about successful outcomes, such as academic papers and patents.

2.7 Policy/Technology Trends

The NTIS provides information on the science and technology policy and agenda in real time. Users can find information on science and technology policies as well as technological and industrial trends ahead of their competitors. The NTIS also presents the key trends of eleven institutes, including STEPI, the MCST, the IITP, and the KDI; collects information published by the government and the National Assembly; and presents integrated information on government policies. Users can quickly take in and understand underlying shifts and changes in science and technology policies and industrial trends.

[Figure 2-12] Policy/Technology Trends Service

Policy · Technology Trends

Search for policy and technology trends

Science and Technology Policy Information | Trend Information | Visualization analysis | Notification Request

National Council of Advisors on Science and Technology | Parliamentary Policy Data | Meetings and Committees | National Policy Research

The National Science and Technology Advisory Council provides information on matters reported and considered by the President's direct body (Article 127(3) of the Constitution and Article 1 of the National Science and Technology Advisory Council Act) established for the innovation of national science and technology. On April 2018, 4, the National Science and Technology Advisory Council Act was amended, including all the materials of the former National Science and Technology Commission, the National Science and Technology Review Council, and the National Science and Technology Advisory Council.

Deliberative body ▶	whole	National Council of Advisors on Science and Technology	National Council for Science and Technology	National Science and Technology Commission		
Councils ▶	whole	Plenary Meeting	Advisory Council	Deliberative Meeting (Plenary Session)	Deliberative Meeting (Steering Committee)	
Buddha ▶	whole	Ministry of Science and ICT	Ministry	National Council of Advisors on Science and Technology	Pentagon	Ministry of Land, Infrastructure, and Transport
Observatory	Ministry of Planning and Finance	Ministry of Agriculture, Food and Rural Affairs	Rural Development Agency	Ministry of Culture, Sports and Tourism	Defense Projects Agency	Department of Health and Welfare
Forest Service	Ministry of Trade, Industry and Energy	Ministry of Food and Drug Safety	Nuclear Safety Commission	Ministry of SMEs and Startups	Patent office	Ministry of Maritime Affairs and Fisheries
Ministry of Public Administration and Security	Ministry of Environment	guitar	Korea Disease Control and Prevention Agency	+ View previous ministries		

Item Type: choice | Date of Round Deliberations: ~

search

Search results 1,252 cases | Download the list | 10 pcs. | application

☐	Waiting	Deliberative body	Councils	Item Name	Ministries under its jurisdiction	Item Type	Episodes	Date of deliberation	file
☐	1252	National Council of Advisors on Science and Technology	Deliberative Meeting (Steering Committee)	National R&D performance evaluati...	Ministry of Science and ICT	report	51	2023-06-26	
☐	1251	National Council of Advisors on Science and Technology	Deliberative Meeting (Steering Committee)	National R&D Integrated Performan...	Ministry of Science and ICT	deliberation51		2023-06-26	

Source: NTIS website.⁶

- **Science and technology policy information:** The NTIS provides information on the agenda of PACST, policy data of the National Assembly, and the agendas of diverse councils and committees.
- **Trend information:** The NTIS provides information on policy, industry and technology trends for each science and technology standard classification.

⁶ www.ntis.go.kr, accessed on 8 November 2023.

2.8 R&D by Issue

The NTIS presents national and social issues through its News feature with R&D data. It selects the national agenda and social issues that are currently hot topics by subject and presents them in the form of a national R&D information package. Users can understand the current status of various issues quickly by accessing a wealth of specialised R&D information related to the issues. Furthermore, users can personally package R&D information in their areas of interest.

[Figure 2-13] R&D by Issue Service

The screenshot displays the NTIS 'R&D as an issue' service interface. At the top, there is a header with the NTIS logo and a navigation bar. The main content area is titled 'R&D as an issue' and includes a description: 'It is a service that selects and quickly provides general national R&D-related information on national issues and social issues.' Below this, there is a section for 'Issue keywords and high-frequency related keywords extracted from original documents such as science news and policy trends are serviced as search results for task names and keyword items in NTIS task information. In order to get a more accurate picture of the situation, an expert analysis of the entire list of tasks is required. Therefore, It cannot be used as official data in various reports or academic materials.' The interface features three tabs: 'R&D as an issue', 'R&D as an issue created by users', and 'Special Issues'. The 'R&D as an issue' tab is selected, showing a list of issues. The first issue is 'Carbon Border Tax' (2023.08.13), which includes a description, a foreign language notation, and related keywords. Below the list, there is a search bar and a section for 'All packages 1103 cases'. The bottom of the interface shows four featured packages: 'Gluten Free', 'Ammonia fuel cells', 'Methanol propulsion vessels', and 'Satellite Navigation', each with a thumbnail image and a brief description.

Source: NTIS website.⁷

⁷ www.ntis.go.kr, accessed on 8 November 2023.

- **R&D issues:** The specialised curator of the national R&D information search at the NTIS provides packaged national R&D information related to national and social issues two to three times per week.
- **My R&D Issue:** This service allows users to package national R&D information in their areas of interest.

2.9 Registration of Research Outcomes and Support for Paper/Patent Verification

- The NTIS platform allows users to efficiently register various outcomes from national R&D projects, such as papers, patents and equipment. This feature simplifies the process for users who have multiple outcomes from a project, enabling them to register everything simultaneously on the NTIS, rather than having to submit each outcome individually to the managing institute. Additionally, the platform provides a verification function for papers (specifically those indexed in the Science Citation Index, or SCI) and patents that are submitted as outcomes of national R&D projects, ensuring the authenticity and validity of the information recorded.
- **Research Outcome Registration:** Users can register research outcomes generated as the results of national R&D projects in which they have participated as a research manager or a member. Users can search and select the project and register its outcomes.
- **Paper/Patent Verification Supports:** Users can verify whether a paper submitted as the outcome of a national R&D project has even been registered as an SCI (Science Citation Index) paper, or whether a domestic paper is pending or registered with the Korean Intellectual Property Office.

[Figure 2-14] Papers and Patent Verification Support

Papers and patent verification support

Thesis Verification

Patent Verification

* When verifying a thesis, the name of the paper, (one of the journal name or ISSN or DOI) and the year of publication are required items.
 * If it is not verified by the ISSN or DOI entry entered during the paper verification, please enter the journal name and verify it.

Thesis Verification

Papers Search

Domestic thesis sample

Overseas Papers Sample

* Title of the paper	Fabrication of gold nanowires (GNW) using aluminum anodic oxide (AAO) as a metal-ion sensor	verification Initialize
* Journal name	KOREAN JOURNAL OF CHEMICAL ENGINEERING	
* Year of issue	2015	

The paper was identified as a 2015 SCI(E) paper.

Title of the paper	Fabrication of gold nanowires (GNW) using aluminum anodic oxide (AAO) as a metal-ion sensor
Journal name	KOREAN JOURNAL OF CHEMICAL ENGINEERING
Year of issue	2015
Validation results	success
SCI(E) / Non-SCI(E) distinction	SCI(E)
Validation key	NTIS:001-SCI-20180718092343-000000610621-rs

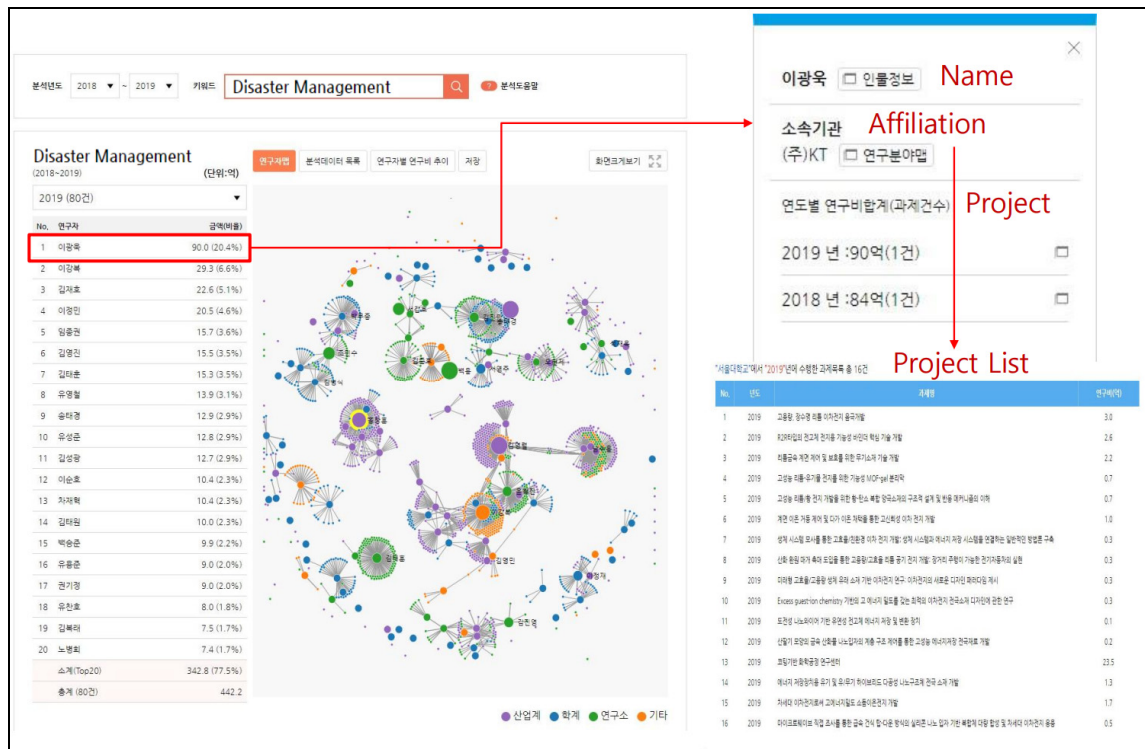
Source: NTIS website.⁸

2.10 Map of the Research Ecosystem

The NTIS matches people with people, enterprises with researchers, and institutes with enterprises. These cooperative networks help users to “see the forest rather than the trees”. The NTIS also provides a cooperative network of researchers and research institute cooperatives, the current status of research in the user’s area of interest, and the key research areas of research institutes in a knowledge map format designed to enable the user to understand them easily and intuitively.

⁸ www.ntis.go.kr, accessed on 8 November 2023.

[Figure 2-15] Map of the Research Ecosystem



Source: NTIS website.⁹

- **Researcher map:** Users can enter their area of interest to view the researcher map showing the top twenty researchers in terms of the research budget and the relationship between researchers intuitively.
- **Research institute map:** Users can enter their area of interest to view the research institute map showing the top twenty research institutes in terms of the research budget and the relationship between research institutes intuitively.
- **Comparison of research areas of institutes:** The NTIS shows a comparison of the research areas of each research institute based on the science and technology standard classification (up to three institutes).

⁹ www.ntis.go.kr, accessed on 8 November 2023.

3. Status of International Cooperation and Success Cases

3.1 Status of International Cooperation

The NTIS shares its expertise in national R&D information management and deployment know-how with many countries. We have cooperated with Cambodia, Kazakhstan, Vietnam, Chile, Belarus, and Costa Rica among other countries. Of these, we actually exported the relevant technology to Costa Rica and established a prototype of Costa Rica's science and technology information system (SINCYT).

[Figure 2-16] Status of International Cooperation of the NTIS



Source: NTIS website.¹⁰

- March 2015, Vietnam, Cooperation on the Establishment of the Science and Technology Information System.
- June 2015, Kazakhstan, Cooperation on the Establishment of the Science and Technology Information System.

¹⁰ www.ntis.go.kr, accessed on 8 November 2023.

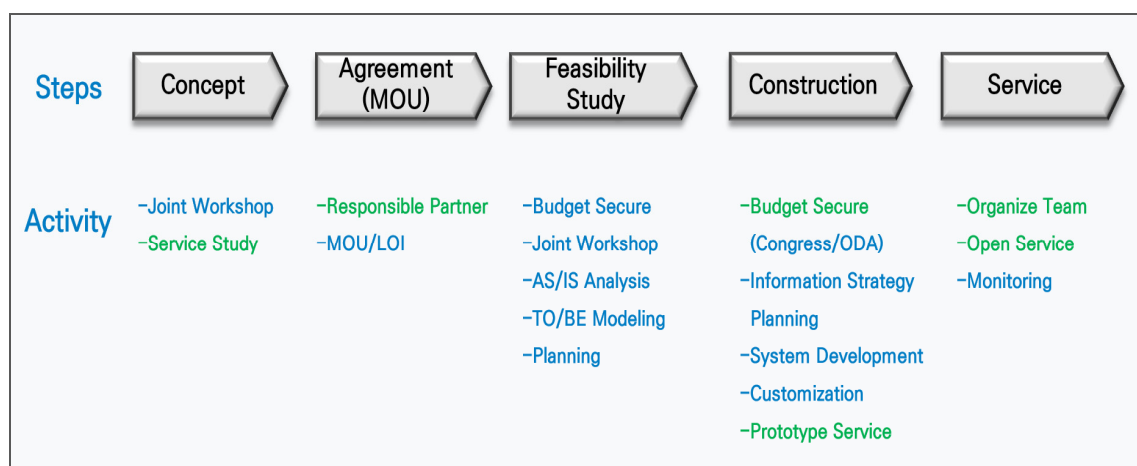
- October 2015, Cambodia, Cooperation on the Establishment of a Comprehensive R&D Information Management System.
- July 2016, Chile, Cooperation on Science and Technology Information Management.
- August 2018, Costa Rica, Cooperation on the Establishment of the Science and Technology Information System.
- May 2019, Belarus, Cooperation on the Establishment of the Science and Technology Information System.
- August 2020, Costa Rica, KISTI-Costa Rica Ministry of Science and Communication (MICITT) LOC Conclusion.

3.2 Success Cases – Costa Rica, SINCYT

An official from Costa Rica's MICITT (Ministry of Science, Technology and Telecommunications) learned about the NTIS, the world's first national R&D knowledge portal, while attending the World Science Summit held in Korea in 2015. At that time Costa Rica was striving to enter the OECD via the development of science and technology and ICT technology, but it had run up against various difficulties and limitations in systematically and comprehensively managing science and technology statistical information, so the meeting with the NTIS proved to be a fortuitous breakthrough in finding solutions to their difficulties.

Accordingly, Costa Rica requested cooperation on the establishment of the Costa Rican Science and Technology Information System (SINCYT), and in 2015 ministers from South Korea and Costa Rica discussed the possibility of revitalising cooperation in the field of science and technology. Figure 2-17 shows the overall process of cooperation between Korea and Costa Rica.

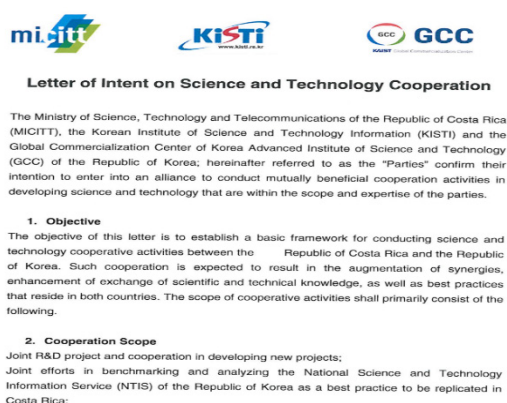
[Figure 2-17] Cooperation Steps



Source: Author's elaboration.

In June 2016, the director of MICITT's R&D visited South Korea to conduct a feasibility study on Costa Rica's national R&D information management system. The Costa Rican delegation stayed in Korea to understand the entire process of establishing an information system, from collecting national R&D information to managing and using it, and ascertained whether it could be applied to Costa Rica. The following month, KISTI, the leading organisation of the NTIS, and the MICITT of Costa Rica signed an LOI (Letter Of Intent) for mutual cooperation and technology transfer in the field of national science and technology information management.

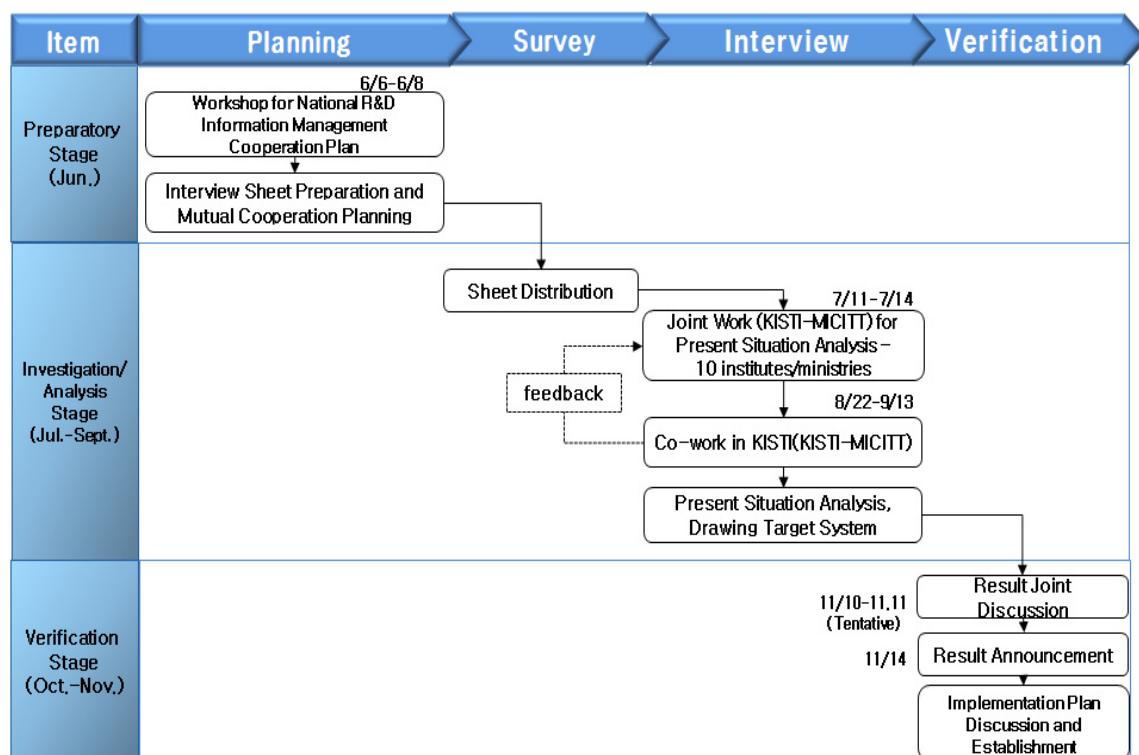
[Figure 2-18] Signing of LOI (Letter of Intent)



Source: KISTI NTIS Centre.

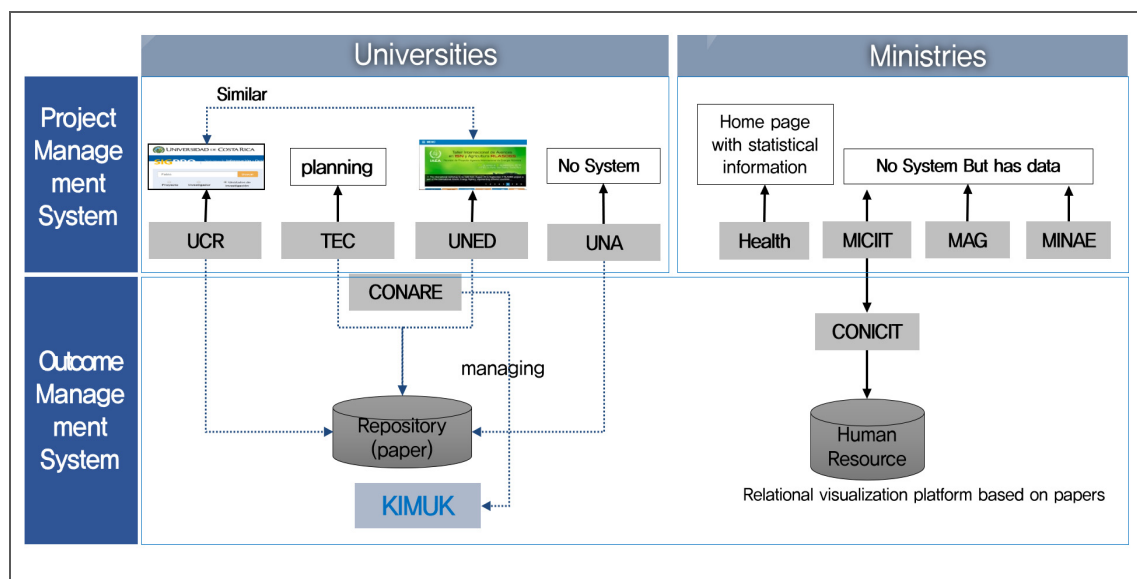
After signing the Letter of Intent (LOI), a collaborative initiative was launched from August 2016 to November 2017 aimed at enhancing Costa Rica's science and technology capabilities. This included transferring expertise in the management and construction of national science and technology information systems, as well as supporting manpower training across Costa Rica's government ministries. The collaboration began with comprehensive in-depth interviews to assess the current state of legal affairs, infrastructure, environmental concerns, standardization, information systems establishment, the scale of R&D projects, and R&D work processes in Costa Rica. Interviewees were carefully chosen from ten industry-academia-research institutes that are pivotal in managing the country's national R&D projects. These interviews were essential for identifying and analyzing the overall state of science and technology information management in Costa Rica.

[Figure 2-19] Status of the National R&D information management survey process



Source: Development of a Platform Model of National R&D Information System for Costa Rica (2016), KISTI Report.

[Figure 2-20] National R&D management systems in Costa Rica



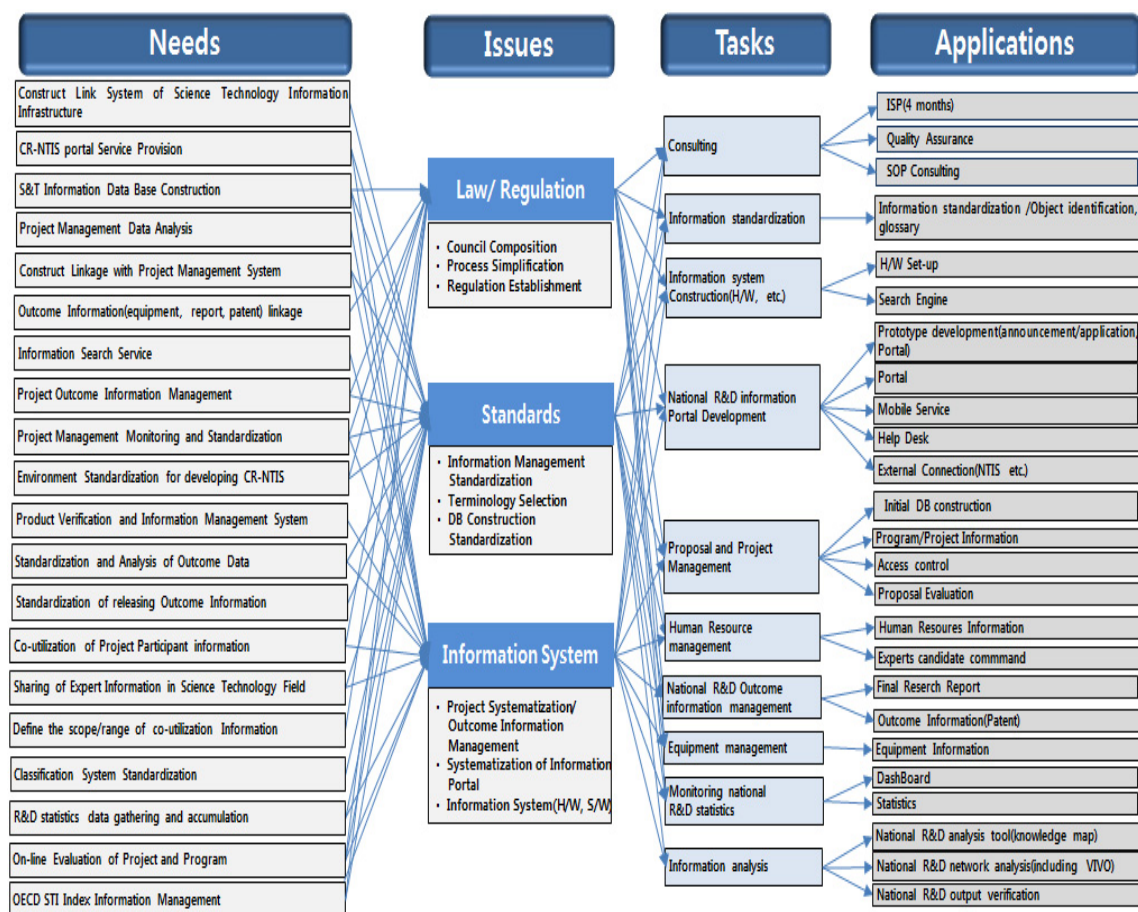
Source: Development of a Platform Model for National R&D Information System for Costa Rica (2016), KISTI Report.

As shown in Figure 2-21, most Costa Rican institutes or universities lack an R&D management system, whereas some do have such a system but have no standard for project management items, resulting in the sporadic generation of data. Most of the interviewees recognised the necessity of an information system such as the NTIS and showed much interest in establishing an integrated national R&D information management system. In addition, although there are regulations on R&D management for each institution, they are not systematised.

Accordingly, the Costa Rican government aimed to establish an integrated R&D management system to integrate and provide comprehensive information on projects, performance outcomes, and researchers. To that end, collaboration with experts in Costa Rica's national R&D project was continued in order to derive improvement tasks necessary for the development of the country's national R&D information management platform model and the creation of a roadmap, and to investigate economic analysis indicators. In detail, twenty-five key tasks were derived in order to prepare a plan for establishing a national R&D information management system; a

roadmap for the next three years was drawn up; and an economic analysis of the cost of constructing the system was conducted. Finally, in order to establish Costa Rica's national R&D information management system, the construction costs for three years were calculated for ten categories such as consulting and standardisation.

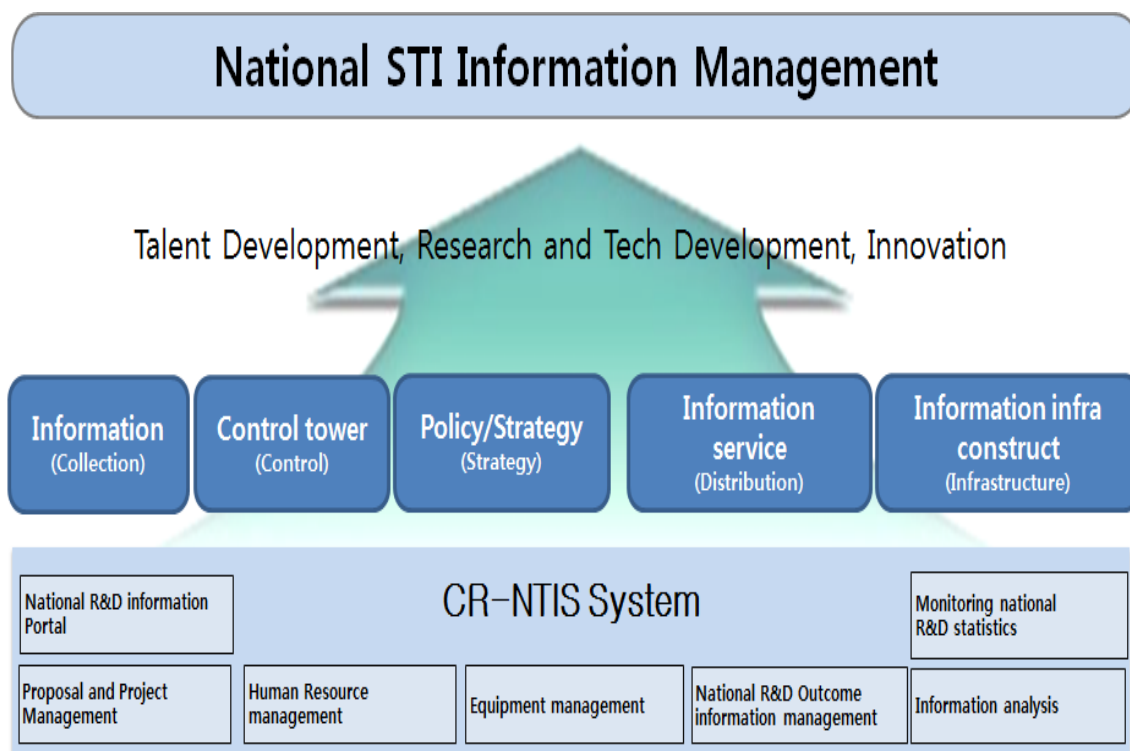
[Figure 2-21] National R&D Management Systems in Costa Rica



Source: Development of a Platform Model of National R&D Information System for Costa Rica (2016), KISTI Report.

Based on this, as shown in Figure 2-22, the integrated management of national science and technology information - through the collection of information, establishment of policies, and provision of an information service based on the information system - was established as the vision for Costa Rica's national R&D information management system.

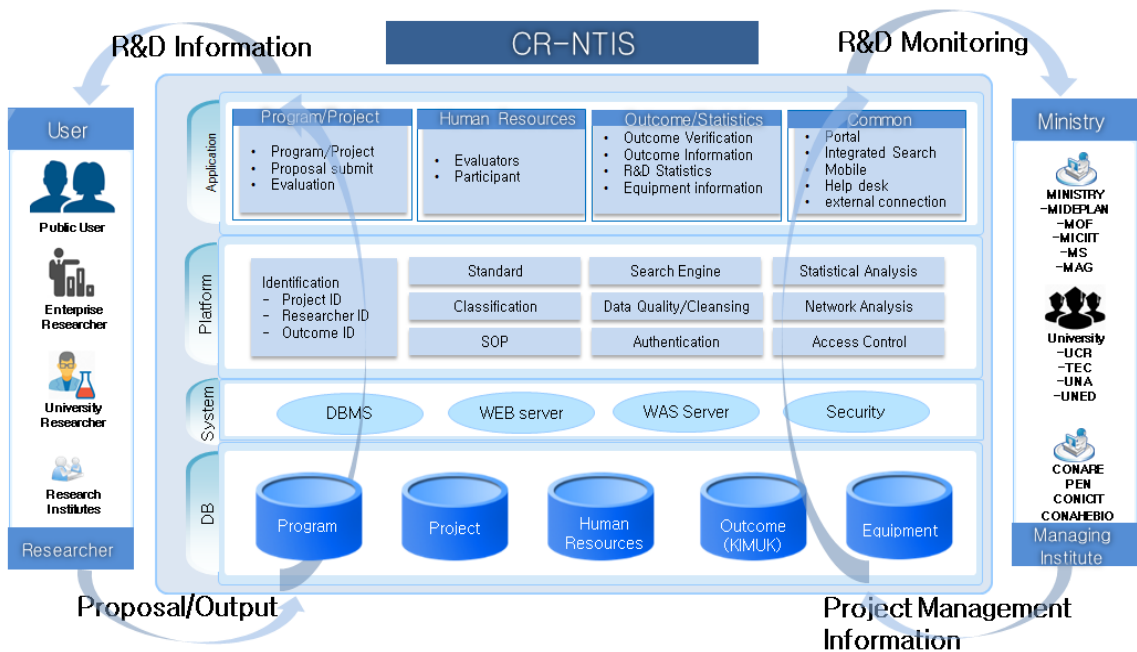
[Figure 2-22] Vision for Costa Rica's National R&D Information Management System



Source: Development of a Platform Model of National R&D Information System for Costa Rica (2016), KISTI Report.

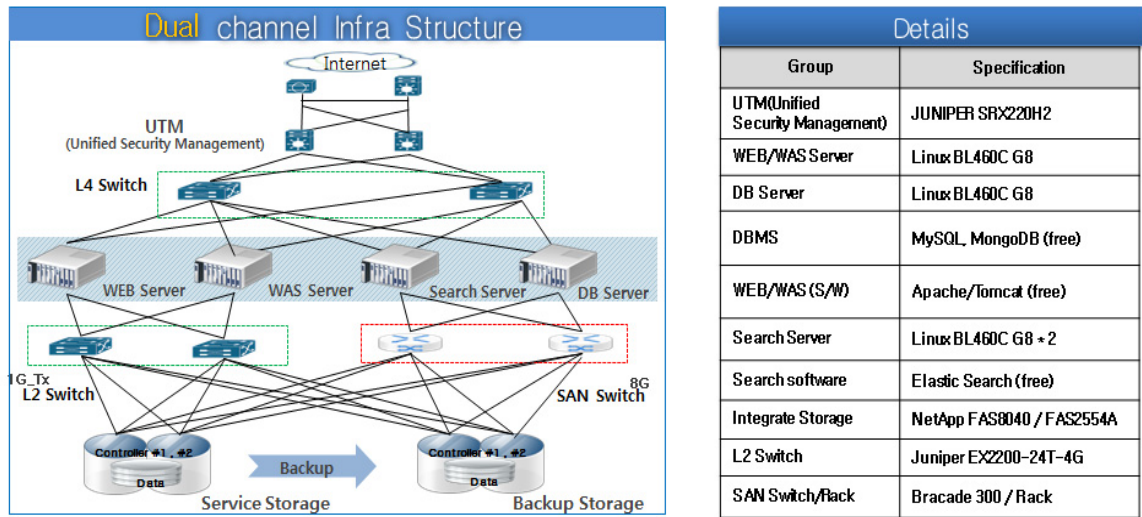
Figures 2-23 and 2-24 provide a schematic representation of Costa Rica's national R&D information management system. This system functions as an integration platform, serving multiple roles including a database, information system, platform, and information service for both administrators and researchers. It is designed to monitor the entire national R&D process effectively. Additionally, from the analysis and system planning, specifications were derived for configuring the information system. Detailed items necessary for establishing the national R&D information management system were also identified and are presented in Figure 2-24. These elements collectively form the backbone of the system, ensuring that it supports the needs of both management and ongoing research activities efficiently.

[Figure 2-23] Schematic diagram of Costa Rica's National R&D Information Management System



Source: Development of a Platform Model of National R&D Information System for Costa Rica (2016), KISTI Report.

[Figure 2-24] Configuration of Costa Rica's National R&D Information Management System



Source: Development of a Platform Model of National R&D Information System for Costa Rica (2016), KISTI Report.

Finally, we shared the most important standards for successful operation of the national R&D management system. Figure 2-25 shows a sample of project standards appropriate for Costa Rica based on an analysis of the status of its national R&D.

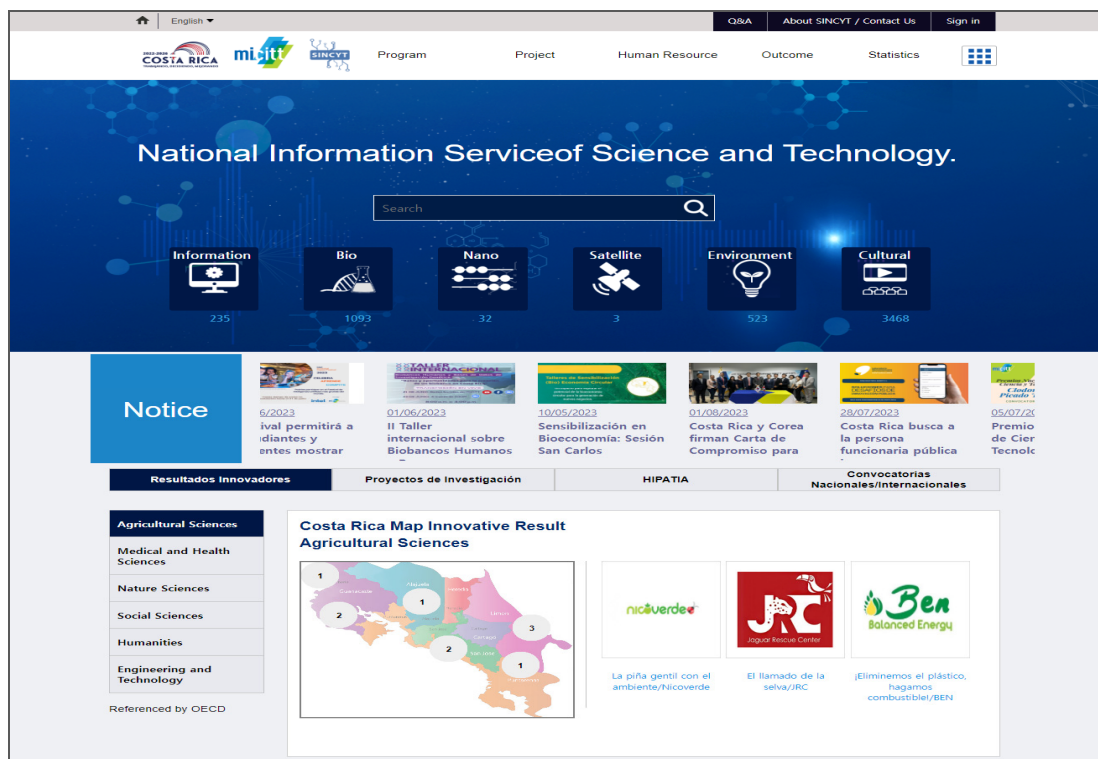
[Figure 2-25] Standard Information Items(Project) for Costa Rica's National R&D Information Management System

Category	Information Item	Number	Information Item Name
Project Information	Program Information	1	Program Name
		2	Name of Ministry
		3	Program period
		4	Program budget
		5	Funding resources
		6	Enforcement Amount
		7	Budget Amount of agreement
		8	Program purpose classification code
	Basic Information	9	Continuous Project Y/N
		10	Project ID
		11	Project Title
		12	Total project Period
		13	Research Subject Code
		14	Region
		15	Science and Technology Standard Classification Code
		16	6T Technonology related Code
		17	Summary of Research Objectives
		18	Summary of Research
		19	Expected effect summary
		20	español Keywords
		21	English Keywords
		22	Confidential Project(Y/N)
		23	Confidential Project End Date
		24	Period of the Project
		25	Project management Organization name
		26	Name of organization
		27	R&D phase code
		28	project progress Status
		29	Government Investment Research budget
		30	Date of Signing
		31	Project Identification Number
	Final Evaluation Result	32	Aborting the project Y-N
		33	Type of the abortion
		34	Reason for the Abortion
		35	Abortion date
		36	Final evaluation result Evaluation Date
		37	Final evaluation result Rating
		38	Final evaluation result Score
		39	Final evaluation result Evaluation Category
	Evaluator	40	Project Identification Number
		41	Date of participation evaluation
		42	Science and Technology registration number
		43	Name of the evaluator
		44	Evaluation step
		45	Personnel Expenses(Salary)
	Project Budget	46	Direct expenses
		47	overhead
		48	budget for commissioned research project
		49	Matching fund
		50	Co-operative research(Y/N)
	commissioned projects / Joint research	51	Commissioned/Co-operative Research project number
		52	Commission project name

Source: KISTI(2016), Development of a Platform Model of National R&D Information System for Costa Rica.

In this way, the derivation of a target system suitable for Costa Rica and the establishment of the establishment plan were completed. Based on these results, Costa Rica requested cooperation on the establishment of the Science and Technology Information System (SINCYT). In October 2016, a memorandum of understanding (MOU) was signed between the two governments and the establishment of the NTIS was put on the agenda of the summit meeting between the two countries. Also, through the approval of the National Assembly of Costa Rica, a budget of US\$ 250,000 was secured for the establishment of a pilot system for promoting the SINCYT's main project. In October 2017, an investment agreement (LOC) was signed between KISTI in Korea and MICITT in Costa Rica, containing the directions for establishing the SINCYT pilot system. Finally, after completing all the processes, we launched the pilot service of SINCYT, an R&D information service in Costa Rica, in August 2018. The opening ceremony was attended by the President of Costa Rica, Ministers and Vice-Ministers of the Ministry of Science and Technology, and KISTI.

[Figure 2-26] Costa Rica's Science & Technology Information Service



Source: SINCYT, www.sincyt.go.cr, accessed on November 8, 2023.

4. Conclusion

4.1 Recommendations for Cambodia

Cambodia needs to introduce more systematic R&D management in to maximise research performance within a very limited R&D investment environment. As the size of the R&D input budget is very small and the promotion of R&D at the national level is important, it is necessary to monitor how efficiently the small-scale R&D budget is invested. Accordingly, it is necessary to establish and operate an integrated national R&D management system at the national level.

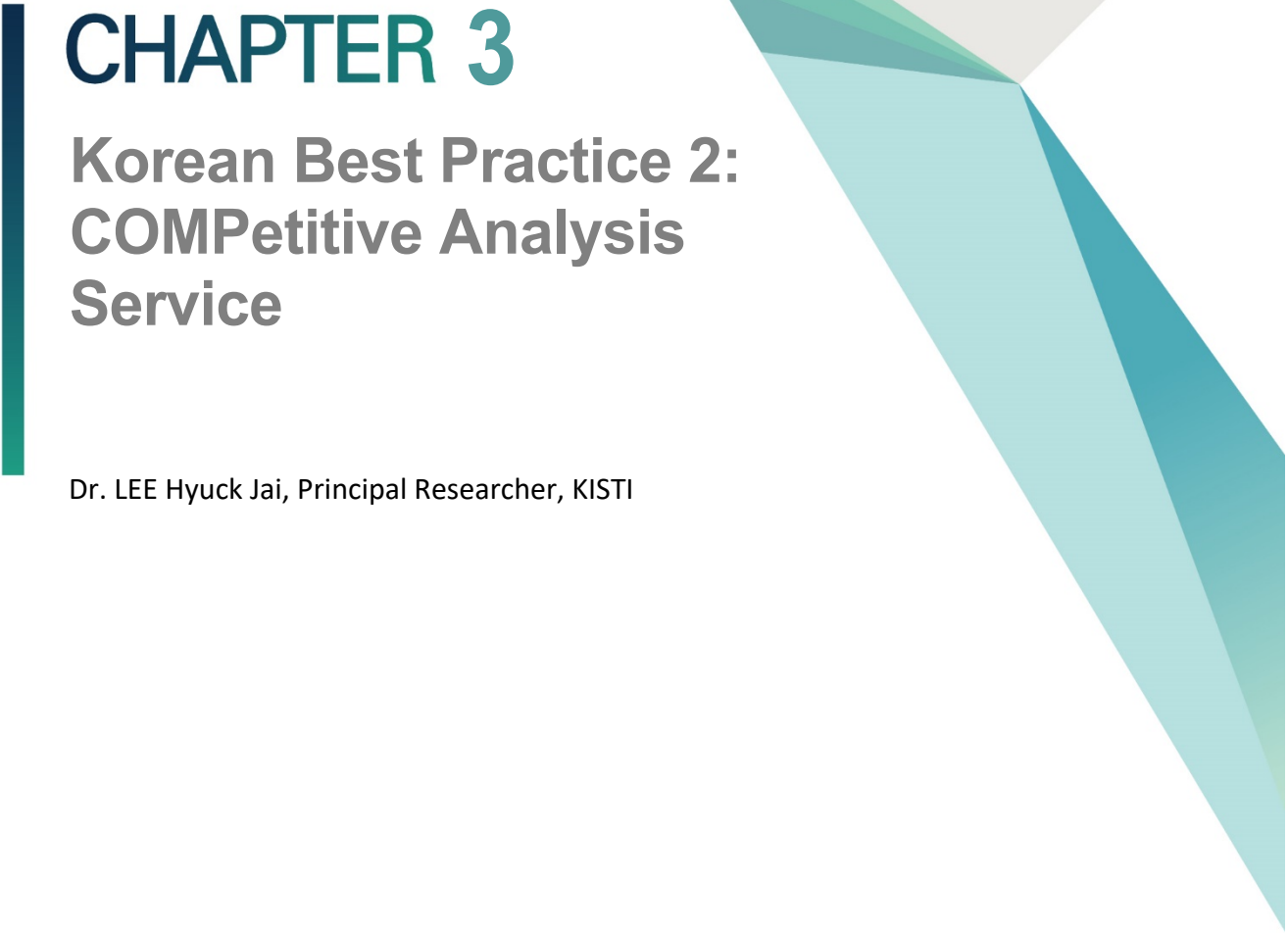
In the case of Costa Rica, national R&D projects are executed on a smaller scale than those of South Korea. As such, the system for carrying out and managing national R&D projects was standardised and integrated into universities that were already operating well. In the case of Cambodia, some universities, including the Cambodian Agricultural University (RUA), were operating an internal project management system. In addition, it seems positive that most of the organisations interviewed agreed on the necessity of establishing an integrated project and outcome management system at the national level.

Therefore, it is strongly recommended to conduct pre-feasibility studies, including environmental analyses, before constructing the system. It is also advisable to develop a pilot system, regardless of its size, and to allow relevant organizations to utilize it. This approach not only demonstrates the system's potential but also supports the case for explaining its necessity and securing a budget for further development. To facilitate this, securing funding in advance for the promotion and implementation of the system based on the results of the pre-feasibility study is crucial. It is recommended to conduct the pre-feasibility study according to the following process :

- Analyse the Status of Cambodia's National R&D Management System.
- Establish a target system and implementation plan for Cambodia's national R&D information management.

- Customise the system, such as the organisation of government departments and the national R&D system, scale, procedure, etc.
- Design a scenario-oriented R&D information management system (portal) screen (draft).
- Establish step-by-step project promotion plans that take into account the feasibility of implementing the project to establish the national R&D information management system and the possibility of effective financing.
- Set the shortest commercialisation target period and perform activities according to the ODA commercialisation promotion process.

Finally, establishing an integrated science and technology information system can be an effective starting point for laying the groundwork for the development of science and technology in Cambodia, i.e. by making timely investments in science and technology based on Cambodia's ICT strategy and utilising the accumulated information.

A decorative vertical bar on the left side of the page, transitioning from dark blue at the top to teal at the bottom. On the right side, there are several overlapping, semi-transparent geometric shapes, including a large light gray triangle and a teal triangle, creating a modern, abstract design.

CHAPTER 3

Korean Best Practice 2: COMPetitive Analysis Service

Dr. LEE Hyuck Jai, Principal Researcher, KISTI



Chapter 3. Korean Best Practice 2: COMPetitive Analysis Service

1. Introduction

This section aims to deepen the reader's understanding of COMPAS (COMPetitive Analysis Service), a system for analysing science and technology information. It provides information on various aspects of COMPAS, including its history, purpose, usage guidelines, available functions, and analysis models. Additionally, this section explores the applications of COMPAS, with a particular focus on its international dissemination and implementation in Vietnam through an ODA (Official Development Assistance) project funded by the Korean government. By providing this information, the reader will gain a deeper understanding of COMPAS and its potential benefits.

1.1 Background to the development of COMPAS

The exponential increase in global digital information calls for the development of advanced technologies designed to manage, analyze, and effectively utilize these massive data volumes. These technologies are essential not just for crafting timely policies and making informed decisions but also for protecting research and fostering advancements in the fast-paced realms of science and technology.

Both small and medium-sized enterprises (SMEs) and individual researchers often encounter difficulties in accessing substantial data or advanced data processing technologies, making support for these technologies crucial for them. In light of this, there is a growing interest in big data processing, and national-level support is also on the rise. COMPAS was conceived amidst this context, with the aim of providing systematic IT-based support from the government side to assist those who lack information infrastructure and find themselves at a disadvantage.

1.2 A brief history of COMPAS

The COMPAS project, initiated in 2008 and funded by the National Research Council, was designed as an "R&D for service" initiative, aiming to develop a public service system. Over its five-year duration, the project team developed and tested various analysis models to be utilized as service models. The initial prototype featured two specific analysis models: "Competitor Identifier" and "Journal Article Analysis." This prototype was first field-tested in 2008 during the Evaluation of National Thrust Technologies by South Korea's Ministry of Science and Technology, which involved the quantitative evaluation of 364 technologies.

Following successful validation of the analysis system's effectiveness and applicability, the COMPAS analysis models were further refined and expanded, and additional databases were developed. In 2013, the COMPAS beta-service was launched, including six analysis models. This beta phase paved the way for the official public service, which launched in 2014 with ten analysis models. Since then, the number of registered users has grown steadily, surpassing 6,000 by 2023. Today, COMPAS stands as a flagship service of the Korea Institute of Science and Technology Information (KISTI), demonstrating significant progress and impact in the field of technology analysis.

1.3 What is COMPAS?

Understanding the competitive environment surrounding a technological area is crucial for firms to survive in the world of fierce competition. COMPAS is an online analysis service designed to help firms make informed decisions by analysing the competitive environment, which is frequently represented as the "big data" of intellectual properties such as patents and journal articles. When planning and conducting R&D in a certain technology field, users can conduct a rapid yet in-depth analysis of (selected) Management of Technology (MOT) questions that are frequently encountered by following the simple steps in COMPAS. Table 1 shows the MOT questions and the corresponding analysis models in COMPAS.

[Table 3-1] The most frequently asked MOT questions and the corresponding analysis models in COMPAS

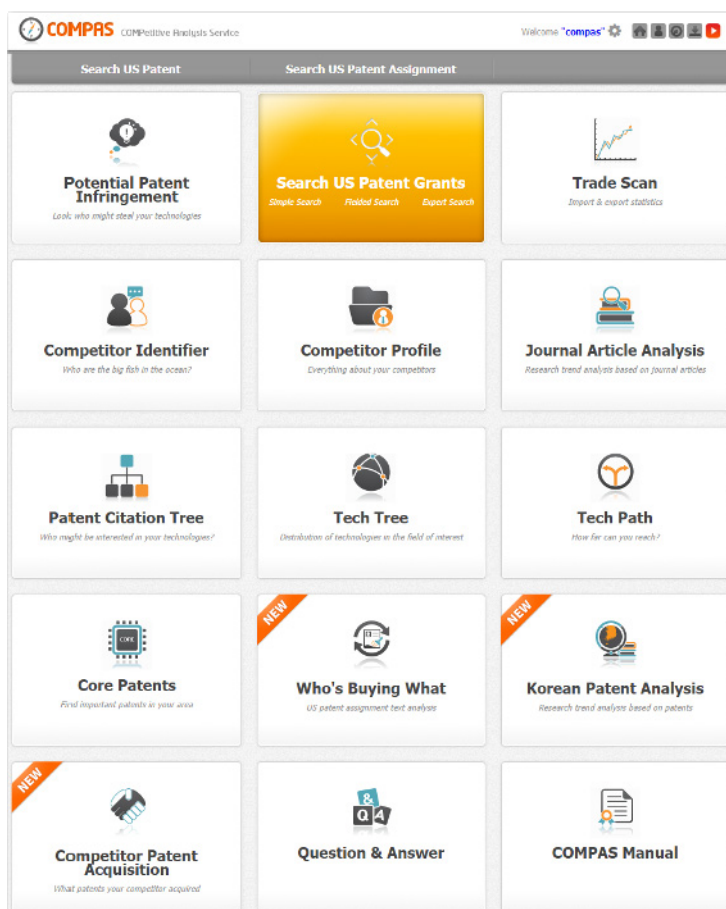
MOT Questions	Analysis Model
Who are the major players?	Competitor Identifier
What are the major players doing?	Competitor Profile
What is going on in the academic field?	Journal Article Analysis
Who is doing something similar to me?	Potential Patent Infringement
Who is interested in my technology?	Patent Citation Tree
What are the most important patents?	Core Patents
What are the subsidiary technologies?	Tech Tree
How far can I extend?	Tech Path
Is there a new market opportunity?	Trade Scan
What gets people's attention?	Promising Patent Technologies

Source: Author.

1.4 Using COMPAS

Users can start using the Korean and English versions of COMPAS shown in Figure 1 at <https://compas.kisti.re.kr>, which is provided by KISTI for domestic public service in Korea. The Vietnamese and English versions of COMPAS provided by the National Agency for Science and Technology Information (NASATI) of Vietnam for domestic users in Vietnam is available at <https://compas.vista.gov.vn>. Detailed information on the Vietnamese service is available in the ODA section of this document. Registration is required to gain access to COMPAS. Restrictions may exist depending on the types of registered users.

[Figure 3-1] Screenshot of the COMPAS service site



Source: Author.¹¹

1.5 Why COMPAS?

COMPAS is designed specifically for users without any expertise in informatics, providing them with information sources of important intellectual properties and the tools needed to manipulate and analyse them effectively.

Users can access COMPAS using conventional web browsers. No additional application is required. The user-friendly design of COMPAS enables easy navigation of the service. By simply specifying what to analyse, the analytical models run automatically. Users do

¹¹ <http://compas.kisti.re.kr>, accessed on 28 June 2023.

not need to spend time on data processing as is the case with conventional analytical tools. All analysis is conducted on the COMPAS server, so high computing power is not required on the user's side regardless of the amount of data to be analysed.

COMPAS differs from conventional search sites for patents (or for other intellectual properties). Besides the search site for US Patent Grants (USPG), COMPAS provides various functions and analysis models that are useful for R&D and R&D management and are designed to meet the needs of users. The essential features of COMPAS are its analysis models, each of which represents the most frequently asked MOT questions. Though COMPAS mostly works on USPG, some models work on other types of information, such as a bibliography of journal articles (Web of Science, Scopus, and PubMed) and trade statistics. Each of the analysis models of COMPAS works individually, except for "Competitor Profile", which must be combined with "Competitor Identifier".

COMPAS provides various functions for the user's convenience. Search results can be stored, viewed, or used afterwards as input materials for the analysis models. The results of analyses are automatically stored on the server and can be viewed or organised when necessary. The "Basic Analysis" function allows the user to conduct a quick review of the technology field for fine-tuning the search conditions, and does not require the user to save the search results or run any of the analysis models. Automatic analysis upon database updates is available for the "Patent Citation Tree" model.

1.6 The structure and composition of COMPAS

Figure 3-2 illustrates the structural layout and components of COMPAS, showcasing how data sources (DBs) form the foundation of the system, akin to other conventional analytical software. The primary data source for COMPAS is the US Patent and Grants (USPG) database from the US Patent and Trademark Office (USPTO). Additionally, PATSTAT from the European Patent Office (EPO) supplements COMPAS with non-US family patents and patent citations, enhancing the breadth of patent data available.

For trade analysis, COMPAS utilizes import and export data from the past five years for countries listed in the UN ComTrade database. This allows for a comprehensive review of trade flows and patterns.

COMPAS also incorporates the ability to analyze bibliographies of journal articles, though these are sourced externally from databases such as Web of Science, Scopus, and PubMed, which are not inherently part of COMPAS but are critical for a thorough bibliographic analysis.

Furthermore, the main database for COMPAS is undergoing expansion to include a global patents database based on the DOCDB provided by the EPO. This world-patent version of COMPAS is expected to be launched by the end of 2023. Notably, this new service will be accessible via a separate website, distinct from the one currently hosting the USPG data, marking a significant enhancement in COMPAS's capabilities and services.

[Figure 3-2] Structure of COMPAS

Visualization & User Interface	
Documentation	Analysis Models
Scheduler	
Meta DB	
DB Modeling	Search Engine
DBs (US Patent Grant, PATSTAT, Trade DB, Journal Articles*, Supplementary DBs)	

* From external resources (WoS, Scopus and Pubmed)

Source: Author.

Users can run search queries and save them to define their TOIs (Technologies of Interest) on the built-in patent search page of COMPAS. The analysis models come into operation when the input data are specified by various methods, such as feeding a saved search, uploading the bibliographic data of journal articles, or specifying the trade item. Once the input data has been specified, the analysis model initiates and begins the analysis, the result of which is shown on the Projects page, which can be

downloaded as MS Word and Excel files.

1.7 Defining TOI (Searching Patents)

Running the analytical models starts by defining a TOI, which refers to a set of patents designated by the user via a patent search, and then saving the search result in COMPAS. COMPAS provides a patent search page for the full-text USPG, which is available in three search modes: Simple, Fielded, and Expert Search. These can be selected depending on each user's search skills.

1.7.1 Simple Search

In the Simple Search mode, the user can run searches simply by addressing the search keywords (words or phrases) describing the TOI. There is also a function for applying the Boolean operation (AND or OR) to the keywords. The target of the search is limited to the TEXT field: a combined field of title, abstract, description/specification, and claims of patent documents. The date of publication can be refined optionally. The user-input search condition is automatically converted to the search query and displayed at the top of the View List (search result page).

[Figure 3-3] Screenshot of Simple Search

Search US Patent

Simple Search | Fielded Search | Expert Search

"Simple Search" searches given keyword (or keywords) from title, abstract, description/specification, and claims. One can specify the Boolean operator (AND/OR) among the keywords. Default operator is set to AND.
Database coverage 1991-01-01 - 2023-06-06

Text
(title, abstract, des/spec, claim)

example
- oled OR amoled
- oled* AND display
- (oled OR amoled) NOT pmoled

☐ Publication Date 2001-01-01 ~ 2023-06-30

SEARCH RESET

Source: Author.¹²

¹² <http://compas.kisti.re.kr>, accessed on 28 June 2023.

1.7.2 Fielded Search

The Fielded Search mode provides more options for a search, enabling the user to refine the TOI more accurately. The search fields available in this mode are as follows: Publication Number, Publication Date, Application Number, Application Date, Title, Abstract, Description/Specification, Claim, Assignee, Assignee Country, Assignee State, Assignee City, Inventor, Inventor Country, Inventor State, Inventor City, IPC (International Patent Classification), and USPC (US Patent Classification). The user-input search condition is automatically converted to the search query and displayed at the top of the View List.

[Figure 3-4] Screenshot of Fielded Search

Source: Author.¹³

1.7.3 Expert Search

In the Expert Search mode, the user can write search queries according to the query syntax defined by the search engine. In this case, the user can tailor the search queries optimally, although learning the query syntax requires additional effort. The Expert Search mode is highly recommended for the best analysis results.

¹³ <http://compas.kisti.re.kr>, accessed on 28 June 2023.

[Figure 3-5] Screenshot of Expert Search

Source: Author.¹⁴

Every patent on the View List is linked to the USPTO so that the original documents can be accessed if needed. However, this linkage may not become unavailable without notice due to a change in the policy of the USPTO. All of the search queries during a log-in session are stored in the server and can be retrieved using the Search History function. A function for applying the Boolean operation to the stored search queries in the Search History is also available.

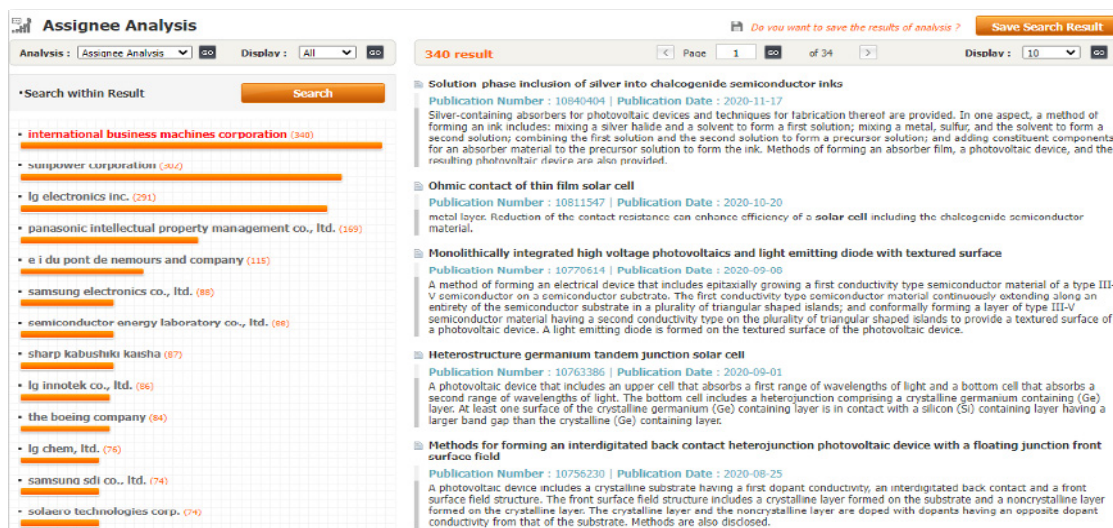
1.7.4 Basic Analysis

COMPAS provides a function for quick and easy analysis called “Basic Analysis”, which enables patents to be analysed on a 'simple counting basis' with several analytical elements, such as date/time (application/publication year), assignee, inventor, assignee country, inventor country, IPC, and USPC. Basic Analysis analyses the patents shown in the search result page without saving it. Users can use the ‘Search within Result’ option to refine the search queries further. Using Basic Analysis is a good way to browse a TOI and fine-tune the corresponding search query. The final search query

¹⁴ <http://compas.kisti.re.kr>, accessed on 28 June 2023.

for a TOI needs to be saved for use as the input material for the analysis models of COMPAS.

[Figure 3-6] Basic Analysis showing the assignee distribution of "solar cell" patents



Source: Author.¹⁵

1.8 Analysis Models in COMPAS shown in the format of *Model Name (Required Data)*

1.8.1 Competitor Identifier (USPG)

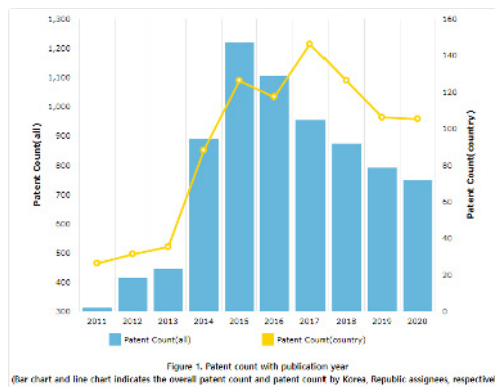
The initial step of planning R&D for a TOI involves gaining an understanding of the TOI's landscape and identifying prominent competitors. The most suitable approach is to use the model Competitor Identifier. This model offers an overview of the R&D activities within the TOI, including the number of patents (with the year of publication) and the distribution of patents among various assignees (competitors) and assignee countries. Additionally, it provides indicators that assess the quality of patent portfolios for major assignees (and assignee countries) based on patent citations and family patent sizes. A compilation of highly cited patents within the TOI is also provided. In conjunction with the aforementioned indicators, the analysis concludes by

¹⁵ <http://compas.kisti.re.kr>, accessed on 28 June 2023.

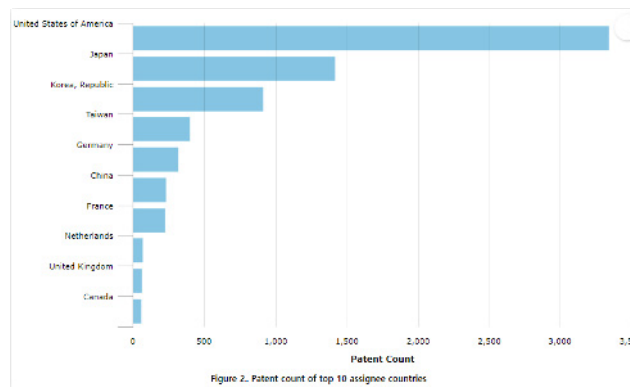
presenting a list of potential core competitors, specifically the top 30 assignees with the highest patent counts. The indicators provided by this model are as follows:

- Overall patent count with the year of publication.
- Patent counts of assignees and assignee countries (with the year of publication).
- Patent quality (Q-factor and average family size) for assignees and assignee countries.
- Patent counts of Vietnamese assignees (with the year of publication).
- Patent quality (Q-factor) for Vietnamese assignees.
- Potential core competitors.
- Highly cited patents.

[Figure 3-7] Selected charts from Competitor Identifier applied to the 'solar cell' technology field: (a) Patent count (overall and by domestic assignees), (b) Quality factor of assignees



(a)



(b)

Source: Author.¹⁶

¹⁶ <http://compas.kisti.re.kr>, accessed on 28 June 2023.

1.8.2 Competitor Profile (USPG)

This model provides a detailed profile of the patent activities of core competitors (assignees) designated by the user from among the competitors that exist within a specific TOI. To run this model, the model "Competitor Identifier" must be conducted in the first place to obtain the list of potential competitors. In the process of running this model, the user needs to select a specific project of "Competitor Identifier" and select the desired core competitors to create a profile (or portfolio). For the core competitors selected by the user, this model analyses and provides the following patent profiles. The user can analyse the detailed profile of each of the core competitors in the TOI, thereby identifying the major technology field, major commercialisation area, and the status of the core inventor of each competitor. The results of analysis can be used as a guideline for setting R&D directions and benchmarking competitors in the TOI. The indicators provided by this model are as follows:

- IPC analysis by competitor.
- USPC analysis by competitor.
- Analysis of the applicant countries by competitor.
- Inventor analysis by competitor.
- Major patent lists by competitor.

[Figure 3-8] Screenshot of Competitor Profile selecting the competitors to be profiled in the 'solar cell' technology field

No	Core competitor(applicant)	Patent count(P)	Q-factor(Q)	Average family size compared to the field average	Specify competitors of interest
1	IBM	341	0.355	0.431	☒
2	LG ELECTRONICS INC	292	0.161	0.986	☒
3	SUNPOWER CORP	288	2.891	1.55	☒
4	DU PONT	138	0.789	1.103	☐
5	SAMSUNG ELECTRONICS CO LTD	92	0.52	0.643	☐

Total analysis period: 2011 ~ 2020

Specify the point of strategic change: 2015 (The criteria for examining the competitor's patent portfolio in both the first and second half of the year (the default setting is the middle year of the total analysis period))

Analysis Application

Source: Author.¹⁷

1.8.3 Journal Article Analysis (Bibliography of Journal Articles)

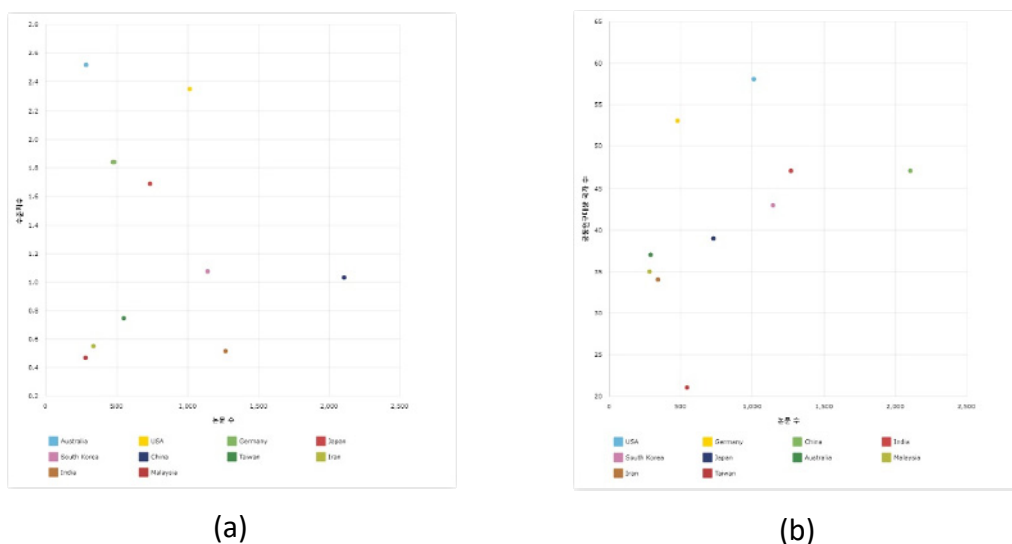
Although Competitor Identifier offers valuable insights into a TOI's "technological" landscape through the examination of patents, it may be necessary for the users to gain a deeper understanding of the TOI from a "scientific" perspective. In such cases, Journal Article Analysis becomes a useful tool. This analysis requires the users to obtain the relevant articles by themselves, but it also provides a means to extract valuable insights and trends from the scientific literature. Users can run analyses of the bibliographic information obtained from Web of Science, Scopus, and PubMed. The indicators provided by this model are as follows:

- Overall article count (with the year of publication).
- Article counts of countries and institutions (with the year of publication).
- Article quality (Q-factor) for authors, author countries, and institutions.
- Activity Index (AI) for author countries and institutions.

¹⁷ <http://compas.kisti.re.kr>, accessed on 28 June 2023.

- International collaboration strength (L-factor) for authors, author countries, and institutions.
- List of core institutions and authors.
- Highly cited patents.
- List of Vietnamese institutions.

[Figure 3-9] Selected analysis results from Journal Article Analysis: (a) Quality Factor vs. the number of journal articles for author countries, (b) International collaboration factor vs. quality factor for institutions



Source: Author.¹⁸

1.8.4 Potential Patent Infringements (USPG)

The objective of this model is to explore patents that share similarities with the Patent of Interest (POI) designated by the user. The technology field to which the model is applied refers to a set of patents designated by the user (TOI). Users can search for the TOI using the COMPAS patent search function, and the results of the search are stored in COMPAS. During the process of model execution, the user designates the TOI and a specific patent as the POI from the designated TOI. The similarities between the POI

¹⁸ <http://compas.kisti.re.kr>, accessed on 28 June 2023.

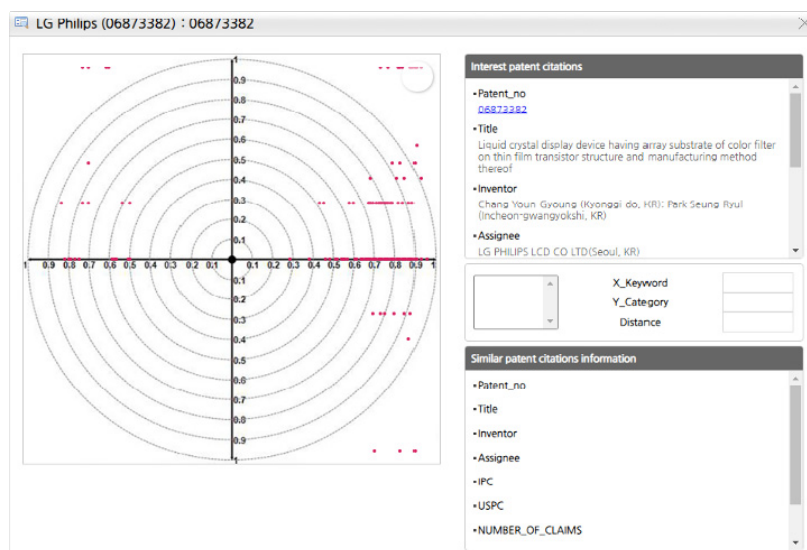
and the remaining patents are then analysed by the model.

The results of the analysis are presented in an easy-to-understand figure wherein the POI is located at the origin (0, 0), and the degree of similarity between the other patents and the POI is given as "distance". In other words, patents that are closer to the origin are more similar to the POI. The similarity between patents is calculated in two ways: i) by measuring how similar the terms extracted from the claims of each patent are, and ii) by measuring how similar the patent classifications given to each patent are. The similarities are represented in terms of distance on the Cartesian coordinate visualisation interface, where the two axes refer to distances in terms of claims and patent classification, respectively. This model provides the following analysis results along with similar patent search results.

- Time-series patent quantitative analysis.
- Similarity measurement and a list of similar patents.
- Comparative analysis of highly similar patents.
- Review of the details of similar patent.

This model can be applied to various tasks such as the investigation of prior patents, determination of duplication of patent research results, comparative analysis of competitors and competitor patents, novelty of invented technologies, and evaluation or diagnosis of inventive technologies.

[Figure 3-10] Screenshot of Potential Patent Infringement with the POIs US6873382 and LCD color filter patents



Source: Author.¹⁹

1.8.5 Patent Citation Tree (USPG)

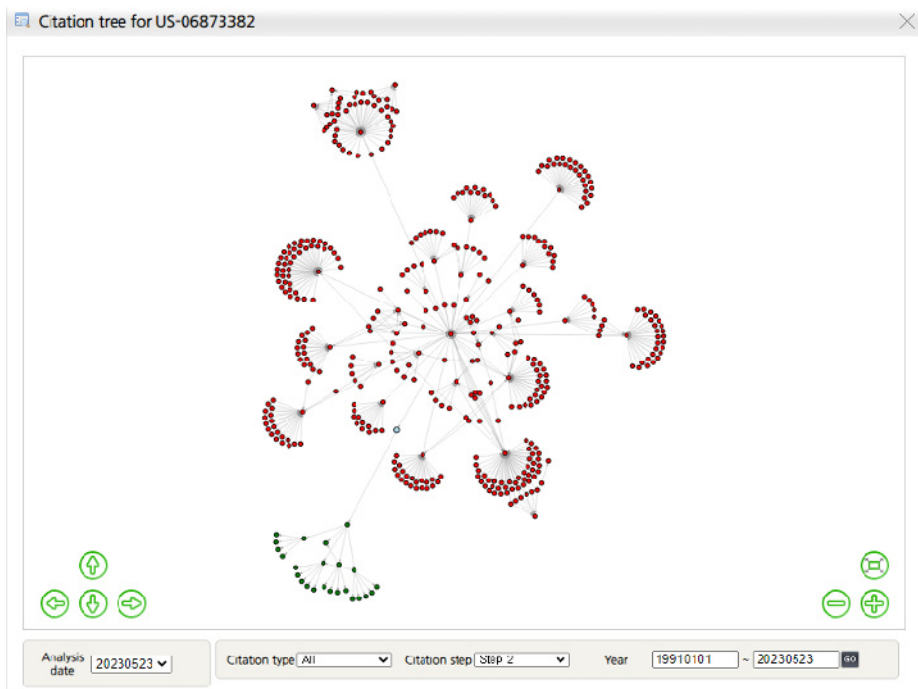
This model analyses and monitors the backward and forward patent citations of a POI. Backward and forward patent citation refers to the patent documents that are cited by and citing the POI, respectively. The result of the analysis is given in a visualised form (i.e. a citation tree). Once the POI has been registered in a project, this model monitors the forward citations and alerts the user when a new citation occurs. This model provides the following features.

- Citation tree: Visualisation of backward and forward citations up to 3 levels, with filters such as backward/forward, level of citation, time period, etc.
- Patent list: List of patents in the citation tree with such features as patent number, IPC, assignee, citation category, backward/forward citation, level of citation, etc.
- Statistics: Statistical analysis of the patents in the citation tree with such analysis items as the list of new forward citation patents, chronological distribution, assignees, etc.

¹⁹ <http://compas.kisti.re.kr>, accessed on 28 June 2023.

Backward citation can be used to search the prior arts of a patent, whereas forward citation can be used to search potential competitors or collaborators in terms of citations. Other applications include potential patent infringement and technology market search.

[Figure 3-11] Patent Citation Tree of US6878382. The green and red dots indicate the backward and forward citations, respectively



Source: Author.²⁰

1.8.6 Trade Scan (Trade Data)

This model provides global trade (import and export) information on all available items within the most recent five years. Items are designated by the HS Code (Harmonised Commodity Description and Coding System) defined by the World Customs Organisation (WCO). The item to be analysed can be specified in two ways: by entering the HS code in a 6-digit format, or by selecting an HS code from the results of a search for items whose trade amounts are larger than the search criteria (amount of export or

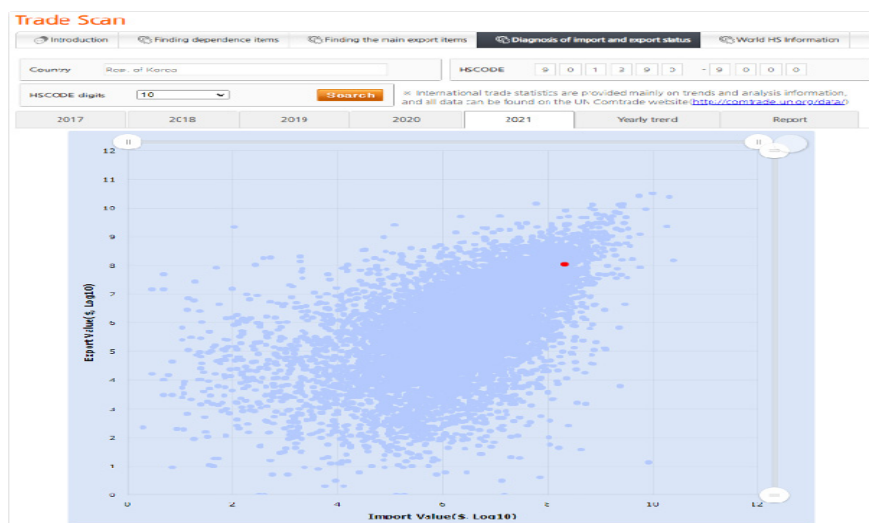
²⁰ <http://compas.kisti.re.kr>, accessed on 28 June 2023.

import in US dollars) as specified by the user. For the specified item, this model provides the following analysis results:

- Yearly trading unit price.
- Yearly trading volume.
- Export/import status by target country.
- Trade Specialisation Index.
- Export concentration.
- Export superiority cross-analysis.

By understanding the domestic and foreign trade status of specified items of interest, users will be able to find new markets, i.e. by finding new commercial products, supporting technology development strategies and technology opportunities linked to market demand, promoting import substitution, and finding main export items.

[Figure 3-12] Import-export chart from Trade Scan for all the items in 2021 with the color filter for LCD (HS Code 90-13-90-9000) displayed in red



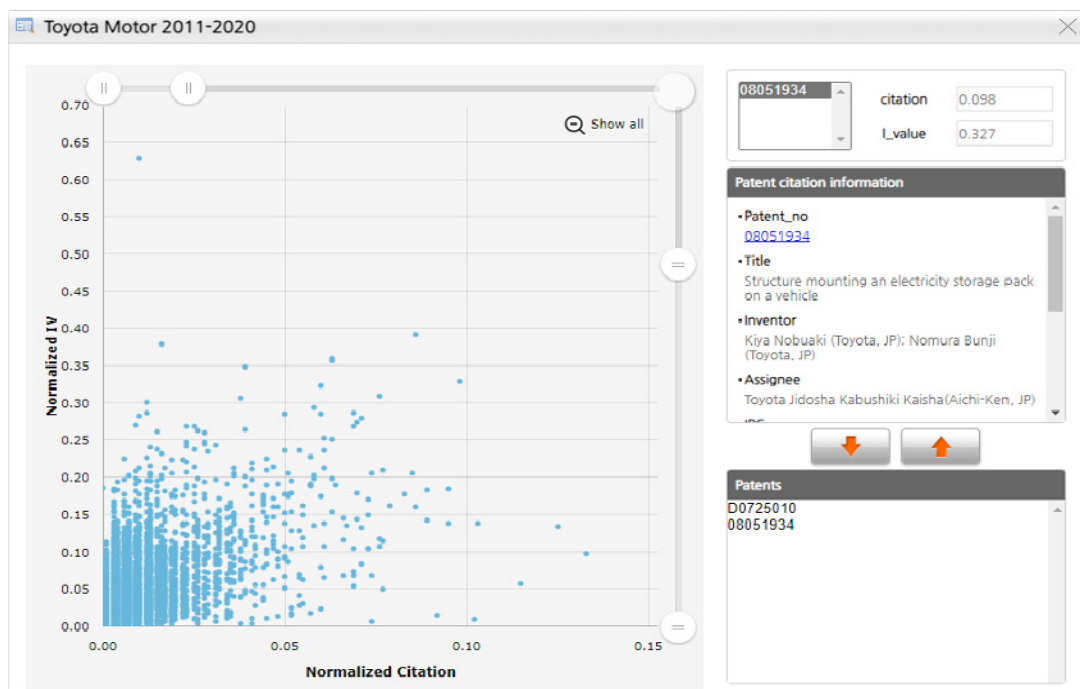
Source: Author.²¹

²¹ <http://compas.kisti.re.kr>, accessed on 28 June 2023.

1.8.7 Core Patents (USPG)

This model aims to identify 'core patents' that have a significant influence within a TOI. During the process of executing the model, the user designates the TOI to be used for analysis. The core patents are identified by two citation-based indicators: traditional citation count and modified citation count. The modified citation count, which is referred to as the 'impact value (IV)', is obtained by modifying the traditional citation count so that the age of the cited patents and the impact of the cited patents are considered simultaneously. By utilising two indicators, the user can avoid overlooking patents with low conventional citation counts with high impact values, which cannot be obtained by conventional citation counting alone.

[Figure 3-13] Screenshot of Core Patents exploring the patents of Toyota Motor published from 2011 to 2020. The X and Y axes indicates the (normalised) conventional citation count and the impact value, respectively. The picture shows patents D0725010 and 08051934 selected as the core patents ready for further analysis



Source: Author.²²

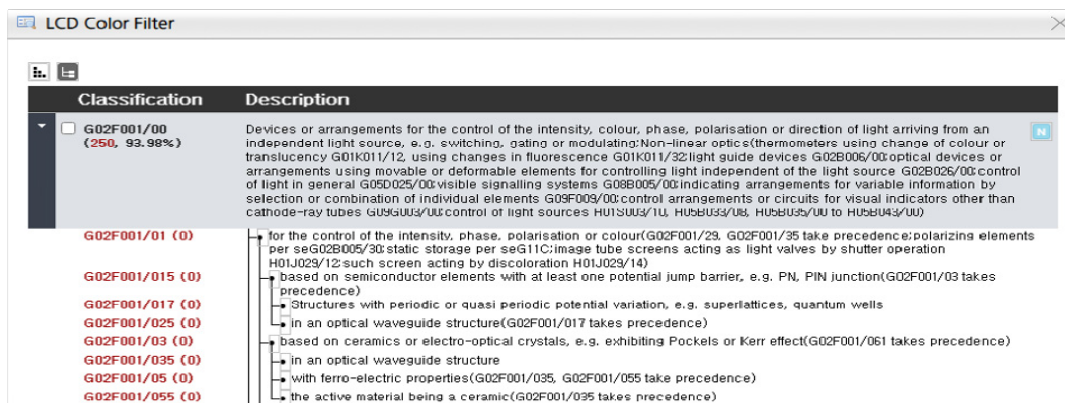
²² <http://compas.kisti.re.kr>, accessed on 28 June 2023.

1.8.8 Tech Tree (USPG)

This model identifies the subsidiary technologies comprising a TOI and gives detailed information on the subsidiary technologies selected by the user. The analysis is conducted based on the International Patent Classification (IPC) by the World Intellectual Property Organisation (WIPO). The IPC follows the hierarchy with the levels of Section, Class, Subclass, Main Group, and Subgroup. During the process of executing the model, the user designates the TOI to be used for analysis. When the user designates a TOI and runs the analysis, a list of IPCs consisting of the TOI is provided. The initial list is made of the IPC main groups, which can be expanded to the subgroup level in a hierarchically visualised structure. The user selects multiple IPCs (User's IPC) for detailed analysis, which contains the following analytical items:

- Patent activities by major IPCs and the User's IPC.
- Major assignees' patent activities by IPC and the User's IPC.
- Vietnamese assignees' patent activities by the User's IPC.
- IPCs related to the User's IPC.

[Figure 3-14] Screenshot of Tech Tree applied to the technology field 'LCD Color Filter' showing the subsidiary technologies within it. The picture shows the IPC G02F001/00, which is included in the LCD Color Filter technology, and its subsidiary technologies are depicted in a hierarchical layout



Source: Author.²³

²³ <http://compas.kisti.re.kr>, accessed on 28 June 2023.

1.8.9 Tech Path (USPG)

This model provides technology paths for the potential applications of a TOI. From a technical point of view 'technology path' means the link from one IPC to another in the technology map, which is obtained based on the co-occurrence network of IPCs. When the technology map is drawn, it is assumed that the IPCs assigned to a patent are the building blocks of the invention in that patent, and that the IPCs occurring in the same patent are co-related. The user selects an IPC as the starting point, which is then extended to its co-occurring IPCs according to the user's choice up to 8 levels. When the ending IPC is selected, the analysis will be conducted on each IPC in the path (step-wise) or on the path from one IPC to another (path-wise):

- Step-wise analysis
- Path-wise analysis

[Figure 3-15] Screenshot of Tech Path showing the path from the IPC G03F007 to B24B037 via H01L021 identified from the paths existing in the patent set published in 2022

The screenshot shows the 'Tech Path' interface with the following components:

- Search Section:** Project: G03F007, Starting IPC: G03F007, Search within (years): 1 (2022), Start Search button.
- Path Visualization:** A sequence of IPCs: G03F007, H01L021, B24B037, followed by Level4, Level5, Level6, Level7, and Level8.
- Results Table:** A table listing IPCs and their associated data.

IPC	Description	Count	Year	Level	
17	B08B003	101	447	2022	2
18	B23K026	98	668	2022	2
19	B24B037	94	188	2022	3
20	G01N021	92	2572	2022	2
21	G11C011	88	1957	2022	3

Buttons at the bottom: Path list, Start Analysis.

Source: Author.²⁴

²⁴ <http://compas.kisti.re.kr>, accessed on 28 June 2023.

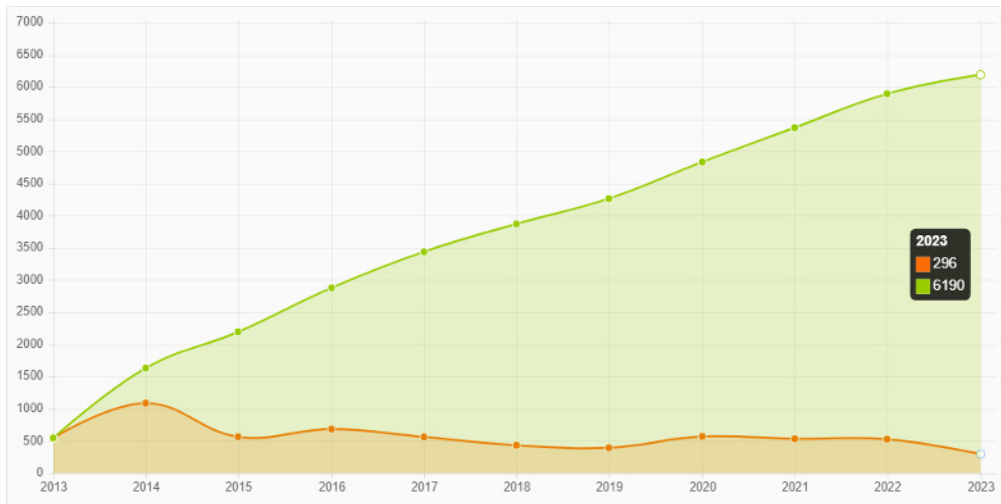
1.8.10 Promising Patent Technologies (USPG)

This model aims to provide users with the technologies related to a TOI that have recently attracted attention from researchers. From a technical point of view, this model groups (clusters) highly cited patents based on the similarities between them. The clusters are pre-made and stored in COMPAS. During the process of model execution, the user designates a TOI, and the model compares the patents in the TOI with those in the clusters. When at least one of the user's patents is found in the clusters, the model returns the list of corresponding clusters. Users can identify trends in research topics by identifying recent technology areas of interest within the technology field and utilising them to set direction for future research and development.

2. COMPAS service and applications in Korea

The official domestic public service of COMPA was launched at <https://compas.kisti.re.kr> on 3 December 2014, approximately fifteen months after the launch of the beta-service (with six analytical models) on 13 September 2013. Since the introduction of the beta service, the registered user count and model run count have been increasing continuously, as shown in Figure 3-16. This rise is attributed to the activities of individual users. Users from various types of organisations, such as universities, SMEs, government research institutions, government funding agencies, patent attorney's offices and so forth, are registered with COMPAS.

[Figure 3-16] Annual user count (orange) and accumulated user count (green) with the service year of COMPAS shows an accumulated user count of 6,190 as of 28 June 2023



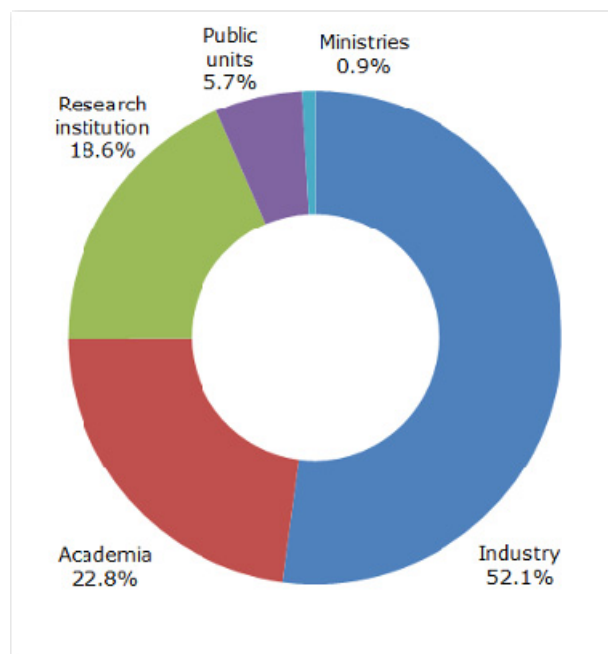
Source: Author.²⁵

A breakdown of the types of users is shown in Figure 3-17. Users from “industry” dominate the pie, accounting for 52.1% of the total, followed by those from “academia” and “research institution” at 22.8% and 18.6%, respectively. Part of the reason why “industry” outnumbers other types of users could be because COMPAS

²⁵ <http://compas.kisti.re.kr>, accessed on 28 June 2023.

mainly deals with patent information.

[Figure 3-17] Breakdown of users according to their types



Source: Author.

Aside from applications by individual users, COMPAS has been applied to much of the strategic planning of government-funded R&D programmes, some of which are shown below:

- National Research Council: planning of Collaborative Allied Projects (CAPs, 5 out of 6 cases) and National Agenda Projects (NAPs, 5 out of 10 cases).
- Korea Technology and Information Promotion Agency for SMEs: identifying promising items for SMEs (62 cases).
- Association of Science and Technology Information: planning of R&D projects (47 cases).

2.1 International Dissemination of COMPAS and the ODA project for Vietnam

KISTI has been trying to disseminate COMPAS to countries such as Kazakhstan, Uzbekistan, and Vietnam among others, with the ultimate goal of implementing COMPAS within an appropriate organisation in each of those countries. Figure 3-18 shows selected dissemination activities, while Table 3-2 summarises the list of dissemination activities. Different approaches have been considered as a means of implementation, such as technology transfer, licensing, sales, and ODA projects.

Among the countries that have been contacted regarding the adoption of COMPAS, Vietnam showed the strongest desire to adopt it. After numerous discussions with candidate organisations in Vietnam, NASATI (National Agency for Science and Technology Information), a government agency whose missions and roles are most appropriate for COMPAS, was selected as the beneficiary organisation in Vietnam. The implementation of COMPAS in NASATI was supported by the ODA project funded by the Ministry of Science and ICT of South Korea.

[Table 3-2] Activities for the international dissemination of COMPAS

Date	Organisation	Country	Activity
May 2015	UIUC	United States of America	• Centre for Informatics Research in Science and Scholarship at UIUC (University of Illinois at Urbana-Champaign) - Introduction of COMPAS. - Discussions on candidate areas of collaboration in information science.
Aug. 2015	NCSTE	Kazakhstan	• International Workshop on the National R&D Information System - Introduction of COMPAS to NCSTE (National Centre for Science and Technology Evaluation). - Discussions on the implementation of COMPAS in Kazakhstan.
Feb. 2016	TLO of KZ National University	Kazakhstan	• Technology Licensing Office of Kazakhstan National University - Introduction of COMPAS.

Date	Organisation	Country	Activity
			- Discussions on the application of COMPAS to technology scouting .
Mar. 2016	VAST	Vietnam	• The 4th Korea-Vietnam International Workshop on Science and Technology Information - Introduction of COMPAS. • Discussions with VAST (Vietnam Academy of Science and Technology) leaders - Discussions on the adoption of COMPAS by VAST, and an international collaboration project. - Vice-President of VAST and leaders of subsidiary institutes in VAST.
Apr. 2016	NCSTE	Kazakhstan	• International Forum on Science and Business - Introduction of COMPAS. - Discussions on the implementation of COMPAS in Kazakhstan.
May 2016	CCSTD	Uzbekistan	• Discussions with CCSTD (Committee for coordination science and technology development) - Discussions on the implementation of COMPAS in Uzbekistan. • International Innovation Forum - Introduction of COMPAS. - COMPAS selected as one of the twelve recommendations to support innovation activities in Uzbekistan.
Oct. 2016	MOST, NASATI, SATI	Vietnam	• Discussions on the implementation of COMPAS in Vietnam - Discussions on the implementation of COMPAS in Vietnam with MOST (Ministry of Science and Technology), SATI (State Agency for Technology Innovation), and NASATI (National Agency for Science and Technology Information). • International Workshop on the “Development of a National Database on Science, Technology and Innovation” - Introduction of COMPAS.

Date	Organisation	Country	Activity
Nov. 2016	SAST	China	• SAST (Shanghai Academy of Science and Technology) - Introduction of COMPAS. - Discussions on the adoption of COMPAS by SAST.

Source: Author.

[Figure 3-18] Selected images of international COMPAS dissemination activities



(a) VAST, Vietnam (March 2016)



(b) CCSTD, Uzbekistan (May 2016)



(c) NASATI, Vietnam (October 2016)



(d) GS-NSTC, Cambodia (June 2018)

[Table 3-3] Activities for the international dissemination of COMPAS (Continued)

Date	Organisation	Country	Activity
Oct. 2017	ACCIÓ	Spain	• International Workshop on “Innovation, Networking, and Entrepreneurship for the Future” - Hosted by the National Research Council of Science and Technology (Korea) and ACCIÓ Catalonia Trade & Investment Agency, Catalonia. - Introduction of COMPAS and discussions on future collaboration between KISTI and Spain.
Apr. 2018	VAST, NASATI, SATI	Vietnam	• VAST-KISTI Joint Seminar for Knowledge Information Analysis - Introduction of the Vietnamese version of COMPAS (V-COMPAS). - Discussions on the adoption of V-COMPAS by VAST and NASATI.
Jun. 2018	GS-NSTC	Cambodia	• Korea-Cambodia International Workshop - Introduction of COMPAS. - Discussions with GS-NSTC on the implementation of COMPAS in Cambodia.
Nov. 2019	Tech Demo 2019	Vietnam	• Tech Demo 2019 - Participation in the Tech Demo hosted by SATI. - Discussions on the adoption of COMPAS by SATI.
Jan. 2020	NASATI	Vietnam	• Discussions on the ODA project - Discussions on the adoption of COMPAS by the ODA project funded by the Ministry of Science and ICT, South Korea. - NASATI agrees to be the beneficiary organisation of the ODA project.
Feb. 2020	MOST	Vietnam	• Discussions on the ODA project - Reporting of the ODA plan to MOST. - Request for approval of the MPI (Ministry of Planning and Investment).

3. ODA project for Vietnam

3.1 Project background

A global endeavor is under way to accomplish the United Nations' "Sustainable Development Goals (SDGs) 2030". To achieve the SDGs, it is necessary to differentiate individual demands according to the situation and growth stage of developing countries, such as economic and industrial development.

In Korea, there has been continuous support for the ASEAN countries under the government's "New South-bound Support Policy."* As such, "enhancing the innovation capacity of developing countries through ODA projects" is one of the main policy tasks of the "2019 ODA Budget Plan of the Ministry of Science and ICT" of South Korea.

* 2019 International Development Cooperation Implementation Plan (Korean government)

Amid the general context of national R&D management, there has been a growing need for the formulation of evidence-based and data-based science and technology policies and R&D support. This is due to the increasing importance of rational decision-making on R&D investment because the impacts of science and technology development are very considerable and innovation requires the establishment and provision of policies and R&D support based on evidence (or data).

Most of the ODA programmes carried out in Vietnam by Korea since 2001 have been focused mainly on such fields as education and social overhead capital. However, cooperation in science and technology started to undergo significant change after the Vietnam-Korea Science and Technology Committee was promoted to the ministerial level in 2018.

Vietnam has made concerted efforts to enact the Technology Transfer Act (2006) and the Science and Technology Act (2013), but it has had difficulty securing advanced technologies due to the poor access to science and technology information experienced by SMEs, which account for 98% of all enterprises.

This ODA project consists of a comprehensive and practical support programme designed to address the abovementioned problems by exploiting the experience and the latest ICT technologies of the supporting country. It is a three-year project with KISTI acting as the donor organisation and NASATI as the beneficiary organisation. The total project budget is KRW 3 billion and the annual project budget is KRW 1 billion, roughly equivalent to USD 2.3 million and USD 765,000, respectively. A summary of the project is presented in Table 3-4.

[Table 3-4] ODA Project Summary

Project Title	Advancement of Digital Innovation via Establishment of the S&T Big Data Analysis System
Keywords	S&T information; information analysis; information service; S&T policy; R&D support; SMEs; big data; database; patents
Donor Organisation	Korea Institute of Science and Technology Information (KISTI)
Project Manager	Hyuck Jai Lee, Ph.D. (hlee@kisti.re.kr)
Beneficiary Country/City	Vietnam/Hanoi
Beneficiary Organisation	National Agency for Science and Technology Information (NASATI)
Project Duration	Three years (2021 to 2023)
Budget	KRW 3.0 billion overall (KRW 1.0 billion/year for 3 years)
Funding Agency	Ministry of Science and ICT, Republic of Korea

Source: Author.

3.2 Project Goal

The goal of the project is to help a developing country transform itself into a powerhouse of digital innovation through big data-based science and technology information analysis systems and the related infrastructures. The project consists of the following three components (tasks):

- (Task 1) Support the establishment of a "National Plan for the Development of S&T Information Analysis Infrastructure in Vietnam."
- (Task 2) Support the development of a "Digital Innovation System in Vietnam" based on the experience and ICT technologies of South Korea.

- (Task 3) Develop the "Digital Partnership between Vietnam and Korea" by sharing knowledge and experience on S&T information.

Throughout the entire project, the participation of the government ministries and agencies of Vietnam was encouraged to maximise the outcome of the project.

3.3 Activities

The activities of the three components (tasks) of this project are described in detail as follows:

Task 1: Support the establishment of a "National Plan for the Development of S&T Information Analysis Infrastructure in Vietnam":

- (Goal) To support the planning and development of the "National Plan for the Development of S&T Information Analysis Infrastructure in Vietnam" for the establishment of evidence-based/data-based S&T policies and R&D support.
- (Action 1) Survey the status of national S&T policy, R&D support, and the infrastructure of S&T information analysis and utilisation in Vietnam.
- (Action 2) Identify ten important issues (or missions) related to S&T information at the national level in Vietnam in order to promote the National S&T Information Analysis System.

Task 2: Support the development of a "Digital Innovation System in Vietnam" based on the experience and ICT technologies of South Korea.

- (Goal) Support the development, operation, and utilisation of the "S&T Big Data Information Analysis System."
- (Action 1) Transfer the Vietnamese version of the COMPetitive Analysis Service (COMPAS), whose beneficial effects on SMEs have been proven in Korea, including all hardware, software, databases, and analytical models required for the immediate provision of a public service in Vietnam.

- (Action 2) Expand the analytical models and relevant resources (databases) to reflect needs on the Vietnamese side.
- (Action 3) Expand all of the analytical models and relevant resources to global coverage by embedding the world patent database in COMPAS.

Task 3: Develop the "Digital Partnership between Vietnam and Korea" by sharing knowledge and experience on S&T information.

- (Goal) Support the Vietnamese government's ministries and agencies in order to improve their competence with regard to sustainable development.
- (Action 1) Run training programmes on the operation and utilisation of COMPAS for policy-makers and system operators in the government ministries and agencies of Vietnam in order to facilitate consistent autonomous operation and utilisation of COMPAS for the establishment of evidence- and data-based S&T policies and R&D support.
- (Action 2) Hold an annual conference (Vietnam-Korea S&T Information Conference) in order to share and disseminate the knowledge and outcomes obtained from this project with the participants from government ministries and agencies and the users of COMPAS in the private sector.

3.4 Expected outputs and outcomes

(Policy aspect) As the need for rational decision-making and evidence-based policies for R&D investment increases, Vietnam will secure the foundation for establishing evidence-based science and technology policies and R&D support.

(Economy aspect) Economic development is expected in both countries by linking "technology import" on the Vietnamese side with "technology export" on the Korean side, using the analysis models provided by COMPAS, such as "Korean Patent Analysis".

(Technology aspect) The accelerated development of science and technology in Vietnam is expected by transferring knowledge to find effective ways of R&D planning and execution. At the same time, a reduction of the time and budget required to

develop the infrastructure for the analysis and utilisation of information for the establishment of S&T policies and R&D support is also expected.

[Table 3-5] Annual action plan

Year	2021	2022	2023
Task 1	○ Survey for planning - Conduct a survey on the current status of S&T information creation, services, and utilisation in Vietnam. - Encourage the participation of government ministries and agencies.	○ Identify issues - Identify ten important S&T information issues at the national level.	○ Develop action plans - Survey the previous/current efforts of Vietnam regarding the ten S&T information issues. - Develop action plans for each of the ten issues
Task 2	○ Transfer COMPAS - Produce English and Vietnamese versions. - Provide all the resources (hardware, software, databases, and analytical models) and updates.	○ Expand COMPAS (1st round) - Create new analytical models/resources that satisfy the needs of the beneficiary.	○ Expand COMPAS (2nd round) - Embed the world patent DB. - Expand all of the analytical models and resources to global coverage.
Task 3	○ Run a training programme on the operation and use of COMPAS: - For technicians and operators in the beneficiary organisation.	○ Run a training programme on the operation and use of COMPAS - For users in “S&T information” organisations in the beneficiary country. ○ 1st V-K S&T Information Conference - COMPAS user-oriented	○ Run a training programme on the operation and use of COMPAS - Provide training on an annual basis. - Provide technical training for database maintenance. ○ 2nd V-K S&T Information Conference

Year	2021	2022	2023
		conference. - Share and disseminate acquired knowledge and outcomes.	- COMPAS users + policy experts. - Share and disseminate acquired knowledge and outcomes.

Source: Author

3.5 Participants

- The Korea Institute of Science and Technology Information (KISTI) is responsible for establishing strategies, designing resources (databases) and analysis models that reflect Vietnamese demand, and providing training on the theory and practical utilisation of the analysis models installed in COMPAS.
- A contracted company is responsible for executing the technical tasks, such as developing, installing, operating, and updating the COMPAS system, developing new databases and models, and providing the related training.
- Maximise the effectiveness of the project by encouraging and supporting the participation of Vietnamese stakeholders (ministries and agencies) -
 - Ensure accurate understanding and correct implementation of the status and requirements of Vietnam in the development of strategies and systems.
 - Disseminate the outcomes by announcing the results of Vietnam's participation in the project at the Vietnam-Korea S&T Information Conference.

3.6 Accomplishments

Task 1: Support the establishment of a "National Plan for the Development of S&T Information Analysis Infrastructure in Vietnam."*

- A survey was conducted to explore the *Science and Technology Data Ecosystem* in Vietnam, where the term *ecosystem* implies a community of organisations similar to that of living organisms, which interact with each other as a system with *Data* as the intermediary in the domain of science and technology in Vietnam. The survey

respondents included organisations existing within the *Science and Technology Data Ecosystem* in Vietnam, which were grouped into three categories, each of which plays a distinctive role in the system: Data Creating, Service, and Utilisation. Forty-seven representative organisations in each category were carefully selected by the experts at NASATI.

- Based on the results of the survey, “ten important S&T information (Tentative) issues” were identified by NASATI and KISTI and categorised into four groups (Data, Service, Governance, and Infrastructure), as shown below:
 - (Data) Creation, collection, management, distribution, and utilisation of domestic and international S&T Information.
 - (Data) Development of the data infrastructure for analysis of domestic and international STI.
 - (Service) Digital transformation of the submission, peer-review, and publication systems of domestic academic journals.
 - (Service) Development of the National R&D Information System.
 - (Service) Advancement of the patent application, examination, registration, and utilisation system.
 - (Governance) Formulation of an “open science” policy.
 - (Governance) Standardisation of the regulations and guidelines.
 - (Governance) Formulation of the National R&D Assessment Act.
 - (Infrastructure) High-performance computing resources.
 - (Infrastructure) Human Resources.

Task 2: Support the development of a "Digital Innovation System in Vietnam" by drawing on the experience and ICT technologies of South Korea.

- The Vietnamese and English versions of COMPAS (V-COMPAS) were developed and implemented in NASATI along with the ten analysis models shown in Table 3-6 and

the databases shown in Table 5. The server hardware and the V-COMPAS service page are shown in Figure 3-19.

[Table 3-6] Detailed information on the databases in COMPAS

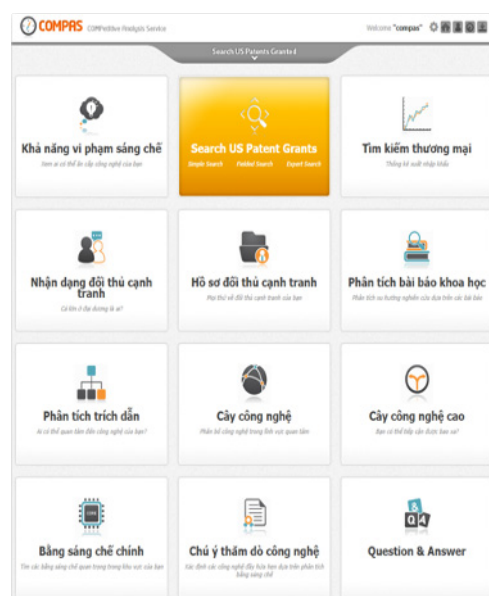
Database	Coverage (from)	Record count (Million)
US Patents Granted	1991	6.2
Korea Patents	1988	2.7
World Patents	1782	139.6
Trade (World)	2017	88.8

Source: Author.

[Figure 3-19] V-COMPAS as implemented in NASATI: (a) server hardware, (b) service page²⁶



(a)



(b)

- The system specifications for V-COMPAS, such as the hardware, software, and additional hardware required for expanding it to the world patent database, are shown in Tables 3-7, 8, and 9, respectively.

²⁶ Available at <https://compas.vista.gov.vn>, accessed on 28 June 28 2023.

- The main difference between the Korean and Vietnamese versions of COMPAS resides in the fact that the former has a “Patent Assignment Text” database, which records changes in the ownership of patents, thus enabling the patent ownership search function and the analysis models specified for patent ownership analysis: “Who’s Buying What” and “Competitor Patent Acquisition”.

[Table 3-7] Hardware list with specifications in brief

Item	Specification	Qty
WEB/WAS Server	HPE DL20G10 (E-2236 6C 3.4GHz / 64GB / 480GB SSD *1, 2TB NLSAS *1 / E208i-a LH/ 500W *1)	1
Search Server	HPE DL20G10 (E-2236 6C 3.4GHz / 64GB / 480GB SSD *1, 2TB NLSAS *1 / E208i-a LH/ 500W *1)	2
DB Server	HPE DL20G10 (E-2236 6C 3.4GHz / 64GB / 480GB SSD *1, 2TB NLSAS *1 / E208i-a LH/ 500W *1)	1
Analysis Server	HPE DL20G10 (E-2236 6C 3.4GHz / 64GB / 480GB SSD *1, 2TB NLSAS *1 / E208i-a LH/ 500W *1)	1
Server Rack	HPE G2 Advanced Series Rack	1

Source: Author.

[Table 3-8] Software list with specifications in brief

Item	Specification	Qty
Analysis Server OS	Windows Server 2019 Standard (16core) / USR 5 Cal	1
DBMS	Oracle 11g Standard Edition Two 20USER	1

Source: Author.

[Table 3-9] Additional hardware for the world patent database (completed within 2023)

Item	Specifications	Q'ty
Server	Dell PowerEdge R350 Server(1U) - Intel Xeon E-2336 2.9GHz, 12M cache, 6C/12T, Turbo (65W) - 2.5" chassis Max 8 hot-plug hard drive, Front PERC - iDRAC9, Enterprise 15G - 64GB(4x16GB) UDIMM, 3200MT/s, ECC - 480GB SSD SATA RI 6Gbps 512 2.5" hot-plug AG drive - 2.4TB 10K RPM SAS ISE 12Gbps 512e 2.5" hot-plug - PERC H755 SAS Front	1

Source: Author.

- The backup system provided with COMPAS is activated in the event that the service stops running due to unexpected causes. The specifications of the backup system are shown in Table 3-10.

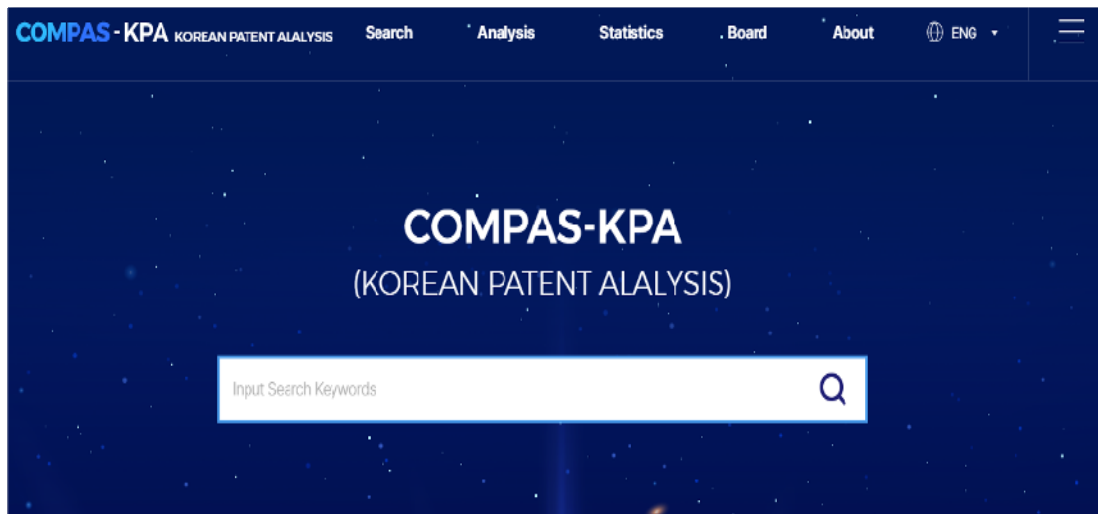
[Table 3-10] Specifications of the COMPAS "Backup" system

Item	Specifications	Qty
Server	Intel Xeon E-2226G (3.4GHz, 80W, L3:12M, 6C), DDR4 16GB Unbuffered ECC 2933MHz 4ea, LSI SAS3 8CH Controller Gen3 Host RAID 0, 1, SSD 480GB SATA III 3D TLC DC, 3.5" 4TB NL-SAS 12G 7.2k Enterprise W	5
Windows server software	Windows Svr Std 2019 64Bit Korean 1pk DSP OEI DVD 16 Core	1
DB software	Oracle Database SE2 – 10 user	2
Server rack	Server Rack Sliding-type, 14 slots INTLPDU * 2ea	1
KVM switch	LCD KVM switch 18" LCD KVM Switch, 16 Port OSD, Hot-key, Push button, KVM USB Adapter * 8ea	1
Network switch	Network switch, 1G 24 PORT switch	1

Source: Author.

- Korea Patent Analysis (KPA), which is a separate service from V-COMPAS, has been developed and implemented at NASATI. KPA aims to search and run relatively simple statistical analyses of English-translated Korean patents. Figure 20 shows the service page of KPA.

[Figure 3-20] Service page of Korean Patent Analysis (KPA)

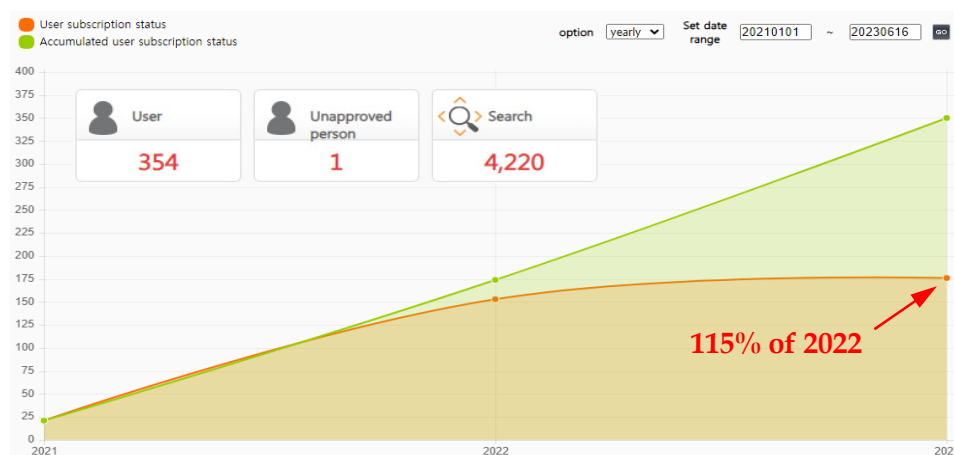


Source: Author's elaboration.²⁷

- The registered user count of V-COMPAS reached 354 as of 16 June 2023, as shown in Figure 3-21. What is noteworthy here is that the registered user count increases rapidly, showing that in 2023 it had already reached 115% of the figure recorded in 2022.

²⁷ <http://kpa.vista.gov.vn>, accessed on 28 June 2023.

[Figure 3-21] V-COMPAS service stats as of 16 June 2023



Source: Author's elaboration.²⁸

Task 3: Develop the "Digital Partnership between Vietnam and Korea" by sharing knowledge and experience on S&T information.

- Providing learning opportunities to potential users is crucial for the dissemination and utilisation of V-COMPAS and, ultimately, for the success of the service in the beneficiary country. KISTI provides technical and utilisation training programmes for service operators and users.
- In 2021 (during the COVID-19 pandemic), online user training and technical training programmes were provided to the users and IT staff of NASATI. In 2022, the scope of the user training programme was extended to six STI-related institutions in Hanoi. In 2023, the user training programme was provided to the leaders and researchers of VAST (Vietnam Academy of Science and Technology) at the request of VAST. Approximately 200 trainees attended the online and offline training. In 2023, a training programme was held in Ho Chi Minh City to disseminate V-COMPAS to southern Vietnam. Some photos of the user training programmes are shown in Figure 3-22.

²⁸ <http://kpa.vista.gov.vn>, accessed on 28 June 2023.

- Based on the user evaluations provided by the participants in the training programmes, V-COMPAS earned high “functional satisfaction” and “willing-to-recommend” scores, as shown in Figure 3-23. Incidentally, the scores are even higher than those obtained by the domestic service in Korea, which implies that V-COMPAS is expected to be widely used by Vietnamese users if it is sustainably maintained.

[Figure 3-22] V-COMPAS training programmes: (a) Offline user training for the six “S&T information” related institutions (2022); (b) On/offline training for VAST (2023)

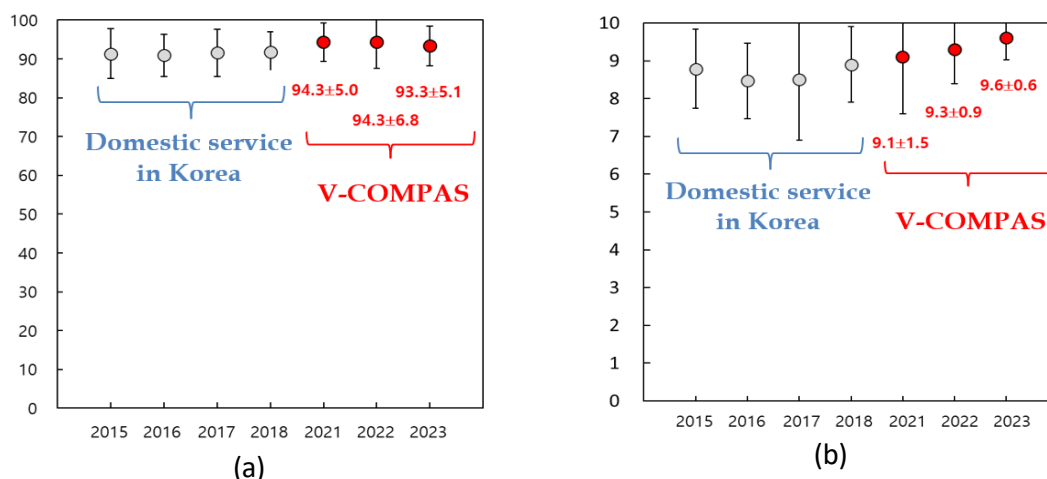


(a)



(b)

[Figure 3-23] User evaluation of V-COMPAS by the training participants: (a) Functional satisfaction and (b) willing-to-recommend scores



Source: Author's elaboration.

- The User's Manual in Vietnamese is available on the V-COMPAS page, while a YouTube Tutorial with subtitles in Korean, English and Vietnamese is provided for those who did not attend the training programme.²⁹

[Figure 3-24] Screenshot of the COMPAS Tutorial on YouTube with Vietnamese subtitles



Source: Author's elaboration, accessed on 28 June 2023.

²⁹ https://youtube.com/playlist?list=PLEtmyaOGmoQ8uwk4C34m4-VCP7Eg_Gbgp.

- The first Korea-Vietnam S&T Information Conference (2022) was held in Hanoi, Vietnam. The main purpose of this event was to disseminate the achievements of the ODA project. During the conference, various applications selected by V-COMPAS users at the STI-related institutions were presented to the approximately 130 attendees.

[Figure 3-25] The first Korea-Vietnam S&T Information Conference (2022) in Hanoi



- The South Korean delegation, led by the Korean Minister of Science and ICT, visited NASATI on 23 June 2023 to look at the achievements of the ODA project and discuss future collaboration between Korea and Vietnam with Vietnam's Deputy Minister of Science and Technology.

[Figure 3-26] The Korean delegation, led by the Minister of Science and ICT, visits NASATI: (a) Deputy Minister of Science and Technology, Vietnam, (b) Minister of Science and ICT, Korea, (c) Visiting the NASATI server room to examine the V-COMPAS servers.



(a)



(b)



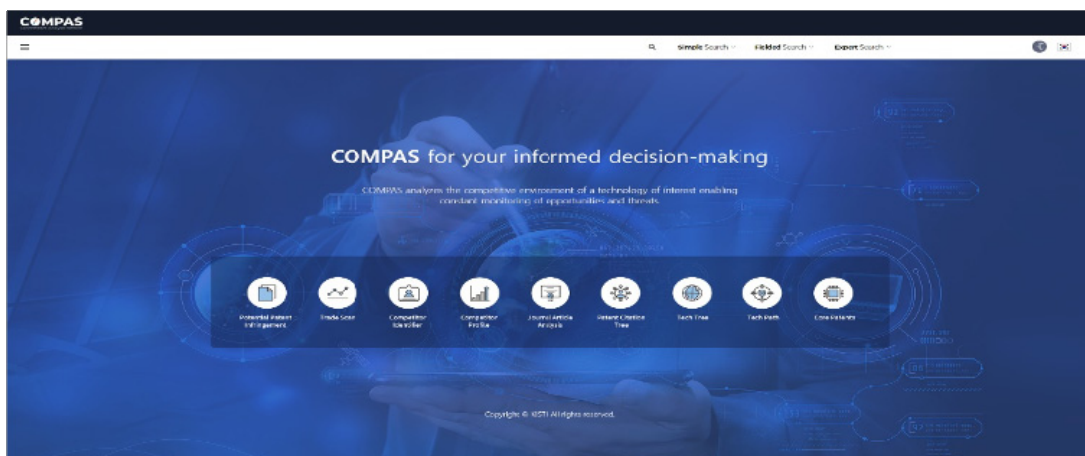
(c)

3.7 Accomplishments expected in 2023

Task 2: Support the development of a "Digital Innovation System in Vietnam" based on the experience and ICT technologies of South Korea.

- The world patent version of V-COMPAS was scheduled for implementation at NASATI at the end of 2023. As such, all the functions and analysis models of the existing V-COMPAS had to be modified to fit the world patent data. Both the US Patent Granted and World Patent versions of V-COMPAS will be operated as separate services. Figure 3-27 shows the expected layout of the World Patent version of V-COMPAS.
- A technical training programme for downloading and processing raw data and updating the databases will be provided to NASATI's IT personnel during the fourth quarter of 2023 to ensure sustainable maintenance and operation of V-COMPAS upon completion of the ODA project.

[Figure 3-27] World Patent version of V-COMPAS to be implemented in NASATI



Source: Author's elaboration, accessed on 28 June 2023.

Task 3: Develop the "Digital Partnership between Vietnam and Korea" by sharing knowledge and experience on S&T information.

- The second Korea-Vietnam S&T Information Conference was due to be held in October 2023 in Hanoi. While the first conference was mainly focused on the "users" of V-COMPAS, the second one is intended for policy-makers from government ministries and agencies.

4. Implications for Cambodia as the candidate beneficiary of COMPAS implementation

4.1 Providing essential resources in order to promote domestic research communities

The establishment of the Cambodian COMPAS through an ODA programme offers a notable advantage in promoting the R&D activities of domestic R&D communities in Cambodia, because it provides essential resources that are crucial for conducting R&D activities. In a time of heightened global competition, it is crucial to have effective measures in place for searching and analysing Intellectual Properties (IPs). With COMPAS, Cambodia will gain access to the resources and tools required for efficient IP search and analysis, thereby empowering researchers in their R&D pursuits. This access to valuable resources will serve to support and foster a thriving R&D ecosystem within the beneficiary country.

While some commercial IP resources are available, it is unfortunate that most of them come with significant subscription or purchase fees. However, obtaining IP resources through an ODA programme eliminates or at least significantly reduces these costs for the beneficiary country, which is a remarkable advantage. Furthermore, it is worth noting that many commercial resources are built upon the same foundation as COMPAS, such as DOCDB of the European Patent Office (EPO), which provides comprehensive worldwide patent information and serves as the backbone of numerous commercial products and services. Additionally, publicly accessible databases such as US Patents and Korean Patents, which are richer in contents than the DOCDB, contribute to the wealth of available resources.

Possessing its own resources will grant Cambodia independence from potentially predatory providers of commercial databases. Such providers often offer their products at a low initial price, only to substantially raise the costs once the customers have become reliant on their offerings. This can lead to significant increases in the expenses associated with maintaining the resources and related services, which is a

scenario commonly observed when relying on commercial databases. By acquiring its own resources through the ODA programme, Cambodia will be able to avoid such cost escalations and maintain greater control over the expenses associated with maintaining the database and related services.

The acquired IP resources will serve not only as a component of COMPAS but also as an infrastructure for diverse applications. Given that Cambodia's science and technology development remains in the early stage, there is a multitude of tasks related to Science and Technology Information (STI) that should be facilitated by the government. These tasks include the provision of support for researchers and SMEs, technology foresight, technology evaluation, and Patent Office responsibilities, among others. Based on the priorities of these tasks, Cambodia and South Korea can collaborate on the development of a series of ODA programmes to address specific areas of focus and promote the sustainable development of STI in Cambodia.

Fortunately, the recipient country has the flexibility required to modify and enhance the contents of the acquired resources and to tackle challenging problem-solving tasks involving cutting-edge technologies like Artificial Intelligence (AI). For instance, the patent text can serve as the input for AI applications, such as Large Language Models like ChatGPT, enabling the development of advanced analysis models and applications. This adaptability will allow the integration of state-of-the-art technologies into existing resources, thereby facilitating ever more sophisticated analysis and expanding the potential applications of the system.

4.2 Providing the Technology Intelligence (TI) system for diverse applications

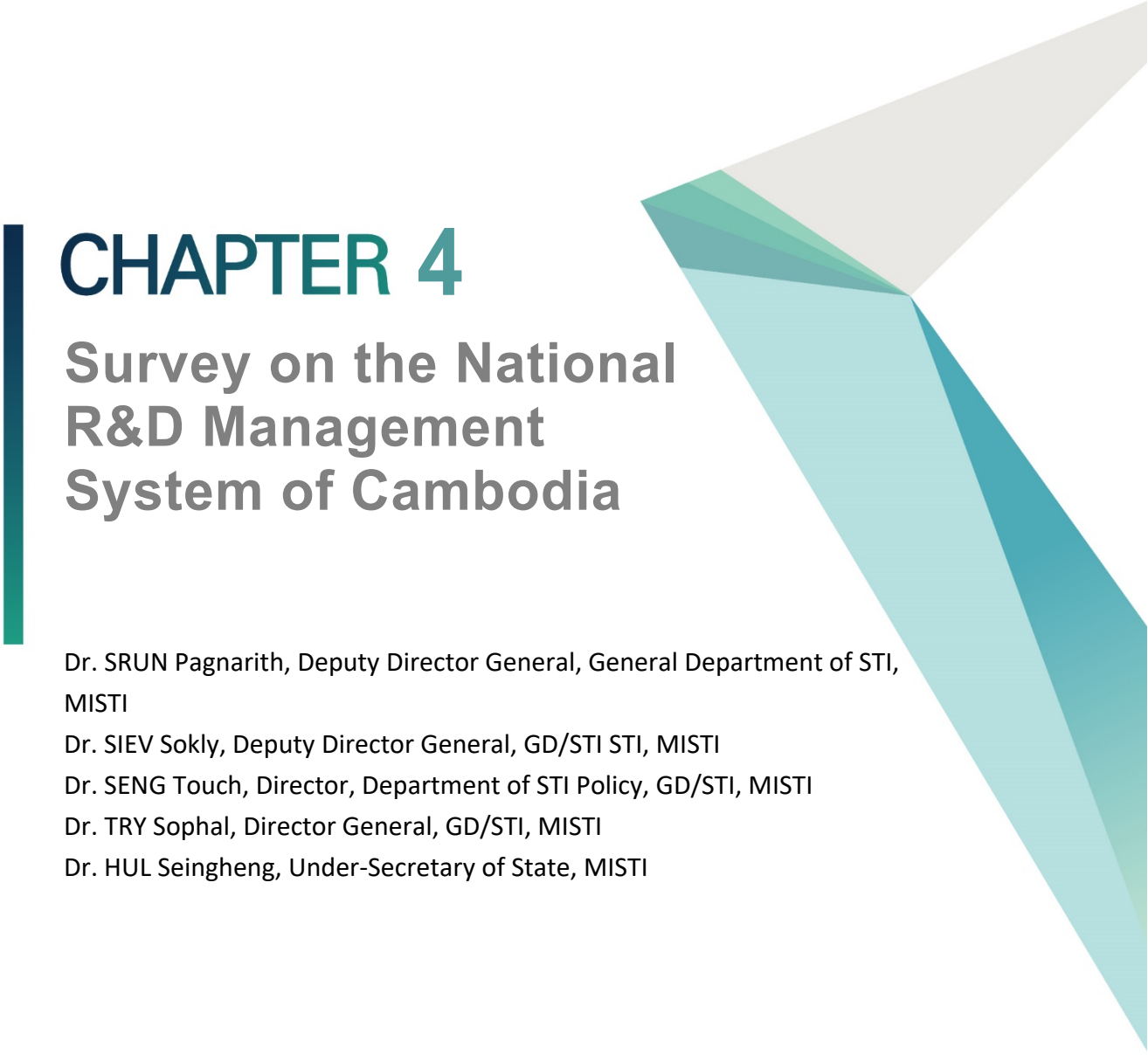
The establishment of Cambodia's own version of COMPAS will also bring the significant advantage of enabling Cambodia to possess its own "technology intelligence" (TI) system. This means that the Cambodian government or a relevant public agency will be able to deliver TI services based on the Cambodian language, catering to domestic researchers and SMEs. Therefore, there will be several benefits in acquiring a Cambodian version of COMPAS.

COMPAS will eventually become accessible to an ever larger number of users, overcoming language barriers in the process. COMPAS possesses the capability to automatically analyse patent information, journal articles, and competitive technology data, all of which are crucial for research and development activities. By building Cambodia's version of COMPAS, all aspects of COMPAS - such as the user interface, user's manual, technical manual, and training programmes, including online tutorials - will be available in the local language. This will ensure that users can fully comprehend and effectively utilise all the features of the system.

Similar to the situation in Vietnam, Cambodia's COMPAS can be tailored to address specific local needs. For instance, it can be enhanced with the capability to analyse domestic journal article information, just like the Vietnamese COMPAS, which was implemented at the request of the beneficiary organisation (National Agency for Science and Technology Information of Vietnam) to fulfill domestic requirements. COMPAS can also be customised to suit other specific requirements of Cambodia, provided that they remain within the constraints of the available data.

The implementation of COMPAS in the beneficiary country also entails the transfer of the valuable knowledge and experience gained by KISTI from years of development and servicing of COMPAS. This transfer will enable Cambodia to minimise trial-and-error processes, reducing both the time and costs that would otherwise be incurred if attempted independently. Moreover, this knowledge and experience will become crucial in the event that Cambodia aims to enhance the system or develop new systems, highlighting their importance in driving continuous improvement.

As the donor of the ODA programme, KISTI brings valuable experience from a similar project implemented in Vietnam, which can greatly enhance the entire ODA process for Cambodia. This experience can positively impact various aspects such as communication, system development/transfer, and capacity building for the beneficiary country. By leveraging the lessons learned from the Vietnam project, KISTI can optimise the implementation of the ODA programme in Cambodia, ensuring effective communication, efficient development and transfer of the system, and comprehensive capacity-building initiatives.

A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are several overlapping, semi-transparent geometric shapes in shades of teal, light blue, and grey, creating a modern, abstract background element.

CHAPTER 4

Survey on the National R&D Management System of Cambodia

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Chapter 4. Survey on the National R&D Management System of Cambodia

1. Introduction

1.1 Project background

Since 2018, the K-Innovation Programme with Cambodia has provided tailored policy consultation to meet Cambodia's Science, Technology, and Innovation (STI) policy needs at different stages of its development.

■ First Phase (2018-2019):

The programme supported the creation of Cambodia's first official science and technology master plan, known as the STI Policy 2020-2030. This plan was designed to align with the current stage of Cambodia's science and technology innovation.

■ Second Phase (2020-2022):

The focus shifted to developing a national technology roadmap (TRM) to implement the STI Policy 2020-2030 established in the first phase.

■ Third Phase (2023-2025):

This final phase is focused on establishing a national R&D system, enhancing the capabilities of R&D stakeholders, and developing strategic technologies in line with the national TRM formulated in the second phase. This phase will be guided by current policy demands to ensure effective implementation and advancement of Cambodia's STI capabilities.

1.2 Rationale

The Royal Government of Cambodia (RGC) has set forth an ambitious vision to elevate Cambodia to a high middle-income country by 2030 and a high-income country by 2050. Central to achieving this vision is the enhancement of the nation's scientific and technological capabilities and the improvement of its innovation performance. Strengthening these areas is crucial for promoting economic diversification, boosting economic dynamism, and fostering sustainable and inclusive development, thereby contributing to national prosperity, peace, security, safety, and an improved quality of life.

Advancing Cambodia's scientific and technological expertise is essential for facilitating the country's transition from a labor-intensive to a knowledge-based economy. In line with this vision, the RGC approved the National Science, Technology, and Innovation (STI) Policy 2020-2030 in 2019. This policy leverages science, technology, research, and development to support inclusive and sustainable development, guided by a strategic framework, objectives, and policies.

A key implementation example of this policy is the establishment of Cambodia's STI Roadmap 2030. One of the roadmap's pillars is to enhance research capacity and quality by strengthening the national higher education and research system to conduct high-quality R&D that addresses both societal and private sector needs.

The STI Roadmap 2030 outlines significant activities, including the development of a "National Research Agenda" and a dedicated national fund for research and development. This initiative aims to advance national scientific and technological capabilities and accelerate the "catching-up" process. The national research agenda provides a framework for knowledge creation and innovation promotion, identifies key scientific challenges for the nation's future, and facilitates thematic prioritization and efficient resource allocation among stakeholders.

To support these goals, robust funding for R&D is essential to unleash the potential of innovation and entrepreneurship. Consequently, this project aims to develop a National R&D Management System, which is expected to lead to increased

government investment in scientific research activities and ultimately the establishment of a national R&D fund.

1.3 Objectives

This study aims to analyze the current state and challenges of R&D management practices in Cambodia and to identify the optimal design for a national R&D management system tailored to the country's specific context and needs. The findings from this study will be instrumental in shaping the design of a national R&D management system that aligns with Cambodia's unique requirements and development goals.

1.4 Methodology Approach

This study employed a combination of quantitative and qualitative methods to thoroughly investigate Cambodia's R&D management system.

■ Desk Review:

The initial phase involved a desk review to brainstorm and acquire a comprehensive understanding of the current R&D management system in Cambodia. This review focused on identifying existing challenges, potential solutions, and the desired functions and challenges of an ideal national R&D management system.

■ In-Depth Interviews:

Informal, in-depth interviews were conducted with four senior experts who possess extensive knowledge and experience in Cambodia's R&D management system. These interviews provided valuable insights into the intricacies and practicalities of the current system.

■ Consultation Workshop:

A consultation workshop was organized and held on 25 October 2023. This workshop included eighteen key stakeholders from various sectors, including line ministries,

higher education institutions, research centers, and the private sector. The workshop aimed to gather qualitative information and inputs on several aspects:

- Existing practices in R&D program/project management.
- Challenges associated with managing R&D programs and projects.
- Desired features and functionalities of an ideal national R&D management system.

The combined insights from the desk review, expert interviews, and stakeholder workshop are intended to inform the design of a national R&D management system that is well-suited to Cambodia's unique context and requirements..

Finally, the online survey was conducted within the researcher community from 28 September to 31 October 2023 in order to explore the prevailing R&D practices and associated challenges, the ideal features of the R&D management system, and the challenges facing the R&D management system.

1.5 Demographic Information

Among the sixty-seven researchers, as shown in Table 4-4, males comprised the majority at 77.61%. In terms of age distribution, the highest percentage of researchers fell within the 35-44 age range, accounting for 43%, while those in the 25-34 age range accounted for 38.81% of the respondents. In addition, 47.76% of the respondents held master's degrees as their highest academic qualification, followed by holders of a PhD at 44.78%. Among the respondents, 35.82% had 5 to 10 years of research experience, while 29.85% were senior researchers with over 10 years of research experience. The majority of their research efforts were concentrated on collaborative projects with other institutions or researchers, accounting for 77.76%. Researchers showed a preference for publishing their research findings in international peer-reviewed journals, followed by international conferences and workshops. Some aimed to publish 1-3 research findings per year, while others were dedicated to publishing more than 3 research findings annually. Furthermore, a majority of the researchers had experience in using digital tools or systems for research management, accounting for 55.22%. Researchers in Cambodia have a wide variety of areas of interest, as shown in Figure 1,

including educational sciences, economics and business, civil engineering, computer and information sciences, environmental engineering, health sciences, biotechnology, electrical and electronic engineering, sociology, agriculture, forestry, fisheries, environmental sciences, social and economic geography, chemical engineering, clinical medicine, food engineering, among others. The respondents' demographic information is shown in the table 4-1 below.

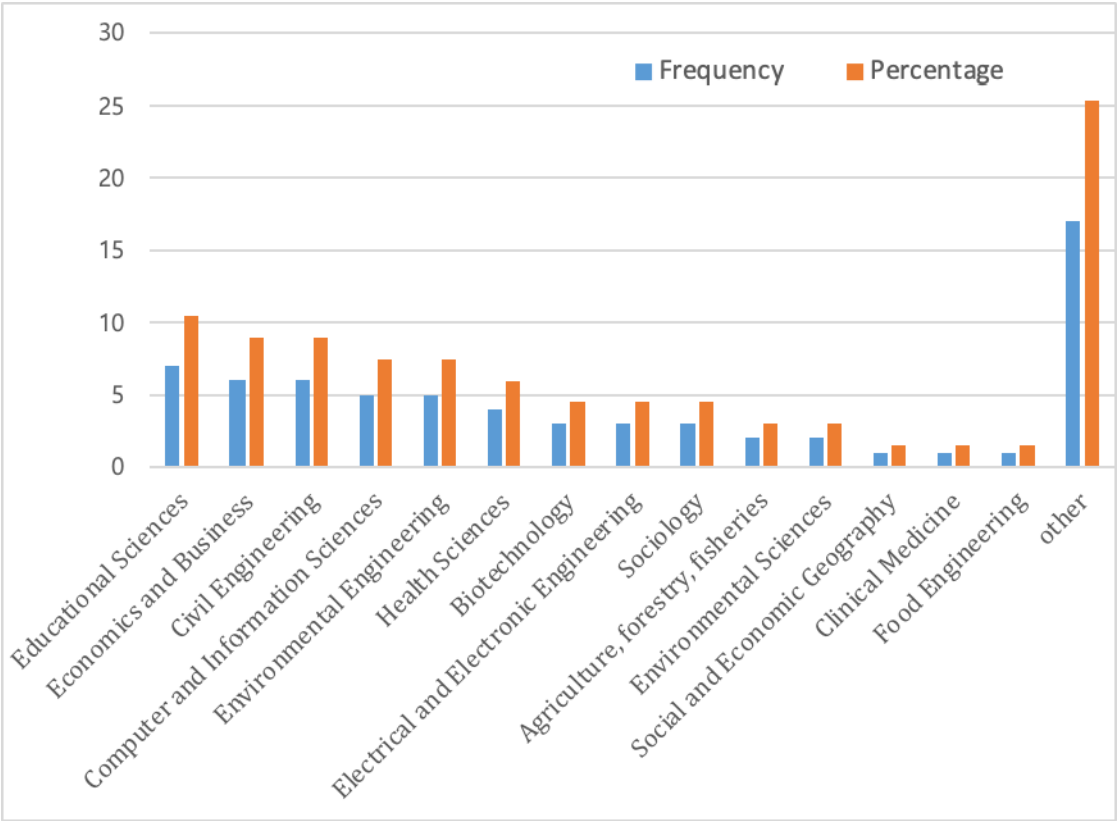
[Table 4-1] Demographic information

Information on Respondents	Category	Frequency (No.)	Percentage (%)
Gender	Males	52	77.61
	Females	15	22.39
Age	Under 25	2	2.99
	25-34	26	38.81
	35-44	29	43.28
	45-54	7	10.45
	Over 54	3	4.48
Highest academic qualification	Bachelor's	2	2.99
	Master's	32	47.76
	PhD	30	44.78
	Post doc	3	4.48
Years of experience in research	Less than 5 years	23	34.33
	5-10 years	24	35.82
	More than 10 years	20	29.85
Involvement in collaborative research projects with other institutions or researchers	Yes	52	77.61
	No	15	22.39
Provision of research grants/funding	Yes	26	38.81
	No	41	61.19
Publication of research findings	International peer-reviewed journals	45	67.16
	International conferences and workshops	38	56.72
	National peer-reviewed journals	32	47.76
	National conferences and	29	43.28

Information on Respondents	Category	Frequency (No.)	Percentage (%)
	workshops		
	Online research networks	10	14.93
	Other	10	14.93
Frequency of publishing research findings	Very frequently (more than 3 per year)	3	4.48
	Frequently (1-3 per year)	21	31.34
	Occasionally (1 every 2 years)	21	31.34
	Rarely (1 every 3-5 years)	14	20.9
	Never	8	11.94
Experience of using digital tools or systems for research management	Yes	37	55.22
	No	30	44.78
Field/area of research	Educational Sciences	7	10.45
	Economics and Business	6	8.96
	Civil Engineering	6	8.96
	Computer and Information Sciences	5	7.46
	Environmental Engineering	5	7.46
	Health Sciences	4	5.97
	Biotechnology	3	4.48
	Electrical and Electronic Engineering	3	4.48
	Sociology	3	4.48
	Agriculture, Forestry, Fisheries	2	2.99
	Environmental Sciences	2	2.99
	Social and Economic Geography	1	1.49
	Chemical Engineering	1	1.49
	Clinical Medicine	1	1.49
	Food Engineering	1	1.49
	Other	17	25.37

Source: Author.

[Figure 4-1] Research areas of respondents



Source: Author.

2. Research Environment and Practices

The Royal Government of Cambodia's "Rectangular Strategy Phase IV (RS IV)" has been superseded by the "Pentagonal Strategy Phase I (PS-I) - for Growth, Employment, Equity, Efficiency, and Sustainability," launched by Prime Minister Hun Manet in August 2023. This strategy aims to protect, preserve, and build upon national achievements, promote national prestige, enhance economic growth, and lay the foundation for realizing Cambodia's ambition of becoming an upper-middle-income country by 2030 and a high-income country by 2050.

In response to the global expansion and significance of emerging technologies and innovation, the Ministry of Industry, Science, Technology, and Innovation (MISTI), together with the National Council of Science, Technology, and Innovation (NCSTI), has led the task of formulating robust policies and frameworks to foster the advancement of Cambodian technology and innovation for the next generation.

As a result of these efforts, Cambodia's Science Technology and Innovation (STI) Policy 2020-2030 was formulated with the aim of promoting STI as a catalyst for inclusive and sustainable socioeconomic development. The policy's principal objectives are to strengthen the foundation for STI, enhance the enabling ecosystem, create an STI environment conducive to sustainable development, and improve the quality of life across all sectors and levels of society. This is to be achieved through the adoption, adaptation, and promotion of technologies and innovation.

The significance of STI promotion is underscored by its integration within various strategic frameworks, such as the aforementioned "Pentagonal Strategy Phase I (PS-I) - for Growth, Employment, Equity, Efficiency, and Sustainability," newly launched by Prime Minister Hun Manet in August 2023. This strategy aims to protect, preserve, and build upon national achievements, promote national prestige, enhance economic growth, and lay the foundation for realizing Cambodia's ambition of becoming an upper-middle-income country by 2030 and a high-income country by 2050.

Frameworks such as the National Strategic Development Plan 2017-2023 and the Industrial Development Policy 2015-2025 present specific measures for promoting science, technology, engineering, and mathematics (STEM) and innovation, highlighting and reinforcing their importance in driving Cambodia's development.

The National STI Policy is focused on five key scientific and technological domains. First, it addresses the need to increase agricultural yields, diversify agricultural produce, and adopt agro-processing to enhance crop productivity and add value to agricultural products. Second, it emphasizes the need to adopt modern production and engineering technologies to enhance industrial productivity, efficiency, and competitiveness. Third, it stresses the need for advancements in the health and biomedical sectors to improve healthcare delivery and address public health challenges. Fourth, it promotes material science and engineering to foster innovative materials and enhance manufacturing processes. Lastly, it targets services and the digital economy, including artificial intelligence, space, and spatial technology, recognizing the growing importance of these areas.

By prioritizing these domains, the National STI Policy provides a comprehensive framework for leveraging STI in Cambodia's socioeconomic development and outlines strategic interventions designed to harness the transformative potential of science, technology, and innovation to achieve sustainable progress and the country's long-term development goals.

Following the endorsement of the STI Policy, the Cambodia Science Technology and Innovation Roadmap 2030 (Cambodia's STI Roadmap 2030) was developed to serve as a strategic framework for implementing the National STI Policy. It highlights several key areas of focus to drive progress in STI within Cambodia. In this regard, enhancing STI governance is crucial, necessitating the consolidation of the Ministry of Industry, Science, Technology, and Innovation (MISTI) to minimize fragmentation. This involves clarifying roles, strengthening the government's capacity, and monitoring progress. Another priority is to build up human capital for STI to promote scientific, digital, and entrepreneurship literacy from basic education. Also important is emphasizing STEM skills in higher education and fostering collaboration with the private sector in

technical and vocational education and training (TVET) institutions.

Meanwhile, strengthening research capacity and quality is seen as a vital step, requiring the development of a national research agenda, collaboration with the private sector, and funding to support the internalization of excellent science and research. Another key element consists of promoting collaboration and networking among various stakeholders; supporting incubation and acceleration facilities, technological platforms, and innovative clusters can enhance collaboration and innovation in SMEs.

The creation of an enabling ecosystem is also crucial to building the absorptive capacities of firms and attracting STI investments. This involves nurturing start-ups, facilitating technology transfer, promoting domestic technologies, and increasing access to finance for innovation activities. Additional measures include incentivizing FDI, supporting technology imports, and protecting intellectual property rights. The formation of public-private partnerships is also vital, and to that end, the government could encourage private sector investment through regulations, incentives, and an innovation-friendly environment. Finally, engaging international development partners and obtaining their technical and financial assistance is also important.

The pursuit of these priorities will drive advancements in STI and contribute to sustainable socioeconomic development in Cambodia.

2.1 Current R&D management system

Research environments and practices are fundamental to the pursuit of scientific knowledge and discovery. These encompass the physical and virtual settings in which research is conducted, the methodologies employed, and the ethical principles adhered to throughout the research process. A well-structured research environment, coupled with established best practices, is critical for ensuring the quality and reliability of research outcomes.

In recent decades, no country has enjoyed socioeconomic growth without strengthening its national science and technology competitiveness, particularly

through investment in research and development (R&D) according to their development agenda. Statistical studies have provided compelling evidence that the more a country invests judiciously in R&D, the more developed it becomes. In fact, a knowledge-based economy will have a strong scientific foundation as a result of R&D, which is a critical prerequisite for realizing Cambodia's vision of becoming a developed country by 2050. Investing in R&D is timely and paramount as it not only leverages the national innovation system but also contributes directly to socioeconomic prosperity in a sustainable, inclusive, and resilient manner.

The development of national research capabilities can benefit virtually all sectors of Cambodia's economy and society. This requires a self-reliant national research ecosystem, with solid institutional support that encourages dynamic collaborations between the public and private sectors, universities and research institutes, civil society, and development partners.

The NCSTI adopted the National Research Agenda (NRA) in 2023 to guide the efforts of these stakeholders toward creating a research ecosystem suitable for Cambodia's ambitious Vision 2030 and 2050. The NRA defines the priority themes to be explored by Cambodian researchers and the necessary policy instruments to support them throughout the next decade.

The national research agenda or "mission" encompasses the eight distinct areas outlined below, which will be the focus of research and investment activities:

- Local food production: 70 percent of Cambodia's food will be produced locally.
- Reliable energy supply: 90 percent of its energy needs will be generated locally.
- Quality education: Education will meet international quality standards.
- Electronic and mechanical spare parts: Cambodia will export 70 percent of the electronic and mechanical spare parts produced in the country.
- Cloud-based services: Cambodia's development of cloud-based services will be on a par with ASEAN.

- Electricity and potable water: All Cambodians will have access to a reliable electricity supply and safe potable water.
- Carbon neutrality: Cambodia will become a carbon-neutral country.
- Digitally-enhanced health: All Cambodians will have access to digitally-enhanced health services.

Additionally, the NCSTI has adopted several key documents pertaining to resilient growth. Some of the major directive documents include the Agri-Tech Roadmap 2030, the Health Tech Roadmap 2030, the Edu-Tech Roadmap 2030, the Digital Tech Roadmap 2023, the Energy Tech Roadmap 2030, and the Tourism Tech Roadmap 2030. The Agri-Tech Roadmap aims to increase the productivity of agricultural commodities and high value-added production/services for the global value chain through technology and innovation by 2030. The Health Tech Roadmap aims to integrate health technologies to benefit individuals, communities, and the country under a single, integrated approach to health; execute a multidisciplinary policy and governance for health technology; and strengthen the country's research and knowledge-sharing capacity. The Digital Tech Roadmap contains in-depth information and strategic documents presenting the vision, goals, strategic products, and services of Cambodia, and the key technologies related to the development of national digital technology.

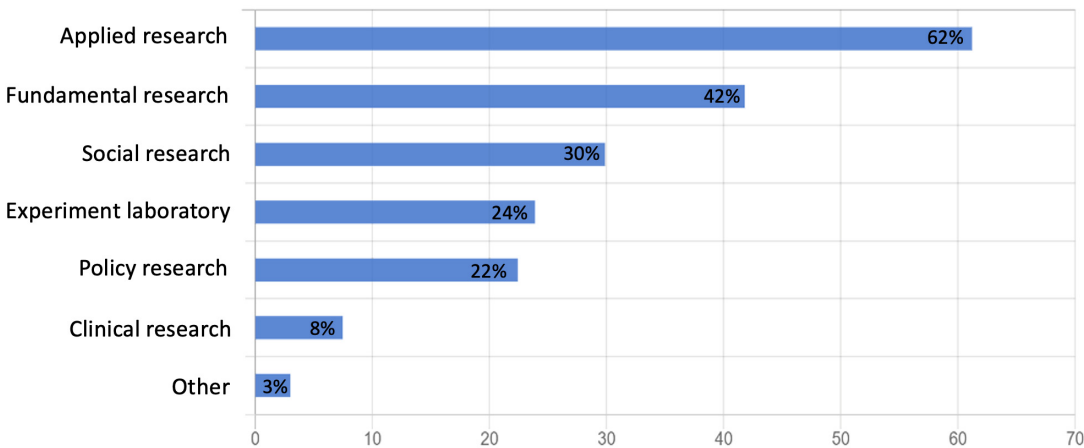
The "Report on the Research Ecosystem of Cambodia" outlines several recommendations aimed at strengthening research capacity and quality based on an analysis of nine building blocks and proposes the following recommendations:

- Align research topics with national developmental needs.
- Introduce a research fund to finance the national research agenda.
- Clarify the roles, mandates, means, objectives, and budgets of government research institutes.
- Select and designate a number of publicly-funded research and training organizations as centres of excellence in research, providing them with adequate funding.

- Introduce an attractive research career system and incentives for researchers.
- Incentivise the private sector to conduct research in collaboration with academia.
- Develop strong international university networks.
- Inform and train policymakers to integrate research into policymaking.

Figure 4-2 shows the results of a comprehensive survey on the diverse landscape of research fields, revealing a rich tapestry of scientific pursuits. Applied research takes the lead, with a substantial 61.9% of respondents actively engaged in this dynamic and practical realm. This category encompasses research efforts aimed at solving real-world problems and developing innovative solutions to pressing issues in various industries. These findings underscore the significant impact and relevance of applied research in today’s knowledge-driven society.

[Figure 4-2] Nature of research projects



Source: Author.

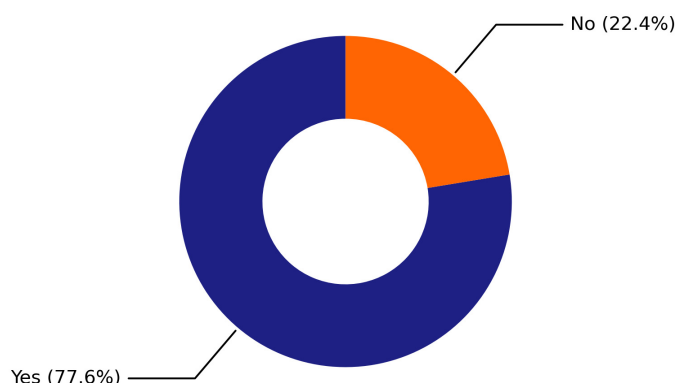
Fundamental research emerged as a prominent contender, with 41.79% of the participants dedicating their intellectual endeavors to unraveling the mysteries of the universe. Characterized by its pursuit of knowledge for its own sake, fundamental research lays the foundational groundwork upon which many other fields build their innovations and serves as the cornerstone of scientific progress, fostering an environment conducive to curiosity and discovery.

Social research, experimental laboratory work, and policy research each claim their own distinctive places in the research landscape, accounting for 29.85%, 23.88%, and 22.39% of the participants, respectively. The diversity within these realms showcases the multifaceted nature of contemporary research endeavors, which range from understanding human behavior and societal dynamics to conducting rigorous experiments and formulating policies that will shape our collective future. Clinical research, though a smaller subset at 7.46%, plays a critical role in advancing healthcare and medical knowledge. Finally, the survey findings demonstrate that 2.99% of the respondents are involved in other fields of research, reflecting the intriguing array of specialized niches and interdisciplinary approaches that contribute to the vibrant ecosystem of research and innovation.

As shown in [Figure 4-3], a striking 86.57% of the respondents in the survey are actively engaged in collaborative research projects. This high degree of collaboration underscores the interconnectedness and interdisciplinary nature of modern research endeavors. Researchers from diverse backgrounds and areas of expertise are joining forces to tackle complex challenges, share their insights, and engage in vibrant exchanges of knowledge. This collaborative spirit is not only driving innovation but also promoting a collective approach to addressing the most pressing issues across diverse fields of study.

[Figure 4-3] Collaboration on research projects

Engagement in Collaborative Research Projects with Institutions and Researchers



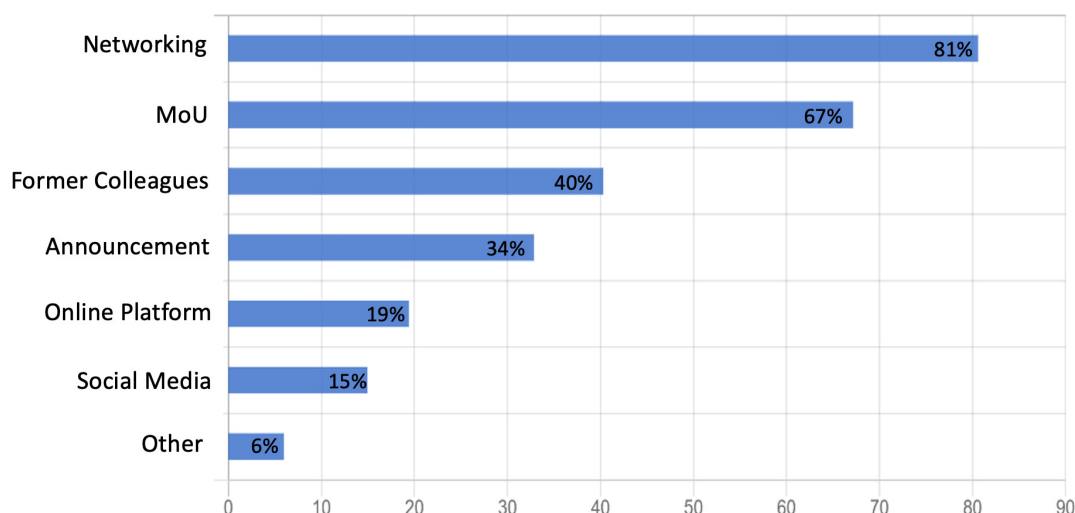
Source: Author.

In the realm of research collaboration and partnership, the survey reveals intriguing insights into the diverse channels through which researchers connect and pool their expertise, as shown in Figure 4-4. A significant 80.6% of the respondents highlighted networking as a pivotal means of forging collaborative bonds. This statistic emphasizes the crucial role of conferences, seminars, and other professional gatherings where researchers converge to share ideas, initiate dialogues, and cultivate partnerships to advance their common scientific goals.

The power of formalized agreements in fostering collaboration is evident, with 67.16% of the respondents relying on Memorandums of Understanding (MoUs). These legally binding documents facilitate structured cooperation, often involving institutions or organizations, and signify a commitment to shared research objectives. MoUs also provide a solid foundation for collaboration, helping researchers and research organizations navigate intellectual property rights, resource allocation, and project responsibilities.

Former colleagues, a source of trust and familiarity, are also integral to the collaborative landscape, as acknowledged by 40.3% of respondents. Their shared history and vision for research projects make them valuable assets for establishing productive partnerships. Moreover, the survey sheds light on the role of online platforms, including ResearchGate, LinkedIn, and Google Scholar (19.4%), as well as social media channels such as Facebook, Instagram, Twitter, Telegram, and Line (14.93%), in facilitating connections. These digital realms provide virtual spaces where researchers can exchange ideas, explore potential collaborations, and stay updated on the latest developments in their respective fields. Lastly, a notable 5.97% of the respondents indicated other means of collaboration, showcasing the endless possibilities that researchers employ to unite their efforts and advance scientific frontiers.

[Figure 4-4] Ways of seeking collaboration on research projects



Source: Author.

In [Figure 4-5], the survey findings provide a comprehensive overview of the diverse funding sources that researchers rely on to sustain their projects and advance the frontiers of knowledge. Notably, a substantial 62.69% of respondents acknowledged the crucial role of development partners in supporting their research initiatives. These partners often include organizations, institutions, and international bodies that invest in research projects to address pressing global challenges, highlighting the collaborative nature of contemporary research efforts.

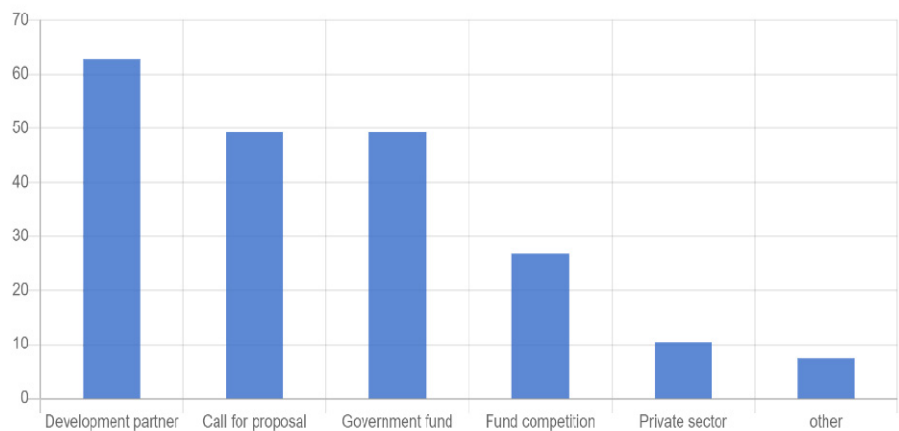
Government funding and calls for proposals are identified by an equal 49.25% of respondents as key avenues for financial support. Government-sponsored research grants, often characterized by their reliability and potential for long-term funding, significantly contribute to scientific advancement. Simultaneously, calls for proposals, which encourage researchers to compete for funding, serve as dynamic platforms where innovative ideas can thrive. Researchers also recognize the competitive nature of this channel, which encourages them to develop strong, compelling project proposals.

The study also reveals that 26.87% of participants seek funding through competitive means, such as fund competitions. These competitions foster a sense of innovation

and drive, pushing researchers to continuously refine their ideas and present them with a strong emphasis on their potential impacts. A smaller yet noteworthy 10.45% of respondents look to the private sector for financial backing, recognizing the value of industry collaboration in propelling research innovation and fostering practical applications of their work.

Lastly, the study underscores the ingenuity of researchers, with 7.46% of participants exploring various other unconventional avenues to secure funding. These diverse approaches may encompass partnerships with foundations, crowdfunding efforts, philanthropic donations, or other unique arrangements that reflect the evolving landscape of research financing. Together, these findings illuminate the multifaceted and adaptive nature of funding in the research community, demonstrating that researchers are actively exploring and leveraging a wide range of resources to support their vital work.

[Figure 4-5] Primary funding source for researchers



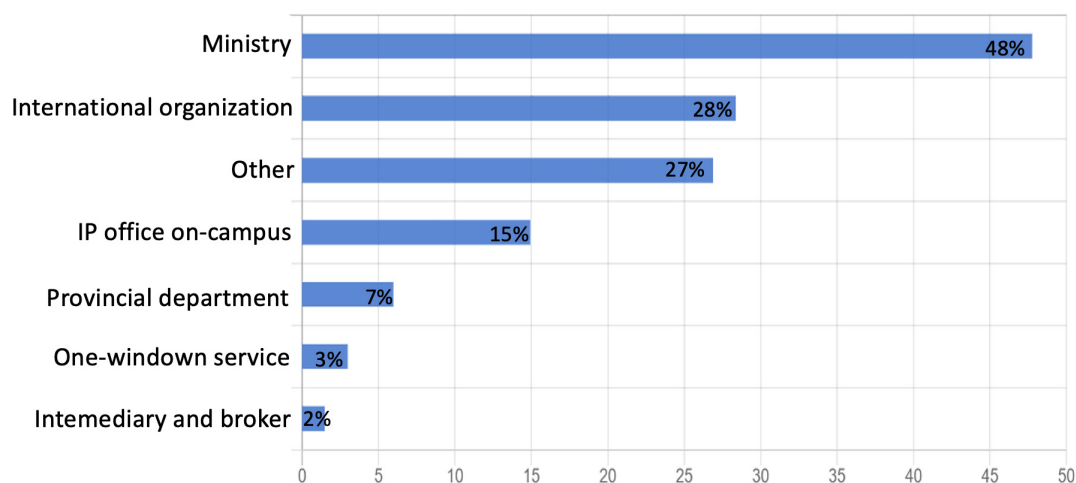
Source: Author.

As illustrated in Figure 4-6, the study offers valuable insights into the diverse array of entities and channels through which researchers seek intellectual property (IP) support and services. Notably, a significant 47.76% of the respondents highlighted the pivotal role of ministries in facilitating the management and protection of intellectual properties. These government bodies often play a central role in shaping IP policies

and regulations, providing a critical foundation for researchers to navigate the intricacies of IP rights and ownership.

International organisations emerge as another influential source of IP-related support, embraced by 28.36% of the respondents. These global entities foster cross-border collaboration and the harmonisation of IP practices, making them instrumental in addressing the increasingly international nature of intellectual property. Additionally, the survey reveals that 26.87% of the participants have explored various other avenues for IP services. These diverse approaches may encompass specialised IP offices on university campuses, regional provincial departments, streamlined one-window service centres, and the assistance of intermediaries and brokers. These findings highlight the dynamic and multifaceted landscape of IP support and services, showcasing researchers' adaptability and resourcefulness in safeguarding their intellectual property assets.

[Figure 4-6] Intellectual property (IP) protection considerations for research outcomes



Source: Author.

2.2 Challenges facing the current management system

Researchers today are confronted with a range of shared challenges that can have a significant impact on their pursuit of knowledge and innovation, as shown in Figure 4-7.

Foremost among these challenges is the formidable difficulty in accessing adequate funding, which is experienced by a staggering 68.66% of the respondents. This pressing concern highlights the constant struggle researchers face in securing the financial support necessary to bring their ideas to fruition and advance their work.

Inefficiencies in resource allocation emerge as another notable challenge, with 61.19% of the respondents grappling with the complex task of optimising the allocation of resources within their projects. These inefficiencies can hinder progress and innovation, emphasising the need for streamlined and effective resource management strategies in research settings.

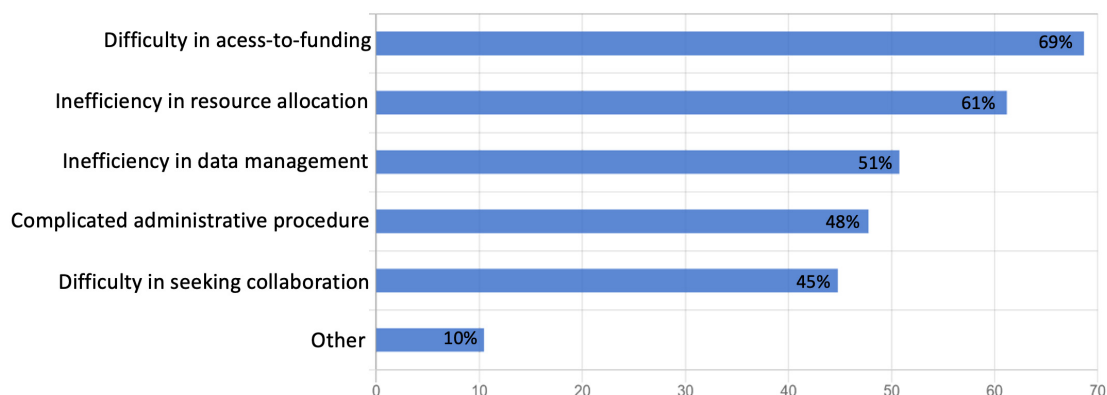
A further challenge, experienced by half of the respondents at 50.75%, pertains to the effective management of research data. In an age of exponential data growth, researchers are often overwhelmed by the volume, diversity, and security considerations associated with their data. This underscores the urgency for comprehensive data management solutions that enable them to harness the full potential of their research findings.

Complicated administrative procedures are a significant impediment faced by 47.76% of the respondents. Cumbersome bureaucratic processes can consume valuable time and resources, diverting their focus from their primary research activities. Simplifying administrative procedures and reducing red tape could enhance researchers' productivity and overall efficiency.

Additionally, 44.78% of the respondents indicated that they struggle with the challenge of seeking collaboration effectively. The ability to establish meaningful partnerships and foster interdisciplinary cooperation is vital for tackling complex research problems. Encountering difficulties in this area may hinder the sharing of knowledge and the progress of scientific endeavors. Lastly, 10.45% of the respondents reported facing various other challenges that, while less common, reflect the diverse and unique hurdles that researchers encounter in their pursuit of academic and scientific excellence. These findings collectively underscore the need for targeted solutions and support structures to address these challenges and empower

researchers to thrive in their pursuits.

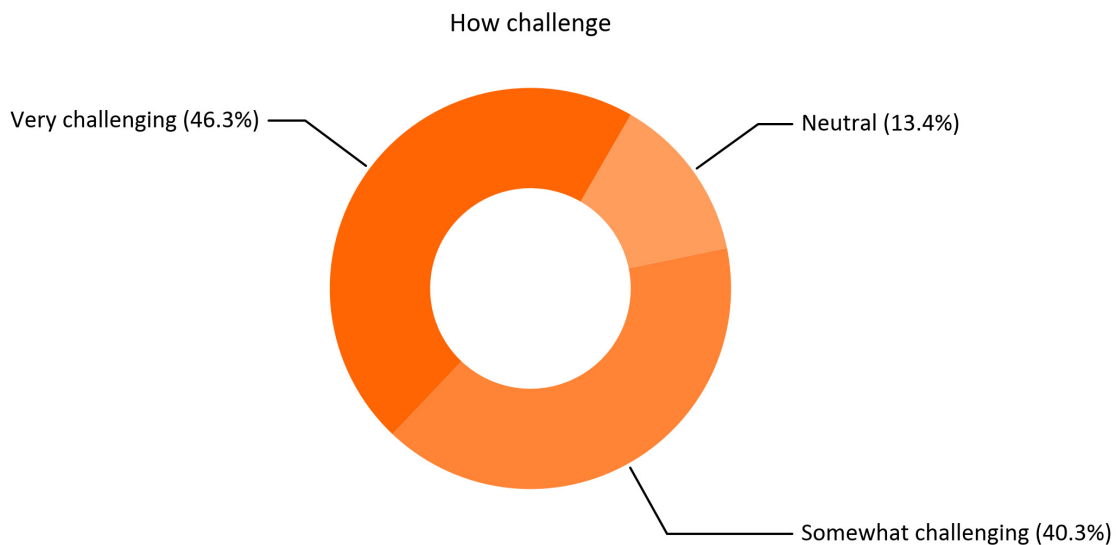
[Figure 4-7] Challenges facing current R&D management



Source: Author.

When it comes to the degree of difficulty researchers encounter in their work, the survey results (see Figure 4-8) reveal that a significant 46.27% of the respondents find their challenges to be very demanding, underscoring the substantial obstacles they must overcome to advance their research endeavors. Additionally, a notable 40.30% of the respondents reported that these challenges are somewhat difficult, indicating that a majority of researchers confront varying levels of complexity in their work. In contrast, a smaller but still noteworthy 13.43% maintained a neutral stance, suggesting that a fraction of researchers may find ways to mitigate or adapt to the challenges they face or may be working in environments with less formidable obstacles. These insights provide valuable context for understanding the range of experiences researchers encounter in their quest for scientific and academic excellence.

[Figure 4-8] Challenges in securing fund for research support



Source: Author.

The method by which research staff submit their reports plays a pivotal role in the dissemination of academic and scientific findings. The survey results, as shown in Figure 4-9, reveal that a significant 43.28% of staff members still rely on manual submission processes, suggesting that a considerable portion of the academic community continues to employ traditional, paper-based means to convey their research results.

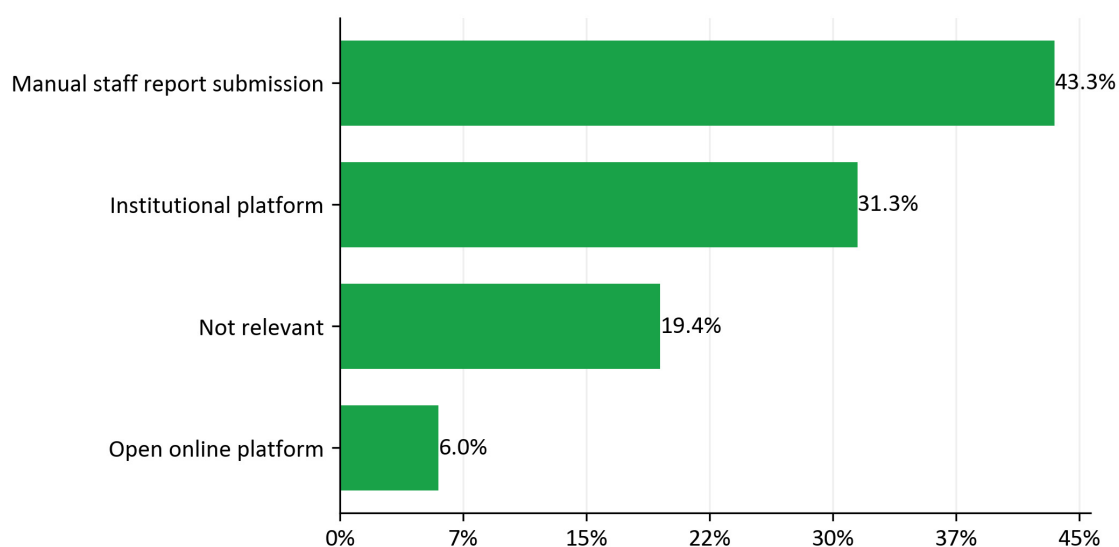
In contrast, a noteworthy 31.34% of the respondents indicated that they use institutional platforms to submit their reports, highlighting the growing trend toward digitisation and the development of tailored systems within academic institutions. These platforms provide researchers with a streamlined and efficient way to submit their work, offering advantages in terms of accessibility, organisation, and data management.

Surprisingly, 19.40% of the respondents indicated that the method of submitting a report is not relevant to their specific roles or fields. This finding suggests that certain of the respondents' research areas or job functions may prioritise other aspects of research work over the mode of report submission, or they may be situated in

environments where these practices are less pertinent to their work.

Lastly, a smaller but still notable 5.97% of the respondents indicated that they tend to opt for open online platforms to submit their reports. These platforms have gained prominence as they enable researchers to share their findings with a broader audience, while promoting transparency, open access, and knowledge dissemination beyond traditional academic boundaries. These results emphasise the evolving landscape of report submission in the research community and reflect the wide range of choices available to researchers, tailored to their specific needs and preferences.

[Figure 4-9] Management of R&D annual report on activities and performance
(Annual report on R&D management activities and performance)



Source: Author.

The survey findings presented in Figure 4-10 indicate that a significant 74.63% of the respondents rely on manual methods, such as spreadsheets, to monitor their research endeavors. This traditional approach, while prevalent, may come with limitations in terms of efficiency and scalability, as researchers invest substantial effort in maintaining and updating these records.

On the other hand, 13.43% of the respondents indicated that they adopt project management tools to track their research projects. These digital solutions offer more

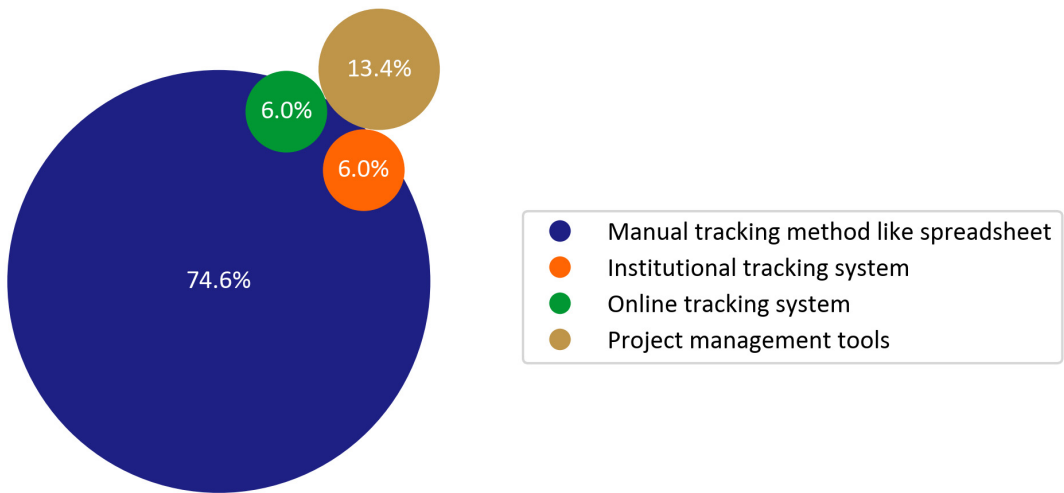
streamlined and collaborative approaches to monitoring research progress, enhancing efficiency, and enabling researchers to manage their tasks and timelines more effectively.

A smaller yet noteworthy 5.97% of the participants stated that they employ online tracking systems to oversee their research activities. These web-based platforms offer convenience and accessibility, allowing researchers to monitor their projects from various locations and collaborate with colleagues and stakeholders more seamlessly.

Similarly, 5.97% of the respondents turn to institutional tracking systems, which are specialised platforms often provided by their academic or research institutions. These systems can offer a tailored and integrated approach to research tracking, aligning with the specific needs and processes within their organisational contexts. These findings showcase the diverse range of methods and tools that researchers employ to manage and monitor their research progress, reflecting a dynamic and adaptable landscape in the research community.

[Figure 4-10] Tool or method for tracking the progress and milestones of research

Tracking the Progress and Milestones

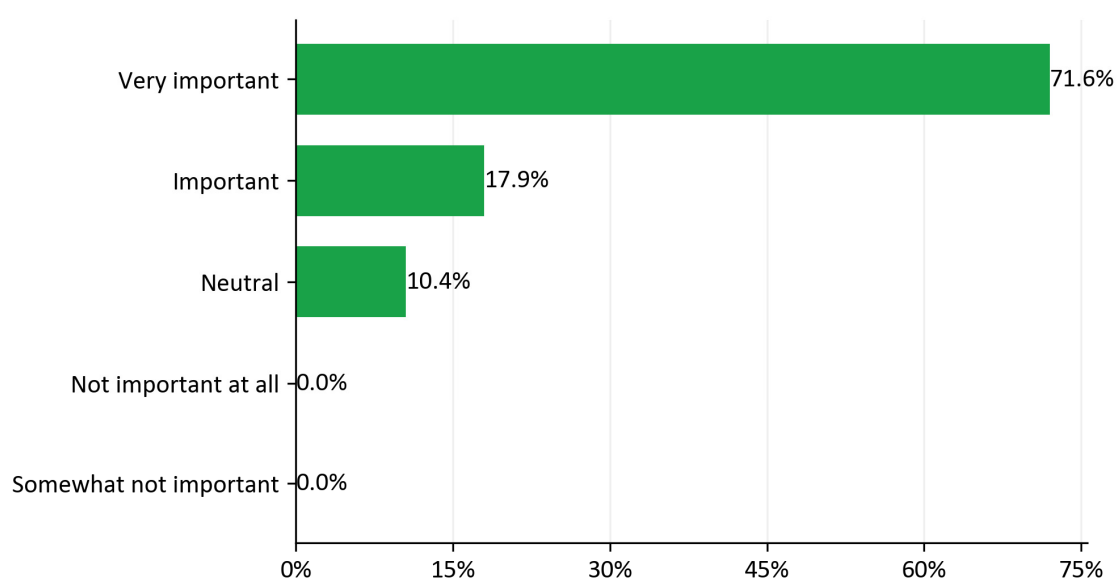


Source: Author.

The significance of a national R&D management system for researchers and research

institutes in Cambodia is overwhelmingly evident in the survey results, as shown in Figure 4-11, with a substantial 71.64% of the respondents affirming that it is "very important." This resounding endorsement underscores the critical role such a system plays in fostering a conducive environment for research, enhancing collaboration, streamlining resources, and promoting innovation within the country. An additional 17.91% find it "important," further emphasising the consensus among researchers regarding the invaluable impact of a robust R&D management framework. Even within the remaining 10.45%, who maintain a neutral stance, their recognition of the system's potential benefits is implicit, hinting at the latent support for its implementation and the opportunities it could unlock for advancing research in Cambodia.

[Figure 4-11] Importance of the National R&D Management System for researchers and research institutes in Cambodia



Source: Author.

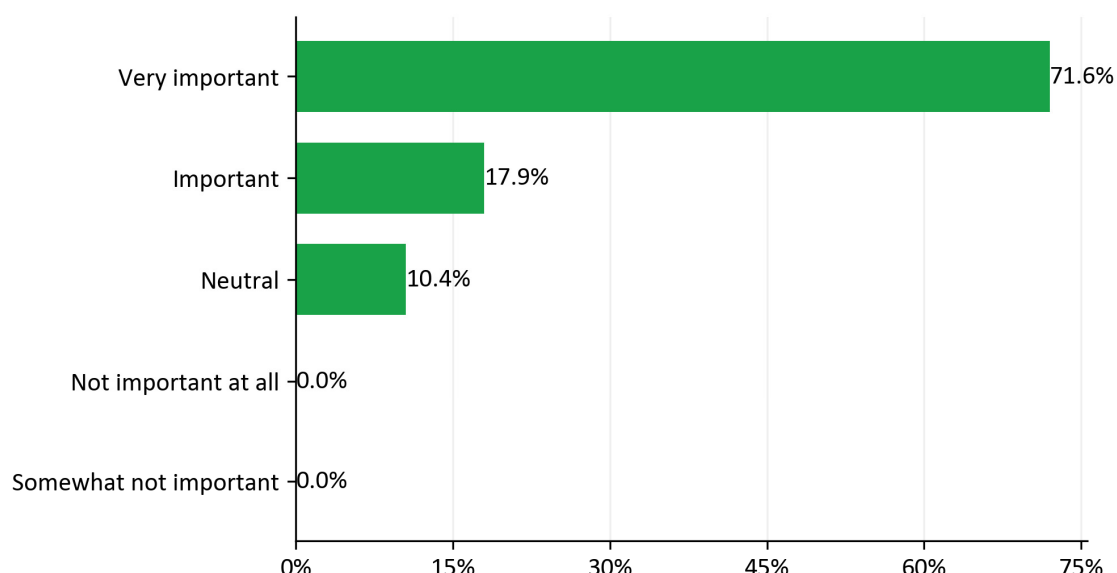
3. The ideal R&D management system

In this section, we explore the collective opinions of the researchers and the scientist community in Cambodia regarding the ideal research and development (R&D) management system. This system is anticipated to provide numerous advantages for boosting research activities and building a vibrant research ecosystem as a foundation for effective and creative research projects, ranging from simplifying project planning and resource allocation to promoting smooth collaboration and communication among researchers. The participants' insight and inputs highlight the elements and functionalities that they believe are essential in such a system, highlighting its potential to completely transform the way that research is conducted, recorded, and disseminated. The goals of the research community are reflected in this section, which depicts their vision of a day when technology will be essential to improving the state of research.

3.1 Level of Importance

The respondents were asked to express their opinions and impressions regarding the level of importance of the national R&D management system for researchers and research institutions in Cambodia. Figure 4-12 shows that the majority, i.e. 70%, of researchers and the scientist community believe that the introduction of a national management system is 'very important', while about 20% believe that it is 'important' and 10% remain neutral on the matter. None of the respondents chose 'unimportant'. Overall, this graph visually represents the strong consensus among the respondents that R&D is highly important, with a small percentage expressing a neutral stance and none at all considering it unimportant or somewhat unimportant. Indeed, this visual representation emphasises the overwhelmingly positive sentiment regarding the significance of research and development.

[Figure 4-12] Level of importance of the R&D management system



Source: Author.

3.2 Features of the system

To ensure that the R&D project management and system can support researchers and boost research activities in Cambodia, the survey gathered the insights and opinions of researchers and the scientist community on the specific features that the R&D management system could offer them. They suggested a diverse selection of features ranging from a research database to intellectual property management. According to the survey results shown in Figure 4-13, a “collection of research databases” ranked first among the most desirable features as it could provide convenient access to previous or existing studies and similar studies for the purpose of comparison. In addition, such a feature could answer their questions on similar projects and researchers in the fields concerned. ‘Project proposal submission’ ranked second among the most desired features, as the respondents considered it both very important for managing government projects and very helpful in reducing the time taken to obtain information on available projects at different ministries in one glance. ‘Resource allocation management’ ranked third, as the respondents felt that such a

feature would enable project leaders or managers to allocate project resources more effectively and efficiently. In addition to these top three features, ‘publication and citation database’, ‘researcher expertise profile’ and ‘integration with funding sources’ followed close behind, with an average score of about 50%, which highlights the importance of these features in the national system. Overall, the features chosen by the respondents were mainly those that would enable them to access information related to their research projects and allow them to manage their projects more efficiently.

[Figure 4-13] Features of the R&D management system



Source: Author.

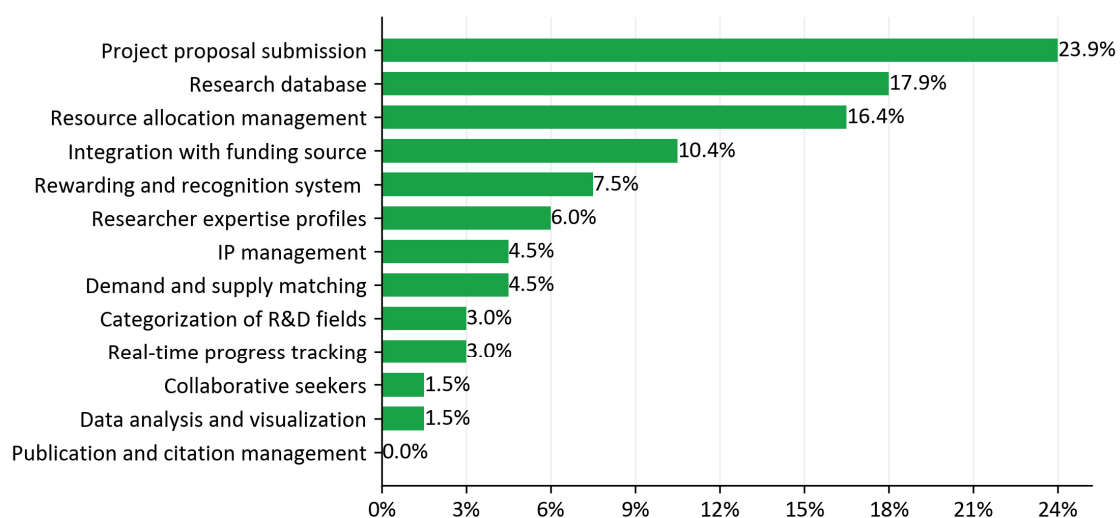
3.3 Ranking the features

The respondents were then asked to rank their top five most desirable features for the system, i.e. those that would best serve their interests and benefit their research activities, on a scale of 1 (most desired) to 5 (least desired). The respondents' top five desired features for the national R&D management system platform are outlined below.

3.3.1 First feature

Based on the survey results, a feature that would enable the submittal of research proposals was the one most frequently chosen as the top priority for inclusion in the system or the platform, as shown in Figure 4-14 below, accounting for about 25%. This was followed by ‘resource allocation management’ and ‘research database’ features, which were also ranked as top priorities.

[Figure 4-14] First feature of the system

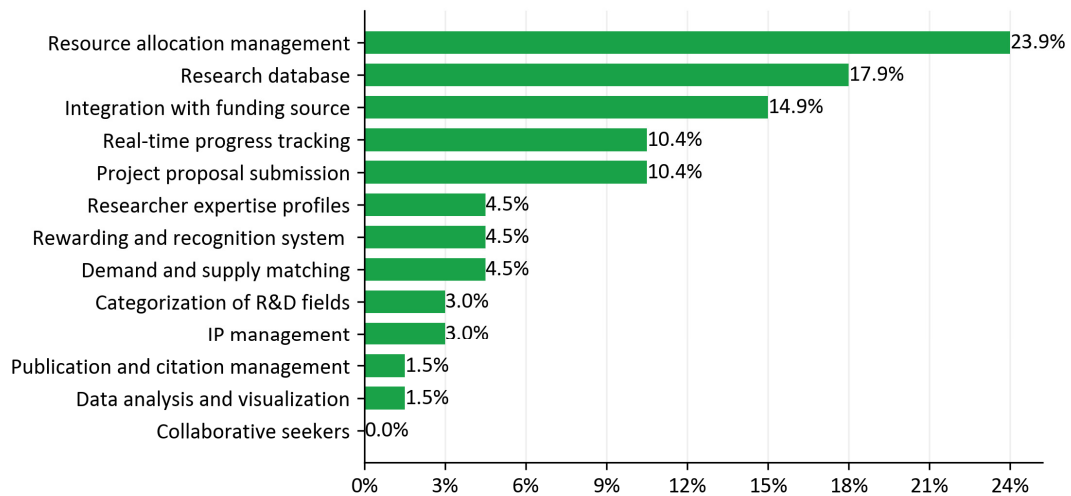


Source: Author.

3.3.2 Second feature

Figure 4-15 below shows the second-ranked feature of the system or platform. The survey results indicate that a “resource allocation management” feature was the most frequently chosen by the researcher and scientific community, accounting for 25% overall. The result is consistent with the previous figure. Meanwhile, ‘research database’ and ‘integration with funding sources’ are also ranked highly.

[Figure 4-15] Second feature of the system

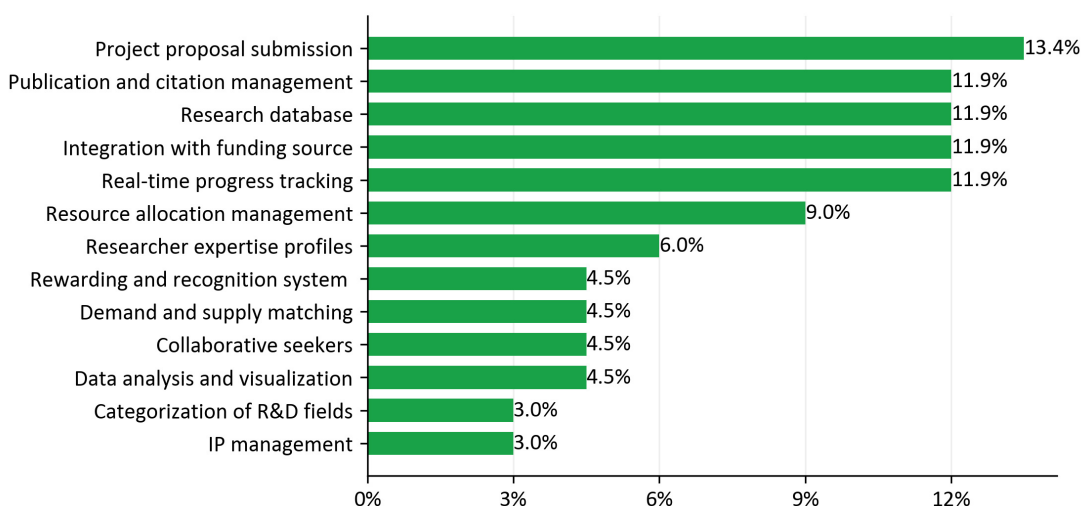


Source: Author.

3.3.3 Third feature

Figure 4-16 below shows a variety of candidate features, including “research database”, “real-time progress tracking”, “publication”, “citation database”, and “integration funding sources”, each of which was chosen by more than 10% of the respondents.

[Figure 4-16] Third feature of the system

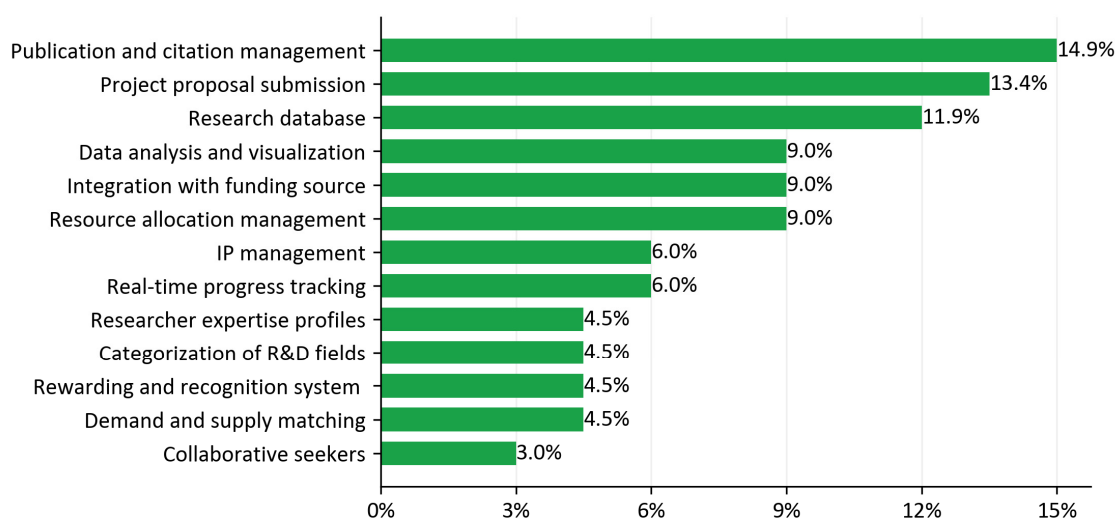


Source: Author.

3.3.4 Fourth feature

According to Figure 4-17 below, the respondents indicated that a “publication and citation database” should be included in the management system or platform above all other features, followed by “project proposal submission” and a “research database”.

[Figure 4-17] Fourth feature of the system

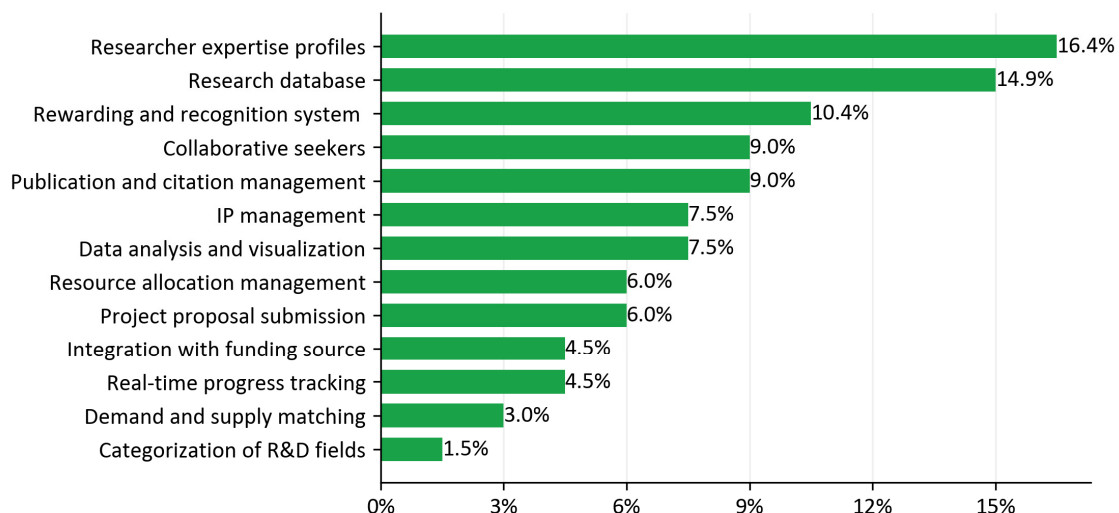


Source: Author.

3.3.5 Fifth feature

Last but not least, Figure 4-18 shows the fifth-ranked feature for the national R&D management system and platform. The results indicate that a “research database” and “researcher expertise profiles” were the most frequently chosen features, each accounting for nearly 20%.

[Figure 4-18] Fifth feature of the system



Source: Author.

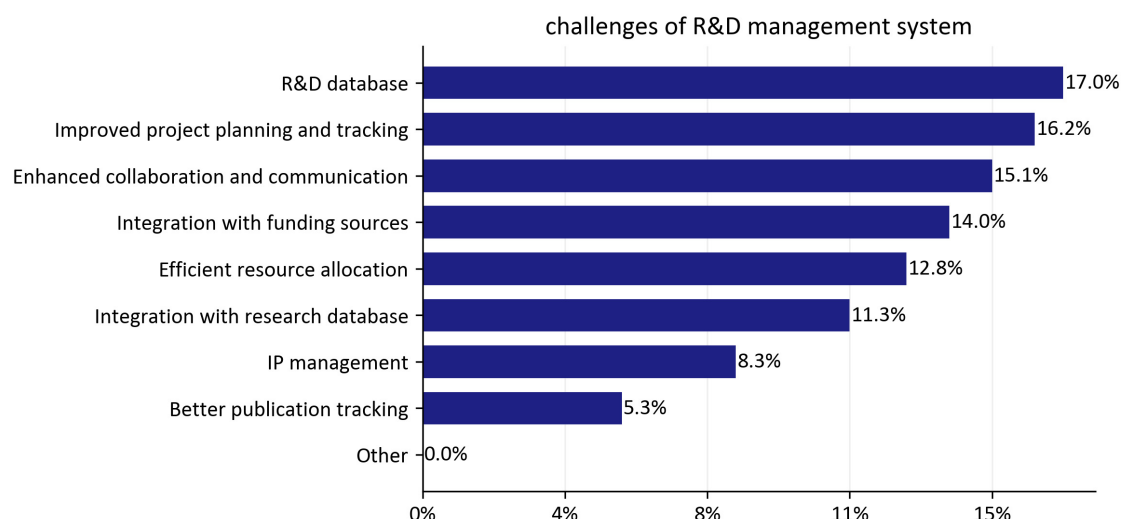
In sum, the top five priority features based on the survey of the researcher and scientific community are as follows: project proposal submission, resource allocation management, research and researcher profile database, publication and citation database, and integration with funding sources. This result also seems to be consistent with the result obtained when the researcher and scientific community were asked to choose their favourite features in the previous question, and also aligns with the inputs received from the roundtable discussions with the government, university, and private sector.

3.4 Challenges that can be addressed by the system

This section aims to identify the potential of the R&D management system to address the challenges facing researchers in terms of project planning and execution. The survey results shown in Figure 4-19 indicate that they could access available research databases, including existing studies, similar projects, expertise in their respective fields, and literature review for the purpose of comparison. Another major impact of both system and platform could be the improvement of project planning and management. As indicated in the previous section, so far most project planning has

been done manually or by using spreadsheets for both project planning and management. As a result, if the system is in place, it could help researchers to plan and manage their projects effectively and efficiently. In addition, enhanced collaboration is another potential advantage offered to researchers by this system as it provides access to researcher profiles, expertise, publication histories, and achievements. In addition, 'integration with funding sources' and 'efficient resource allocation' can also provide great benefits to researchers and scientists from the system. In sum, researchers and scientists anticipate that the system will make a positive contribution to their research activities.

[Figure 4-19] Challenges the R&D management system can address



Source: Author.

3.5 Integration of external sources

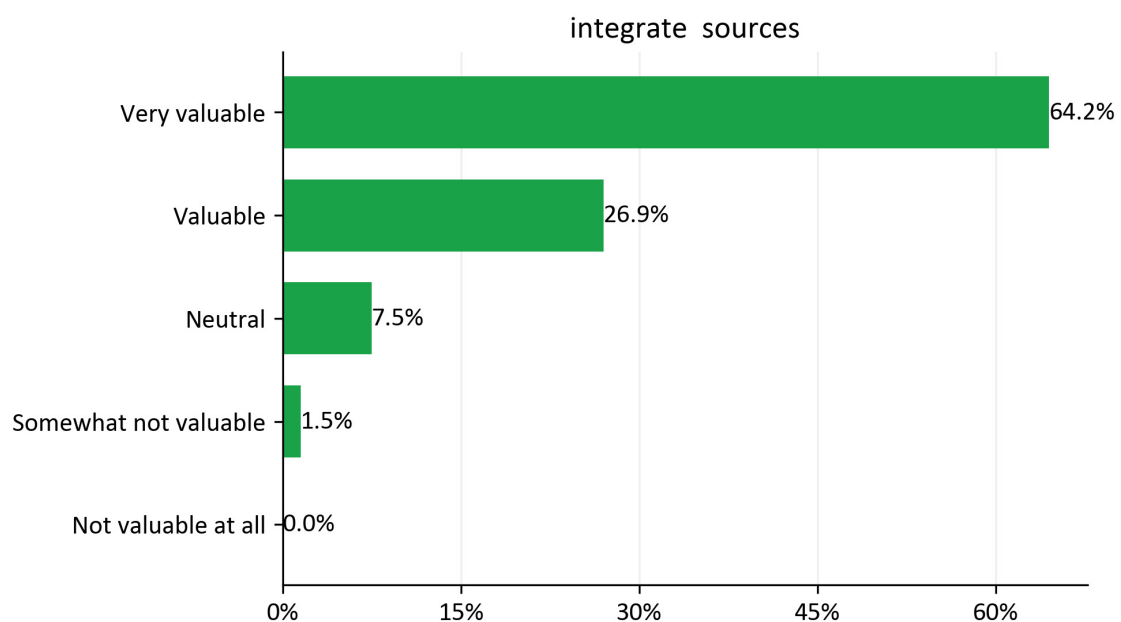
This section explores into the need for integration with external sources and the potential benefits accompanying it.

3.5.1 Integration of funding sources

The survey investigated whether the integration of external funding sources and grant management systems would be valuable for research activities or not. The survey

results shown in Figure 4-20 below suggest that ‘funding source integration’ would be very beneficial, and that more than 60% of the respondents considered it to be ‘very valuable’, while 30% regarded it as ‘valuable’ and less than 10% where either neutral and considered it more or less non-valuable. As indicated in the previous section, most of the respondents have experienced difficulties in securing research funding.

[Figure 4-20] Integration of external funding sources and the grant management system feature

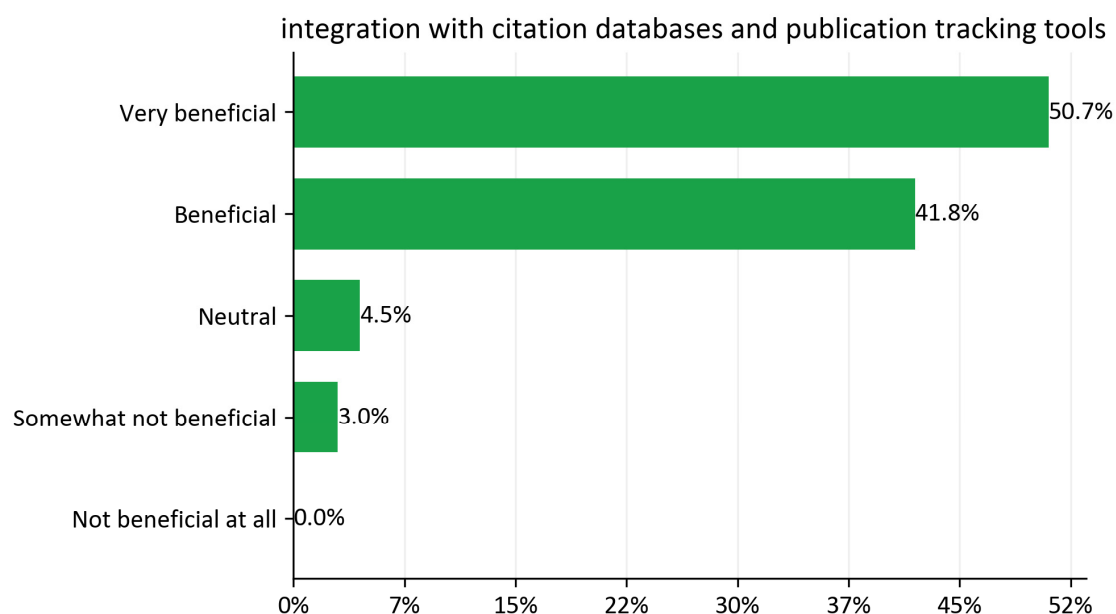


Source: Author.

3.5.2 External citation database

Similar to funding sources, integration with the external citation database is also welcomed by many researchers and scientists as well. The survey results presented in Figure 4-21 indicate that more than 50% and 40% of the respondents suggested that “integration with a citation database” and “publication tracking features” would best serve their interests and be very beneficial their research, while only a small percentage felt otherwise.

[Figure 4-21] Integration with citation database and publication tracking features

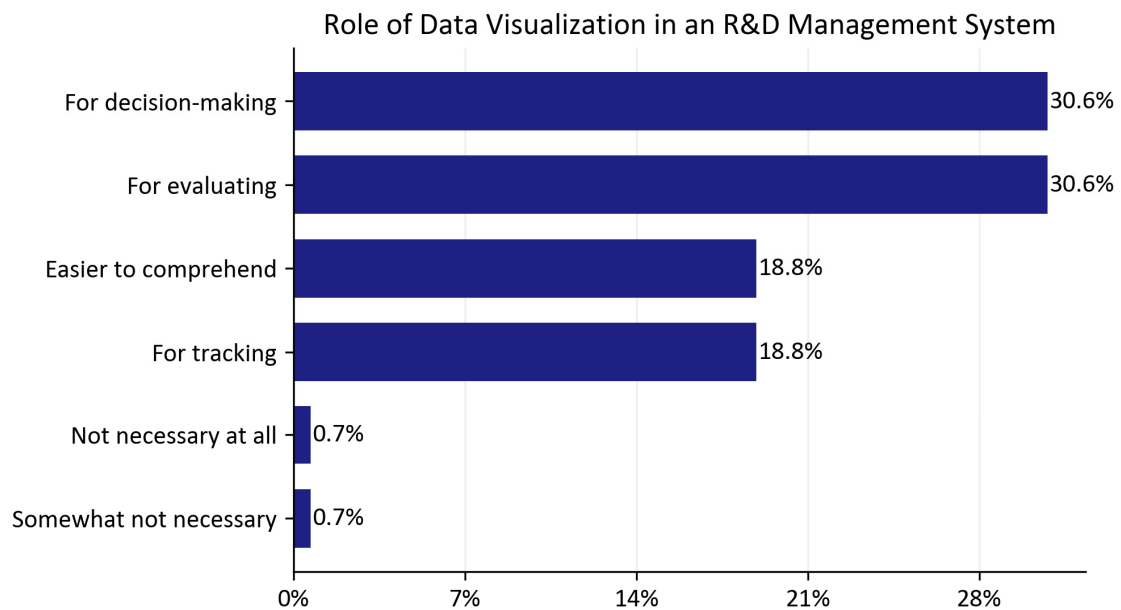


Source: Author.

3.6 Data visualisation

Data visualisation and analytical capabilities were also investigated to determine their importance for the R&D management system platform. The results indicate that most of the researchers and the scientist community strongly agree on the importance of this function for their research activities. As shown in Figure 4-22, more than 60% of the respondents suggested that this functionality would be crucial for evaluating research performance, and that it could also be indispensable as a decision-making tool, as well as for research, business, and policy-making purposes. Some of the respondents suggested that this feature would enable easier comprehension of data analytics as well as making tracking easier. Finally, less than 5% of the respondents asserted that this function is not very necessary or not necessary at all.

[Figure 4-22] Data visualisation and analytical capabilities

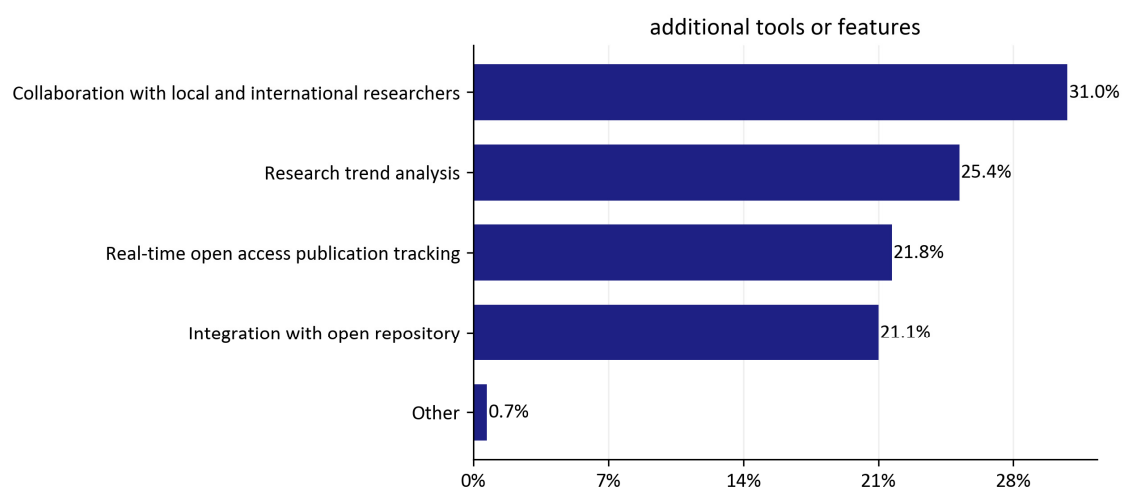


Source: Author.

3.7 Features that support open science

The researchers were also asked to identify any other additional tools or features that the system could incorporate in order to support their research activities, as well as open science initiatives. The results presented in Figure 4-23 suggests that collaboration with local and international researchers is the most desirable feature in this respect one, selected by almost 70% of the respondents, while research trend analysis, real-time open publication tracking, and integration with an open repository were selected by 53%, 46% and 45% of the respondents, respectively, as features to include in the research management system platform.

[Figure 4-23] Additional features for supporting open science initiatives

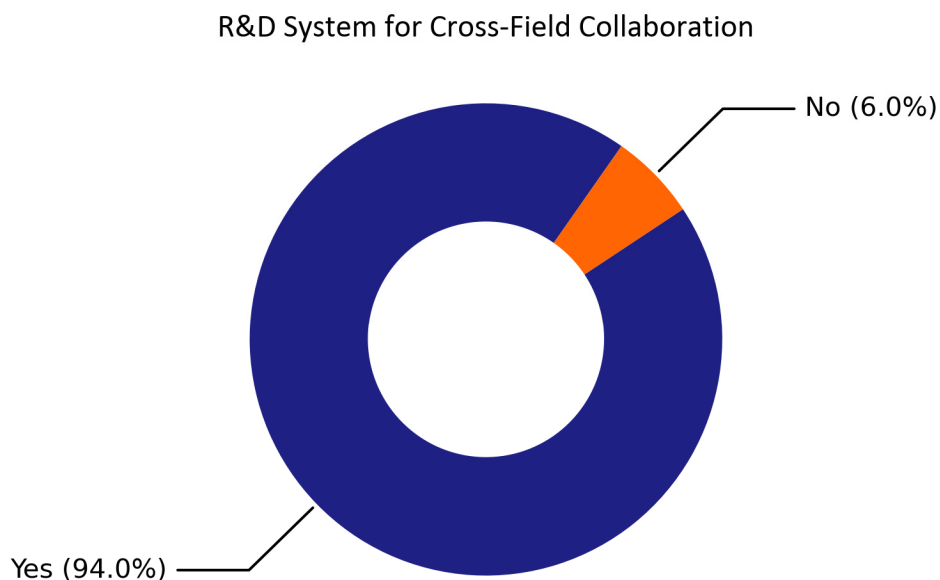


Source: Author.

3.8 Seeking collaborators

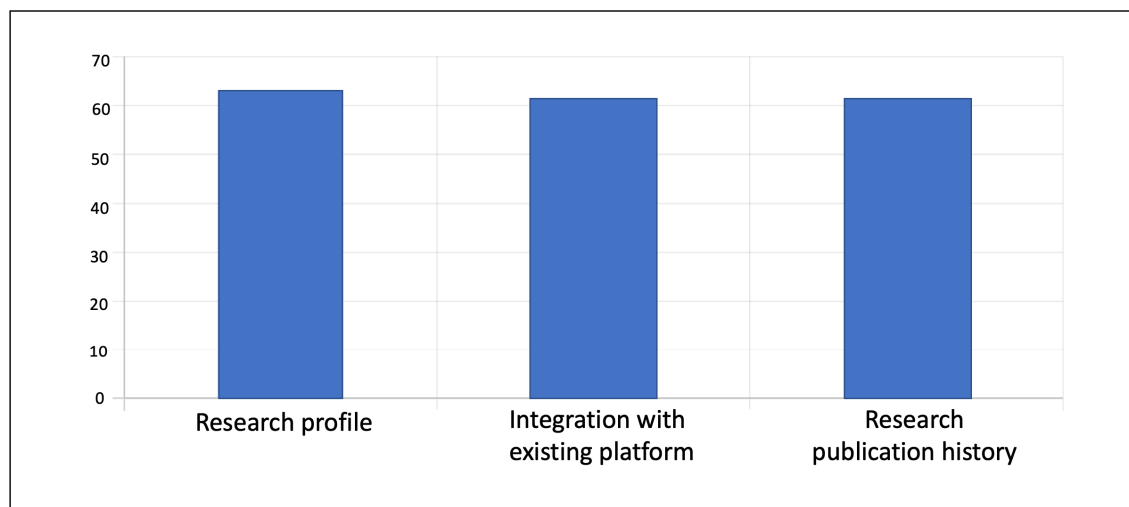
Another feature investigated by the survey is the capability of the system to find collaborators. In fact, the National Research Agenda 2025 states that collaborative research must be promoted in order to enhance innovation. The results shown in Figure 4-24 indicate that more than 90% of the survey respondents positively believe that the R&D management system could assist with the identification of potential research partners or collaborators across different fields. Asked to specify how the system could assist with identification, the respondents suggested, as shown in Figure 4-25, that research partners could be identified through a wide range of data of research profiles, as well as through integration with researcher platforms such as LinkedIn and ResearchGate, and through the researcher publication history (as shown in the figure below).

[Figure 4-24] How the R&D management system assists the identification of potential partners and collaborators



Source: Author.

[Figure 4-25] Methods of seeking collaborators



Source: Author

4. Implementation and Adoption of the NRDMS

4.1 Concerns and Barriers

The implementation of a national research and development (R&D) management system can encounter various concerns and obstacles, as revealed by insights from researchers in the field. These challenges include technical issues, data privacy and security concerns, lack of training, integration challenges, and resistance to change, all of which are clearly illustrated in Figure 4-26.

■ Technical Challenges

Implementing a robust R&D management system often involves using sophisticated software, data analytics tools, and other technical solutions. The figure below shows that a significant 65.67% of respondents expressed concerns about technical challenges, emphasizing that adopting advanced tools and technologies could impede the seamless integration of the R&D system. These technologies can also be complex to set up and maintain, requiring substantial investment in infrastructure and expertise.

■ Data Privacy and Security Concerns

Research and development activities often involve handling sensitive and confidential data, making data privacy and security paramount. Ensuring compliance with data protection regulations and safeguarding against breaches or unauthorized access are significant challenges. The study shows that 61.19% of respondents ranked data privacy and security as the second major concern, especially when handling sensitive research data.

■ Lack of Training

The introduction of a new R&D management system typically requires comprehensive training for the staff and researchers who will use the system. Inadequate training can result in under-utilization, errors, and inefficiencies. About half of the respondents, 49.25%, expressed concerns about the inadequacy of training for staff, researchers,

and public users, which could hinder the effective operation of the national R&D management system.

■ Integration Challenges

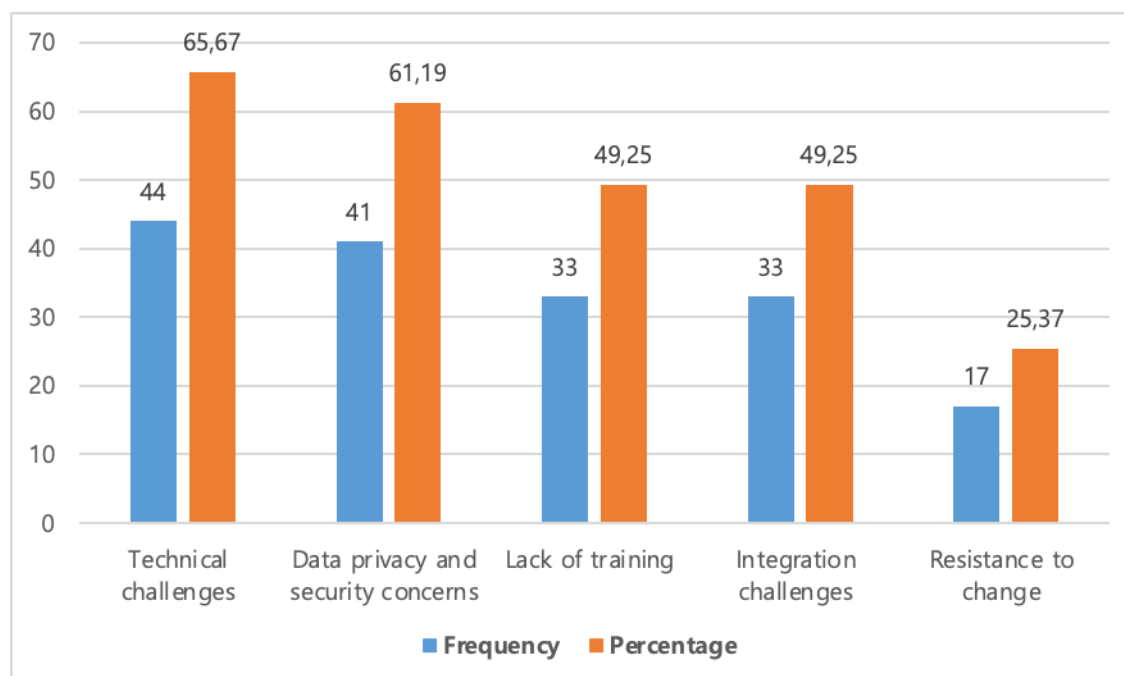
Researchers and users from different organizations, departments, and research teams, including government, higher education, and private sector institutions, use their own processes and systems. Integrating these diverse systems and processes into a unified R&D management system is highly complex. Achieving seamless communication and data exchange between these components is a significant challenge. The findings reveal that 49.25% of respondents recognized integration challenges as a substantial barrier, given the need to harmonize diverse R&D processes and systems across various organizations and departments.

■ Resistance to Change

People and organizations often resist change, especially when it involves departing from established practices and adopting new ways of working. The results indicate that 25.37% of respondents considered resistance to change a significant hurdle. They expressed concern that individuals and institutions might hesitate to embrace new approaches and depart from established practices, particularly when related to new technology and systems.

To overcome these concerns and barriers, it is essential to establish a comprehensive strategy for addressing each issue, promote awareness, and foster collaboration among all stakeholders in the R&D community. This strategy should balance the need for data sharing and collaboration with security measures and provide comprehensive training programs, effective change management strategies, communication, and leadership.

[Figure 4-26] Concerns about and barriers to the R&D management system

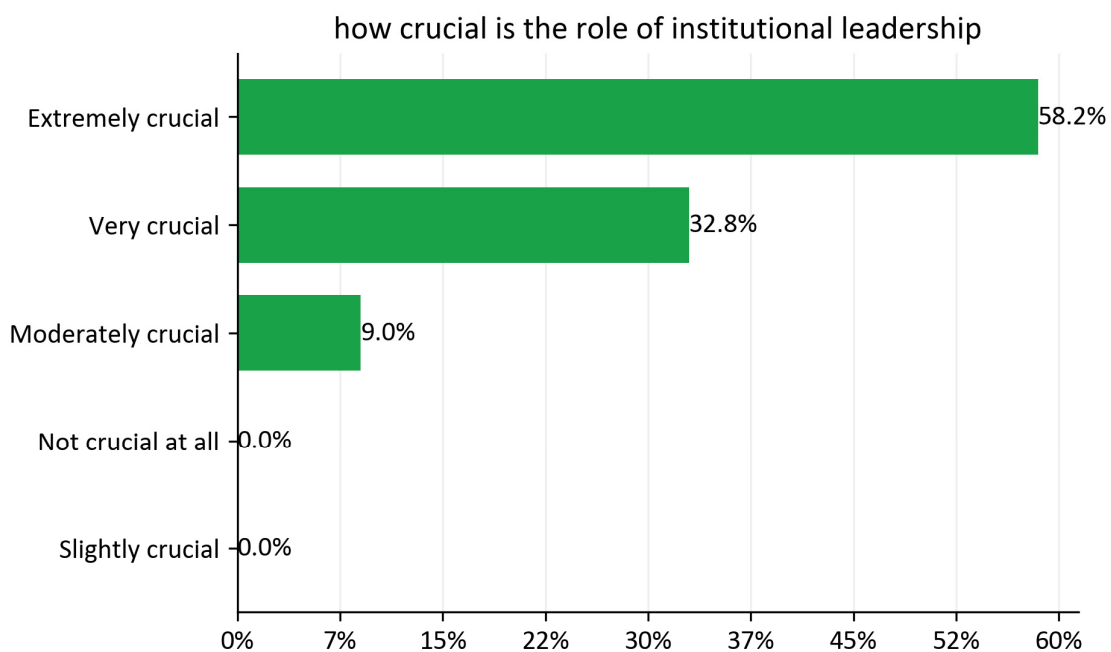


Source: Author.

4.2 Institutional leadership in promoting the adoption of an R&D management system

Institutional leadership plays an indispensable role in championing the adoption of a national research and development (R&D) management system. Institutional leadership will act as the driving force behind the successful implementation of such a system in the future. As indicated by the research findings shown in Figure 4-27, there is evidently a substantial consensus among the respondents, with 58.21% of them deeming 'institutional leadership' to be extremely crucial, 32.84% considering it 'very crucial', and 8.96% regarding it as 'moderately crucial'.

[Figure 4-27] Institutional leadership in promoting the system



Source: Author.

5. Consultation Workshop

After the online survey to collect input from the researcher and scientific community, a consultative workshop was held on 25 October 25, 2023 to have a consultation on the survey results and to discuss the R&D ecosystem in Cambodia. The participants invited to this workshop were experts from diverse organisations including line ministries, higher education institutions and research centres, and the private sector, all of whom have a great deal of expertise, are experienced in research and development, and are very familiar with the research landscape of Cambodia. Three main topics were discussed during the workshop in an effort to gain qualitative information and inputs related to the existing R&D programme/project management practices at the participants' organisations, the main challenges for managing R&D programmes and projects facing their organisations and, last but not least, the features of the ideal national R&D management system.

In each main topic, there are various sub-topics that the experts have delved into, as well as some concerns that need to be taken into account when developing the national R&D management system for Cambodia for the first time. Most of the inputs from the consultative workshop regarding the challenges, and the proposed features of the optimal R&D management system, are aligned with the findings obtained from the survey results. However, the experts also discussed other matters including intellectual property management, platform sustainability, platform promotion, and the challenges and concerns that we may eventually face when the system rolls enters into operation. The inputs of this consultative session could be beneficial when designing the R&D platform or system. Figure 4-28 shows some photos of the activities of the workshop.

[Figure 4-28] Consultative workshop activities



6. Recommendations

This section looks at the research community's conception of the perfect or ideal research and development (R&D) management system, emphasising the essential qualities and functions that the participants held to be essential. These revelations highlight how such a system could completely transform the field of research. According to the results of the study, the following characteristics emerged as the top priorities:

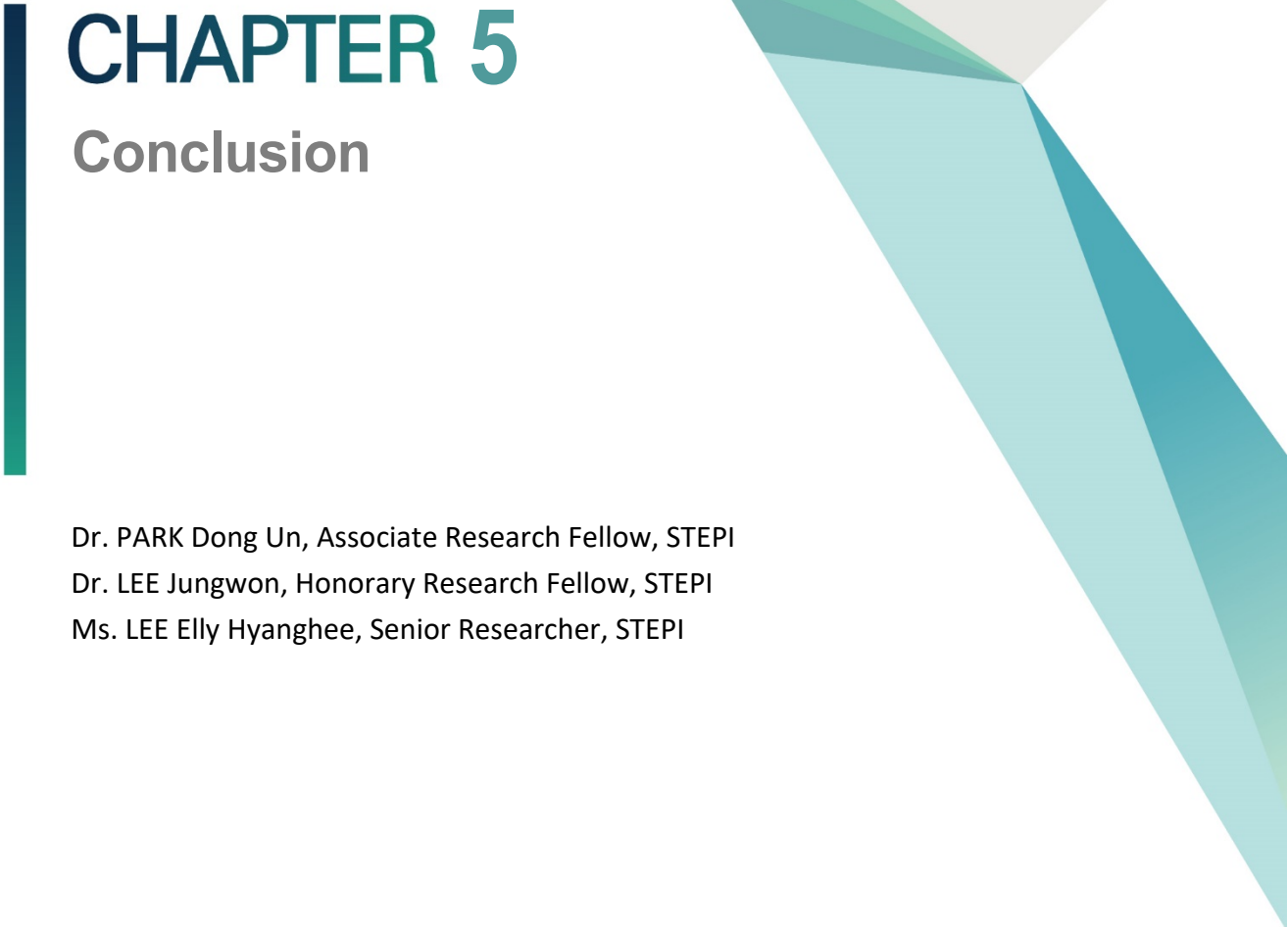
- **Research Database:** Most of the respondents emphasised how crucial it is for the R&D management system to have a comprehensive research database. A centralised repository for keeping and organising study data, methods, conclusions, and all related documentation would be provided by this database. Scholars imagine a dynamic, intuitive platform that would allow them to archive their research findings in a safe and efficient manner. In order to facilitate seamless data retrieval and cooperation, the system should ideally enable the structured storing of diverse types of data ranging from experimental results to book references. This feature is thought to be essential for maintaining research knowledge and encouraging cooperation both within and across research teams and organisations.
- **Project Proposal Submission:** According to the survey respondents, the ideal R&D management system should include an integrated project proposal submission system, as such a feature would make it easier to submit, evaluate, and manage proposals for research projects. Researchers envisage a platform where they can draft and submit proposals for new initiatives, monitor the status of their proposals, and gain inputs from funding organisations or other researchers. It is anticipated that the incorporation of this feature would augment transparency, expedite financing prospects, and encourage the effective distribution of resources towards inventive research undertakings.
- **Resource Allocation Management:** One important component of the R&D management system is resource allocation management. The need to have a structure in place that can distribute resources such as funds, personnel, equipment,

and lab spaces effectively was emphasised by the respondents. Real-time tracking and resource optimisation should be possible with such a system, thus guaranteeing that researchers have all they need to do their work efficiently. They also felt that a dynamic resource allocation system would reduce the waste of resources, improve project efficiency, and eliminate bottlenecks.

- **Publication and Citation Database:** The inclusion of a comprehensive publication and citation database was also seen as fundamental. Researchers desire a system that seamlessly tracks their publications, manages citation data, and supports the referencing of their own work. Such a feature would help to keep researchers informed and up to date about the impacts of their research, simplify citation management, and ensure accurate attribution of intellectual contributions. Furthermore, it could foster collaboration by identifying synergies with other researchers and their work.
- **Researcher Expertise:** The respondents to the survey highlighted the significance of a feature that would allow the system to profile and showcase individual researchers' areas of expertise. With this feature, researchers could highlight their contributions, interests, and abilities, thereby making it easier for potential partners to find experts in a given subject matter. This function could also increase researcher synergy and knowledge sharing by promoting links and multidisciplinary collaboration.
- **Integration with Funding Sources:** One of the top priorities in the opinion of the respondents would be to integrate the R&D management system with different financing sources, given that one goal of research is to optimise the grant application, financing management, and project financial tracking procedures. With such integration, financial resources would flow smoothly, compliance with funding regulations would be made easier, and researchers would have direct access to funding options. It is thought to be necessary for the long-term viability of inventions and research initiatives.

In summary, the research community's ideal R&D management system would include these essential components, each of which would contribute to the establishment of a

smooth, effective, and cooperative research environment. It is strongly believed that, through the survey results and the roundtable discussion with the line ministry, universities, and the private sector, these features will empower researchers, improve research results, and stimulate innovation in a broad range of fields.

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CHAPTER 5

Conclusion

Dr. PARK Dong Un, Associate Research Fellow, STEPI

Dr. LEE Jungwon, Honorary Research Fellow, STEPI

Ms. LEE Elly Hyanghee, Senior Researcher, STEPI

Chapter 5. Conclusion

1. Outcomes and Expected Benefits

The first outcome of the 2023 Programme is the advocacy for the Technology Roadmaps (TRMs) launched in Cambodia on 26 March during the country's first-ever "National STI Day" event. According to Fresh News Asia, "At the opening ceremony, H.E. PM Mr. Hun Sen underlined the importance of the TRMs in serving as a compass for the public and private sectors and the Cambodian people to plan and develop strategies to acquire the right technologies and create technology products that suit the demands of Cambodia." The six TRMs were developed with the support of STEPI's 2nd phase project of the K-Innovation Programme. The first TRMs (Education, Health, and Agriculture) were launched in 2022 during the Covid-19 pandemic. The STEPI team also participated in advocacy activities for three additional TRMs (Digital, Energy, and Tourism), which were launched by the Cambodian government in 2023.

The second major outcome of the 2023 Programme is this policy consulting report, which includes:

- The results of the survey on the national R&D management system in Cambodia, and
- South Korea's best practices in the national R&D management system, particularly concerning the NTIS and COMPAS. The survey is crucial for understanding the current situation and challenges associated with R&D management practices in Cambodia and identifying the local demand for and requirements of the national R&D management system that the Cambodian government aims to create. The

survey results will be used as inputs for designing an optimal national R&D management system that meets the needs and context of Cambodia.

1) Survey

The survey outlined the challenges and barriers encountered in implementing a national research and development (R&D) management system in Cambodia, as identified by researchers in the field. These challenges include technical hurdles, data privacy and security concerns, lack of training, integration challenges, and resistance to change.

To address these concerns, a comprehensive strategy is essential, comprising measures such as promoting awareness, providing comprehensive training programs, and implementing effective change management strategies. Institutional leadership is also crucial in driving the successful implementation of the R&D management system.

Furthermore, the document explores the ideal characteristics and functionalities of the perfect R&D management system as envisaged by the research community. The top priorities include a comprehensive research database for organizing study data, an integrated project proposal submission system to expedite financing prospects, resource allocation management for efficient distribution of resources, and a publication and citation database to track publications and manage citation data. Additionally, features such as researcher expertise profiling, integration with funding sources, and support for collaboration were highlighted as crucial for promoting knowledge sharing and multidisciplinary collaboration within the research community.

In short, the research community's ideal R&D management system would include these essential components to facilitate a seamless, effective, and cooperative research environment. It is strongly believed that through the survey results and the roundtable discussions with the line ministry, universities, and the private sector, these features will empower researchers, improve research outcomes, and stimulate innovation across a wide range of fields.

2) Best practice 1: NTIS

The need for systematic research and development (R&D) management in Cambodia to enhance research performance within a limited investment environment was discussed, comparing Cambodia's situation to that of Costa Rica and South Korea. The importance of establishing a national R&D integrated management system was highlighted. While some universities in Cambodia have their own internal project management systems, there is a broad consensus on the need for a national-level integrated system.

The main recommendations include conducting a pre-feasibility study, building a pilot system, securing funding, and customizing the system to meet Cambodia's needs. The pre-feasibility study should analyze Cambodia's R&D management status, establish target systems and customize them for Cambodia, design a scenario-oriented R&D information management system, plan step-by-step project promotion, and set commercialization targets.

Lastly, establishing an integrated science and technology information system is proposed as a starting point to advance science and technology in Cambodia, aligning it with the country's ICT strategy and leveraging accumulated information.

3) Best practice 2: COMPAS

The potential significance of the Cambodian COMPAS (Cambodian Patent Information System), facilitated through an ODA (Official Development Assistance) program, in promoting domestic R&D activities was highlighted. COMPAS can provide essential resources for R&D activities, particularly in searching and analyzing Intellectual Properties (IPs), which is crucial in a globally competitive environment. Unlike commercial IP resources with subscription fees, COMPAS obtained through ODA can significantly reduce costs for Cambodia while fostering a thriving R&D ecosystem.

Moreover, COMPAS can grant Cambodia independence from potentially exploitative commercial database providers, ensuring control over expenses and avoiding cost escalations. These IP resources will not only enhance COMPAS but also serve as the infrastructure for various applications, supporting tasks such as technology foresight,

evaluation, and Patent Office responsibilities.

Collaboration with South Korea can further enhance STI development in Cambodia through tailored ODA programs. The flexibility of the recipient country in modifying and enhancing acquired resources will enable addressing complex problem-solving tasks, including those involving cutting-edge technologies like Artificial Intelligence (AI). For example, patent text can be used as input for AI applications, enabling advanced analysis and expanding the system's potential applications.

2. Policy Implications and Future Plans

The preliminary study presents the results of the research and development (R&D) management system and provides an overview of the transformative future envisioned for the scientific landscape by the research community in Cambodia. The consensus-building process among survey respondents and the collective insights from experts in the consultative workshop underscore the critical importance of establishing an ideal R&D management system that effectively addresses evolving needs and challenges faced by researchers in the country. Key insights from this survey reflect the collective demand of the research community for a dynamic and collaborative platform that integrates various aspects of the research process.

The emphasis on elements such as a robust research database, a project proposal submission function, a resource allocation management function, a real-time publication and citation tracking function, a database of researcher expertise profiles, and integration with funding sources demonstrates a clear vision for a robust R&D ecosystem that caters to the needs of diverse researchers and scientists. Such an ideal system will serve as a catalyst for creativity, encourage interdisciplinary collaboration, increase transparency, and promote effective resource utilization.

As we navigate the future of the research landscape in Cambodia, it is clear that the ideal R&D management system, as envisioned by the research community, will play a critical role in determining how information is generated, shared, and implemented. The survey results have provided valuable information that will enable stakeholders and developers to align their efforts with the real needs and goals of researchers. To achieve optimal results, widespread use, and long-term sustainability, the proposed system must be adaptable to changing technologies, capable of supporting diverse research methodologies, and, most importantly, user-friendly.

The findings of this report provide a roadmap for developing and implementing an R&D management system aligned with the dynamic and collaborative framework of

research that reflects the needs of the researchers. The shared vision of the research community provides a basis for fostering innovation, promoting technology transfer, leveraging scientific progress, and contributing to the ultimate goal of socio-economic development.

Based on the survey results, the STEPI research team plans to provide consulting on designing the framework of the national R&D management system in 2024. In 2025, STEPI will assist MISTI with developing an operational plan for the national R&D management system. Korean experts from KISTI and public and private experts in R&D management will also be invited to participate in the consulting to meet the program's goals. Additionally, the STEPI team, in cooperation with MISTI, plans to support the development of draft guidelines for national R&D management for the Cambodian government, aiming to improve the governance and ecosystem of R&D management in Cambodia.

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APPENDICES

1. R&D Management System Questionnaire

The Ministry of Industry, Science, Technology & Innovation (MISTI), in collaboration with the Science & Technology Policy Institute (STEPI) of the Republic of Korea, is conducting the survey on the research and development (R&D) management system for Cambodia. We are looking to build the optimal research management platform to support and promote research activities in Cambodia. As such, this survey has been designed to gather your valuable insights and opinions regarding the management of research projects and the systems that support research activities.

The main research objectives are as follows:

1. Assess the current practices and challenges in managing research projects.
2. Identify the features and functions that are most crucial for an ideal R&D management system.
3. Gather your ideas and opinions on how such a system can better serve the research community.

Your participation in this survey is vital. Your experience, preferences, and suggestions will help shape the development of R&D management systems that are tailored to the specific needs of researchers, research teams, and institutions. Please rest assured that your response will be kept confidential and used solely for research purposes.

Thank you for your time and participation.

General Department of Science, Technology and Innovation, MISTI

Section 1: General Information

1. Name:
2. Gender:

- 2.1. Male
 - 2.2. Female
3. Age:
 - 3.1. Under 25
 - 3.2. 25-34
 - 3.3. 35-44
 - 3.4. 45-54
 - 3.5. Over 54
4. Title:
 - 4.1. Mr.
 - 4.2. Ms.
 - 4.3. Mrs.
 - 4.4. Dr.
 - 4.5. Assist. Prof.
 - 4.6. Assoc. Prof.
 - 4.7. Professor
5. Current Position:
6. Email:
7. Affiliated Institution/Organisation:
8. Field/Area of Research:
 1. Agriculture, Forestry, Fisheries
 2. Biotechnology
 3. Chemical Engineering
 4. Chemical Sciences
 5. Civil Engineering
 6. Clinical Medicine
 7. Computer and Information Sciences
 8. Economics and Business
 9. Educational Sciences
 10. Electrical and Electronic Engineering
 11. Environmental Sciences
 12. Environmental Engineering
 13. Food Engineering
 14. Health Sciences
 15. Politics

- 16. Social and Economic Geography
- 17. Sociology
- 18. Other (Please specify:)
- 9. Years of Experience in Research:
 - 9.1. Less than 5 years
 - 9.2. 5-10 years
 - 9.3. More than 10 years
- 10. Highest Academic Qualification:
 - 10.1. Bachelor's degree
 - 10.2. Master's degree
 - 10.3. PhD
 - 10.4. Postdoc researcher
- 11. Are you involved in collaborative research projects with other institutions or researchers?
 - 11.1. Yes
 - 11.2. No
- 12. Does your institution/organisation provide research grants and/or funding?
 - 12.1. Yes
 - 12.2. No
- 13. Where do you publish and/or present your research findings?
 - 13.1. In international peer-reviewed journals
 - 13.2. In national peer-reviewed journals
 - 13.3. At international conferences and workshops
 - 13.4. At national conferences and workshops
 - 13.5. On online research networks
 - 13.6. Other (please specify.....)
- 14. How frequently do you publish your research findings?
 - 14.1. Very frequently (more than 3 times per year)
 - 14.2. Frequently (1-3 times per year)
 - 14.3. Occasionally (once every 2 years)
 - 14.4. Rarely (once every 3-5 years)
 - 14.5. Never
- 15. Have you previously used any form of digital tools or systems for research management?
 - 15.1. If Yes, please specify:

- 15.1.1. Institutional management system
- 15.1.2. Available website/online platform
- 15.1.3. Other (please specify)
- 15.2. No, never

Section 2: Current Practices in Research Systems and Challenges

- 16. What is the nature of your research projects? MC
 - 16.1. Fundamental research
 - 16.2. Applied research
 - 16.3. Experiment laboratory
 - 16.4. Policy research
 - 16.5. Social research
 - 16.6. Clinical research
 - 16.7. Other (please specify)
- 17. Do you or your organisation engage in any collaborative research projects?
 - 17.1. Yes
 - 17.2. No
- 18. How does your organisation seek collaboration on a research project? MC
 - 18.1. Announcement
 - 18.2. Networking
 - 18.3. Former Colleagues
 - 18.4. Online Platform (Researchgate, LinkedIn, Google Scholar)
 - 18.5. Social Media (FB, IG, Twitter, Telegram, Line etc.)
 - 18.6. MoU
 - 18.7. Other (please specify.....)
- 19. What is the primary source of funding for your research?
 - 19.1. Government funds
 - 19.2. Development partner
 - 19.3. Private sector
 - 19.4. Call for proposal
 - 19.5. Fund competition
 - 19.6. Other (please specify))
- 20. How do you proceed with intellectual property (IP) considerations related to your research outcomes?

- 20.1. Ministry
- 20.2. IP office on-campus
- 20.3. Provincial department
- 20.4. One-window service
- 20.5. Intermediary and broker
- 20.6. International organisation
- 20.7. Other (please specify.....)
- 21. What challenges do you face in your current R&D management?
 - 21.1. Inefficiency in resource allocation
 - 21.2. Difficulty in gaining access to funding
 - 21.3. Difficulty in seeking collaboration
 - 21.4. Complicated administrative procedure
 - 21.5. Inefficient data management
 - 21.6. Other (please specify.....)
- 22. How challenging is it to secure research funding for your projects?
 - 22.1. Very challenging
 - 22.2. Somewhat challenging
 - 22.3. Neutral
 - 22.4. Not very challenging
 - 22.5. Not challenging at all
- 23. How does your organisation report its annual R&D activities and performance?
 - 23.1. Manual submission of report by staff
 - 23.2. Open online platform
 - 23.3. Institutional platform
 - 23.4. Not relevant
- 24. Are there any procedures for managing (e.g. registering, verifying) research outcomes in your organisation?
 - 24.1. Yes
 - 24.2. No

If Yes, please explain:
- 25. What tools or methods do you use to track the progress and milestones of your research projects?
 - 25.1. Manual tracking method like spreadsheet
 - 25.2. Institutional tracking system
 - 25.3. Online tracking system

- 25.4. Project management tools
- 25.5. Other (please specify.....)
- 26. In your opinion, do you think that the National R&D Management System is important for researchers and research institutions in Cambodia?
 - 26.1. Very important
 - 26.2. Important
 - 26.3. Neutral
 - 26.4. Not very important
 - 26.5. Not important at all

Section 3: Ideal R&D Management System

- 27. In your opinion, what specific features could an R&D management system offer to researchers like yourself (please rank them from 1-5 in order of importance)?
 - 27.1. Project proposal submission
 - 27.2. Resource allocation management
 - 27.3. Real-time progress tracking
 - 27.4. Integration with funding source
 - 27.5. Integration with research databases
 - 27.6. Data analysis and visualisation
 - 27.7. Publication and citation management
 - 27.8. Seekers of collaboration
 - 27.9. Demand and supply matching
 - 27.10. IP management
 - 27.11. Reward and recognition system
 - 27.12. Categorisation of R&D fields
 - 27.13. Researcher expertise profiles
- 28. Please rank the abovementioned functions from 1-5 according to their importance.
 - 28.1. Project proposal submission
 - 28.2. Resource allocation management
 - 28.3. Real-time progress tracking
 - 28.4. Integration with funding sources
 - 28.5. Integration with research databases
 - 28.6. Research database

- 28.7. Data analysis and visualisation
- 28.8. Publication and citation management
- 28.9. Seekers of collaboration
- 28.10. Demand and supply matching
- 28.11. IP management
- 28.12. Reward and recognition system
- 28.13. Categorisation of R&D fields
- 28.14. Researcher expertise profiles
- 29. What challenges do you believe an R&D management system could address in terms of project planning and execution(MC)?
 - 29.1. Improved project planning and tracking
 - 29.2. Enhanced collaboration and communication
 - 29.3. Efficient resource allocation
 - 29.4. Integration with funding sources
 - 29.5. Better publication tracking
 - 29.6. R&D database
 - 29.7. IP management
 - 29.8. Other (please specify.....)
- 30. Would integration with external funding sources and grant management systems be valuable for your research work?
 - 30.1. Very valuable
 - 30.2. Valuable
 - 30.3. Neutral
 - 30.4. Not very valuable
 - 30.5. Not valuable at all
- 31. What role should data visualisation and analytic capabilities play in an R&D management system?
 - 31.1. Tracking
 - 31.2. Evaluation
 - 31.3. Decision- making
 - 31.4. Easier to comprehend
 - 31.5. Not very necessary
 - 31.6. Not necessary at all
- 32. Would integration with citation databases and publication tracking tools be beneficial for your research?

- 32.1. Very beneficial
 - 32.2. Beneficial
 - 32.3. Neutral
 - 32.4. Not very beneficial
 - 32.5. Not beneficial at all
33. What additional tools or features could an R&D management system offer to support open science initiatives? (MC)
- 33.1. Integration with open repositories
 - 33.2. Real-time open access publication tracking
 - 33.3. Collaboration with local and international researchers
 - 33.4. Analysis of research trends
 - 33.5. Smart search
 - 33.6. Other (please specify.....)
34. Could an R&D management system assist with the identification of potential research partners or collaborators across different fields?
- 34.1. Yes, through integration with platforms like LinkedIn and Researchgate
 - 34.2. Yes, through researcher profiles
 - 34.3. Yes, through researcher publication histories
 - 34.4. No, not at all
35. Are there any features you consider essential that have not been covered in the previous questions?
36. Is there anything else you would like to share about your expectations and hopes for an R&D management system?

Section 4: Challenges Facing the R&D Management System

37. What concerns or barriers might researchers and institutions face when transitioning to the R&D management system?
- 37.1. Technical challenges
 - 37.2. Resistance to change
 - 37.3. Lack of training
 - 37.4. Data privacy and security concerns
 - 37.5. Integration challenges
 - 37.6. Others (please elaborate.....)

38. In your opinion, how crucial is the role of institutional leadership in promoting the adoption of an R&D management system?
- 38.1. Extremely crucial
 - 38.2. Very crucial
 - 38.3. Moderately crucial
 - 38.4. Slightly crucial
 - 38.5. Not crucial at all

Do you have any more suggestions or recommendations about the national R&D management system?

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2. Respondents of the Survey

No	Position	Organization/Affiliation
1	Project Lead	University California San Francisco
2	Lecturer	Royal University of Phnom Penh
3	Member	Ministry of Commerce, Cambodia
4	Researcher	Cambodia Academy of Digital Technology
5	Research Coordinator	Cambodia Academy of Digital Technology
6	Deputy Chief Officer	MPWT
7	Researcher and Lecturer	CADT
8	Dean	Pannasastra University of Cambodia
9	Director	CDRI
10	Researcher	Water and Environment Research Unit
11	Deputy Director	NISTI/MISTI
12	Head research unit	ITC
13	Researcher/Lecturer	CADT
14	Senior Research Fellow	ADIC
15	Lecturer	Institute of technology of Cambodia
16	Director	Green Move Consulting
17	Data Science Researcher	CADT
18	Director	Institute of Police Science Research, Police Academy of Cambodia
19	Deputy Chief Bureau of National	Ministry of Public Works and Transport
20	Post-Doc researcher	Chonnam National University
21	Chief of Research and development	National Institute of Science, Technology and Innovation (NISTI)
22	Chief Officer	National Institute of Science, Technology and Innovation
23	Assistant to President	Royal Academy of Cambodia
24	iMBA Student	Panyapiwat Institute of Management
25	GIS and Remote Sensing Specialist	Wildlife Conservation Society (WCS)
26	Lead Lecture Research Methodology	University of Puthisastra
27	Deputy Head of Science Department	Phnom Penh Teacher Education College
28	Lecturer researcher	Institute of Technology of Cambodia
29	Researcher	Institute of Technology of Cambodia
30	Officer	CCCA3
31	Data Analyst	University Research Co., LLC
32	Vice President	CamTech
33	Lecture	Phnom Penh Teacher Education College
34	Lecturer	Planning and Management Département
35	Director of Research	Cambodia Development Resource Institute
36	Program coordinator	Royal University of Phnom Penh
37	Lecturer-researcher	ITC
38	Lecturer-Researcher	Institute of Technology of Cambodia
39	Lecturer and Researcher	Institute of Technology of Cambodia
40	Lecturer	Paññāsāstra University of Cambodia
41	Program and Admin Officer	The Center for Khmer Studies
42	Assistant Professor	Institute of Technology of Cambodia
43	Lecturer	Royal University of Agriculture
44	researcher	University of Hong Kong
45	Assistant Professor and Group Leader	Royal University of Phnom Penh
46	Professor of French Literature and	Hamilton College

47	Senior Economist	N
48	English Instructor	Army Institute
49	Deputy head of department	RUPP
50	Head of Research Unit	Institute of Technology of Cambodia
51	Person in Charge, Graduate School	Royal University of Phnom Penh
52	Vice dean	Royal University of Phnom Penh
53	Lecturer and researcher	Royal University of Phnom Penh
54	student	RUPP
55	dean	ITC
56	Research Fellow	Research Fellow
57	Under Secretary of State	Ministry of Planning
58	Government Employee	Ministry of Planning
59	PhD Candidate	University of California, Riverside
60	Technical Manager	SchneiTec
61	Executive Director	Center for Sustainable Water
62	Researcher	MISTI
63	Deputy Secretary General	Ministry of Agriculture Forestry and Fisheries
64	Uss	Misti
65	Head Dynamics and Control Laboratory	Institute of Technology of Cambodia
66	Deputy Director General	MISTI
67	Director of Department	MISTI